

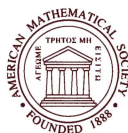
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Gauge Theory on Compact Surfaces

Ambar Sengupta



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Abstract

The quantum Yang-Mills field measure for gauge theory with bundles of specified topology (possibly non-trivial) over compact surfaces, with compact structure group G , is constructed using a Gaussian measure conditioned to satisfy constraints imposed by the topologies of the surface and of the bundle. This is a probability measure which, in a heuristic sense, lives on the space of gauge equivalence classes of connections on the bundle. In this setting, the holonomy around a loop should become a random variable with values in G . We define this stochastic holonomy and compute expectation values of functions of the holonomies associated to a very general class of loops. These values involve topological invariants associated to the bundle and the system of loops on the surface. The effect on loop expectation values of area-preserving homeomorphisms of the surface is determined; it is found that such a homeomorphism transforms loop expectation values into the corresponding values for the pullback bundle. For the construction of the quantum field measure we develop techniques to classify principal bundles over compact surfaces and connections on such bundles up to bundle automorphisms.

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1. Introduction

In this paper we develop a concrete description of connections on principal bundles, possibly non-trivial, over compact surfaces and use this description to construct the Yang-Mills measure which underlies the Euclidean quantum theory of gauge fields, involving compact gauge groups, on compact connected two-dimensional Riemannian manifolds (possibly with boundary). Using this measure we compute expectation values of important random variables, the Wilson loops variables, corresponding to a broad class of configurations of loops on the surface.

The problem of constructing the Yang-Mills measure is as follows. A probability measure μ is to be constructed on an infinite dimensional manifold $\mathcal{C}$ (whose points are gauge-equivalence classes of connections on a principal bundle) starting from a formal expression describing μ . The strategy we use for constructing μ is to view $\mathcal{C}$ as being a submanifold of $\mathcal{H} \times G^p$, where $\mathcal{H}$ is a Hilbert space and G^p is a suitable number of copies of the structure group G . Then the formal expression for $d\mu$ looks like the product of Gaussian measure on $\mathcal{H}$ with Haar measure on G^p . Taking this as motivation, we define the Yang-Mills measure to be the product measure on $\mathcal{H} \times G^p$ conditioned to lie on the “submanifold” $\mathcal{C}$. The embedding of $\mathcal{C}$ in $\mathcal{H} \times G^p$ involves cutting up the surface, and the conditioning is specified by the topology of the surface and of the bundle. Having constructed μ , we are able to compute explicitly expectation values $\int_{\mathcal{C}} f d\mu$ of important functions f on $\mathcal{C}$ (these f are functions of holonomies around various loops; i.e. they are the Wilson loop variables). The expectation values involve certain topological invariants associated to systems of curves on the surface and to the bundle. The expectation values themselves are invariant under (well-behaved) area preserving orientation-preserving homeomorphisms of the surface. The effect of orientation reversing, but area-preserving, homeomorphisms is essentially to map the Yang-Mills measure on the given bundle to the corresponding measure for another bundle, which we specify.

To give a simple example, let G be a compact connected Lie group whose Lie algebra is equipped with an Ad-invariant metric, and consider a principal G -bundle $\pi : P \rightarrow \Sigma$ over the torus Σ . Fix a point $o \in \Sigma$ and a point $u \in \pi^{-1}(o)$. Denote by $g_u(C; \omega)$ the holonomy, with initial point u , of a connection ω on P around a loop C based at o . Suppose, for the sake of illustration, that C is a contractible (and well-behaved) loop on Σ . Under the Yang-Mills measure μ , the holonomy $g_u(C; \omega)$ is a G -valued random variable in ω . Let $\tilde{G} \rightarrow G : \tilde{x} \mapsto x$ be the universal cover of G . Let $\tilde{Q}_t(\tilde{x})$ denote the density, with respect to a Haar measure $d\tilde{x}$ on $\tilde{G}$, of the standard Brownian motion $b_t(\tilde{x})$ on $\tilde{G}$ at time t ; thus $\tilde{Q}_t(\tilde{x})$ is a multiple of the heat kernel on $\tilde{G}$. Our loop expectation value results imply, in this situation, that if f is a bounded measurable function on G , then :

$$\int f(g_u(C; \omega)) d\mu(\omega) = Z_h^{-1} \int_{G^2 \times \tilde{G}} f(x) \tilde{Q}_A(\tilde{x}) \tilde{Q}_{|\Sigma|-A}(h\tilde{b}^{-1}\tilde{a}^{-1}\tilde{b}\tilde{a}\tilde{x}^{-1}) dadbd\tilde{x}$$

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where da and db are the Haar measure of unit total mass on G , A is the area of the contractible region enclosed by C , $|\Sigma|$ is the area of Σ , Z_h is a normalizing constant, and h is an element of $\ker(\tilde{G} \rightarrow G)$ which specifies the topological class of the bundle $\pi : P \rightarrow \Sigma$. If C is replaced by ϕC (joined to o by an appropriate ‘leg’), where ϕ is a nice area-preserving orientation-preserving homeomorphism $\Sigma \rightarrow \Sigma$, then there is no change on the right side of the above formula; if ϕ is orientation reversing then h on the right is replaced by h^{-1} .

More generally, if $C_1, \dots, C_k$ are good loops on Σ , all based at o , and G is semi-simple or if Σ is not orientable, then Theorem 8.4 provides an explicit formula for the expectation value $\int f(g(C_1; \omega), \dots, g(C_k; \omega)) d\mu(\omega)$. This expression involves the areas of the regions on Σ cut out by $\{C_1, \dots, C_k\}$ and topological invariants of $\{C_1, \dots, C_k\}$ on Σ , and the topology of the bundle. For instance, suppose Σ is compact oriented of genus g and without boundary, and G is semisimple with universal cover $\tilde{G}$ equipped with Haar measure of unit total mass. Let $\pi : P \rightarrow \Sigma$ be a principal G -bundle as before. Let $\{\Delta_1, \dots, \Delta_n\}$ be the positively oriented two-simplices of a triangulation of Σ . Let $\mathcal{E} = \{e_1, \bar{e}_1, \dots, e_M, \bar{e}_M\}$ be the set of oriented one-simplices of the triangulation. We shall use maps $\mathcal{E} \rightarrow \tilde{G} : b \mapsto \tilde{x}_b$ for which $\tilde{x}_{\bar{b}} = \tilde{x}_b^{-1}$, where $\bar{b}$ denotes the reverse of the oriented 1-simplex b . If C is a closed curve which is the composite $b_r \dots b_1$ of 1-simplices then we write $\tilde{x}(C)$ to mean $\tilde{x}_{b_r} \dots \tilde{x}_{b_1}$. Thus $\tilde{x}(\partial\Delta_i) \in \tilde{G}$, where $\partial\Delta_i$ is the boundary of Δ_i with some choice of initial point. We write x_b for the element of G covered by $\tilde{x}_b \in \tilde{G}$; so $x(C) \in G$ is covered by $\tilde{x}(C) \in \tilde{G}$. Suppose now that $C_1, \dots, C_k$ are loops, all based at $o \in \Sigma$ (which is assumed to be a 0-simplex of the triangulation), are each composites of oriented 1-simplices. Assume also that all curves involved form a well-behaved collection (in a sense explained in Definition 8.1). Then Theorem 8.4, specialized to this situation, says :

$$\int f(g(C_1; \omega), \dots, g(C_k; \omega)) d\mu(\omega) = \frac{1}{Z_h} \int_{\tilde{G}^M} f(x(C_1), \dots, x(C_k)) \prod_{i=2}^n \tilde{Q}_{|\Delta_i|}(\tilde{x}(\partial\Delta_i)) \tilde{Q}_{|\Delta_1|}(h\tilde{x}(\partial\Delta_1)) d\tilde{x}_{e_1} \dots d\tilde{x}_{e_M}$$

where

$$Z_h = \int_{\tilde{G}^{2g}} \tilde{Q}_{|\Sigma|}(|h\tilde{b}_g^{-1}\tilde{a}_g^{-1}\tilde{b}_g\tilde{a}_g \dots \tilde{b}_1^{-1}\tilde{a}_1^{-1}\tilde{b}_1\tilde{a}_1|) d\tilde{a}_1 \dots d\tilde{b}_g.$$

The element $h \in \tilde{G}$, specifying the topological class of P , is the ‘evaluation’ of an element of $H^2(\Sigma; \pi_1(G))$ on a suitably chosen generator of $H_2(\Sigma; Z)$. If Σ is unorientable then $\tilde{G}$ is replaced by $\tilde{G}/K^2$ where $K^2 = \{x^2 : x \in \ker(\tilde{G} \rightarrow G)\}$. If Σ has boundary then the bundle is trivial, $\tilde{G}$ is replaced by G and $h = e$.

If each loop C_i is replaced by ϕC_i , where $\phi : \Sigma \rightarrow \Sigma$ is an area-preserving homeomorphism which maps the C_i ’s into a well-behaved family of loops, then, according to Theorems 8.4 and 10.1, the effect on the expectation value formula above is that the Yang-Mills measure μ for the bundle $P \rightarrow \Sigma$ is replaced by the Yang-Mills measure for the pullback bundle $\phi^*P \rightarrow \Sigma$.

Expanding $\tilde{Q}_{|\Sigma|}$ in terms of the irreducible characters of $\tilde{G}$ we obtain (see Remark 6.5 for a detailed argument) the following, perhaps more explicit, expression for the normal-

izing constant Z_h :

$$Z_h = \sum_{\alpha} \frac{e^{-C_{\alpha}|\Sigma|/2} \chi_{\alpha}(h)}{(\dim \alpha)^{2g-1}},$$

where the sum is over all inequivalent irreducible unitary representations α of $\tilde{G}$. The formula for Z_h given above is similar to, and agrees in its dependence on $|\Sigma|$ with the expression for the ‘partition function’ of quantum gauge theory found by Witten (formula (4.42) in [Wi 2]; u^{-1} in [Wi 2] corresponds to our h).

This paper is organized as follows. In Section 2 we state the basic problem precisely and set up some notation. In Section 3 we classify principal G -bundles over a compact surface and then show that connections on such bundles are specified, up to bundle automorphisms, by their curvatures and holonomies around a basic set of loops. In Section 4 we describe quantum gauge theory on the disk. The basic conditional probability measure is constructed in Section 5, along with tools for computing expectation values. The Yang-Mills measure is defined in Section 6. In Section 7 we construct topological invariants associated to systems of curves on surfaces; these are needed in Section 8. In Section 8, loop expectation values are computed under certain hypotheses on the surface and on the group. In Section 9, tools are developed for the calculations in Section 10. In Section 10, the behavior of loop expectation values for general compact groups (including the ones not covered by Section 8) under homeomorphisms is determined. In the Appendix we prove a large number of technical results which are used at various points in this work.

The strategy used in the present work for constructing the Yang-Mills measure and for computing loop expectation values evolved from the ideas developed in the papers [Se1,2]. In [Se2] this theory was developed in the case the gauge group is simply-connected; in that setting all the bundles are trivial. In [Se1] the Yang-Mills measure was constructed for non-trivial bundles over the two-sphere; the results there, as well as those of [Se2], become special cases of the present work. In the present work we are able to combine the topology of the bundle with the probabilistic techniques developed in [Se1,2]. There is considerable overlap with [Se 1,2] but we have tried to make the present work fairly self-contained. Thus generalizations of results of [Se1,2] which are needed in the present work are proved here except in a few cases. An examination of [Se 1,2] will reveal the similarities and differences in the places of overlap and we will not spell out these in detail anywhere.

This work and [Se 1,2] evolved from the works [GKS] and [Dr] which treat the planar gauge theory (this is the content of our Section 4 where we prove the results of [GKS] and [Dr] by different methods). While the fundamental measure in that situation is Gaussian, computing loop expectation values is a substantial problem. Some of the techniques we use in computing loop expectation values were motivated by the method used in [Dr] for the planar case. The concept of stochastic parallel translation used in this context was introduced in the works [GKS] and [AHH]. Other works on planar gauge theory have been based on the lattice approach; in this approach one essentially guesses what the loop expectations should be. A brief survey along with a list of earlier works appears in [Se2]. In [Fi1,2] Fine has studied quantum gauge theory on compact surfaces for trivial bundles using a continuum approach which is different from ours; his results agree with ours where they overlap. Witten [Wi1] and Rusakov [Ru] have described quantum gauge theory for compact surfaces using the lattice approach. In [Wi2] Witten takes into account non-trivial

bundle topology, but the approach and goals are different from ours. Recently, D. Fine has informed me that in the paper [Fi3] he has extended his methods to the case of non-trivial bundles. In this approach, Fine has analysed the formal expression for the Yang-Mills functional integral and deduced loop expectation values for a family of loops including in particular the standard generators of the fundamental group of the surface. The values determined by Fine appear to agree, up to choice of scale, with the values implied by the results of our approach specialized to the loops considered in [Fi3].

As far as we know, the present paper is the only one in which :

- (i) the infinite-dimensional continuum quantum gauge field measure is constructed in the presence of non-trivial bundle topology (as mentioned above, [Se1] deals with such bundles over the two-sphere);
- (ii) loop expectation values are derived from the quantum field measure for a class of loops broad enough that the holonomies around such loops specify connections up to gauge equivalence (these expectation values are consistent with certain formulas in [Wi1,2] and [Fi3]).

The formal expression for the Yang-Mills measure is $d\mu(\omega) = Z^{-1}e^{-S(\omega)}[D\omega]$ (Section 2 explains the notation). An interesting problem concerning the Yang-Mills measure is the investigation of the limit of the measures $d\mu_T(\omega) = Z_T^{-1}e^{-S(\omega)/T}[D\omega]$, as $T \downarrow 0$. The works [Fo, KS1,2, Se 3,4,5, Wi 1,2] relate to this. The work [Fo] is devoted to showing that if G is simply-connected and Σ is oriented, without boundary, and has genus bigger than one, then μ_T converges, in a certain sense (comparable to the convergence of finite dimensional distributions), to a probability measure μ_0 on the moduli space of Yang-Mills minima on the bundle; using results from [Wi1], the measure μ_0 is identified in [Fo] with a symplectic measure on this moduli space. In the case where G is semisimple but not simply connected, it is natural to expect that μ_T , as constructed in the present work, should converge to the measure μ_0 for each particular topological class of bundles. This is proven, for the case of $SO(3)$ -bundles and oriented surfaces of genus ≥ 1 , in [Se 4,5] using an explicit description of a natural symplectic structure, and of the corresponding volume form, on the moduli space of flat connections obtained in [KS1,2].

The reference list for this paper includes mostly works which have direct use in or influence on the present paper. More comprehensive bibliographies are available in [GKS, Se2, Wi1].

A preprint version of this work contained a section (section 11) on Yang-Mills minima which is expected to appear as a separate paper.

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2. Terminology and Basic Facts

We set up notation and review some basic facts concerning surfaces, bundles and connections. Let G be a connected Lie group. The Lie algebra of G will be denoted $\underline{g}$, and we will assume that it is equipped with an Ad-invariant metric $\langle \cdot, \cdot \rangle_{\underline{g}}$. We shall work with a compact connected two-dimensional Riemannian manifold (possibly with boundary) Σ and a principal G -bundle $\pi : P \rightarrow \Sigma$. The affine space of all smooth connections on P will be denoted $\mathcal{A}$. The group of all smooth automorphisms (covering the identity $\Sigma \rightarrow \Sigma$) of the bundle P will be denoted by $\mathcal{G}$, and $\mathcal{G}_m$ will denote the subgroup consisting of all $\phi \in \mathcal{G}$ for which $\phi(x) = x$ for some (hence all) x in the fiber over $m \in \Sigma$. The groups $\mathcal{G}$ and $\mathcal{G}_m$ act on $\mathcal{A}$ by pulling back connections. The quotient $\mathcal{C} = \mathcal{A}/\mathcal{G}$ and the quotient $\mathcal{C}_m = \mathcal{A}/\mathcal{G}_m$ are of fundamental interest. If p is a point on the fiber $\pi^{-1}(m)$ over $m \in \Sigma$, C is a piecewise smooth loop in Σ based at m , and $\omega \in \mathcal{A}$, then we denote the holonomy of ω , with initial point p , around C by $g_p(C; \omega)$. The value $g_p(C; \omega) \in G$ depends only on the equivalence class $[\omega] \in \mathcal{C}_m$. For fixed $p \in P$ and connection ω , the values $g_p(C; \omega)$, as C runs over all piecewise smooth closed loops in Σ based at m , specifies $[\omega] \in \mathcal{C}_m$ uniquely. In later discussions we will use $\mathcal{C}_m$ with various values of m (for example, $m = O$ and $m = o$).

The metric on Σ and the inner-product on $\underline{g}$ induce an inner-product on the tangent spaces of $\mathcal{A}$. This induces, informally, a “volume measure” $\mathcal{D}\omega$ on $\mathcal{A}$, and this may be pushed down to the quotients $\mathcal{C}$ and $\mathcal{C}_m$ to induce “measures” (denoted by the same symbol) $[\mathcal{D}\omega]$ on these quotients. The curvature of a connection ω on P is the $\underline{g}$ -valued 2-form Ω^ω on P defined by :

$$\Omega^\omega(X, Y) = d\omega(X, Y) + [\omega(X), \omega(Y)]$$

for every pair of tangent vectors $X, Y \in T_s P$ and every $s \in P$. If $e_1, e_2 \in T_s P$ project to an orthonormal basis of $T_{\pi(s)} P$ then $\|\Omega^\omega(e_1, e_2)\|_{\underline{g}}$ depends only on $\pi(s)$ and ω , not on the choice of s over $\pi(s)$ or on (e_1, e_2) . It thus defines a function $\|\Omega^\omega\|_{\underline{g}}$ on Σ . The *Yang-Mills action* $S(\omega)$ is defined by :

$$S(\omega) = \frac{1}{2} \int_{\Sigma} \|\Omega^\omega\|_{\underline{g}}^2 d\sigma$$

where $d\sigma$ is the area measure on Σ induced by the Riemannian metric. The value $S(\omega)$ depends only on $[\omega] \in \mathcal{C}$.

The *Yang-Mills measure* μ on $\mathcal{C}_o = \mathcal{A}/\mathcal{G}_o$ is ‘defined’ heuristically by the formal expression :

$$d\mu(\omega) = \frac{1}{Z} e^{-S(\omega)} [\mathcal{D}\omega]$$

where Z is the ‘normalizing constant’ ensuring that the total mass $\mu(\mathcal{C}_o) = 1$.

One goal of this paper is to construct μ rigorously (actually, it will turn out that μ lives on a space $\overline{\mathcal{C}_o}$, which is larger than the space $\mathcal{C}_o$ of smooth connections). Granted that μ exists, consider piecewise smooth closed curves $C_1, \dots, C_n$ all based at $o = \pi(u) \in \Sigma$ and a bounded measurable function f on G^n . The values $f(g_u(C_1; \omega), \dots, g_u(C_n; \omega))$ for all choices of f and all choices of $C_1, \dots, C_n$, as above, specifies $[\omega] \in \mathcal{C}_o$ uniquely (at least for compact groups). Thus, the expectation values

$$\int_{\mathcal{C}_o} f(g_u(C_1; \omega), \dots, g_u(C_n; \omega)) d\mu(\omega)$$

would essentially specify the measure μ .

2.1. *Notation* ($q : D \rightarrow \Sigma$, x_t , $\tilde{K}_i$). We shall represent the surface Σ as a quotient of the closed unit disk D :

$$q : D \rightarrow \Sigma.$$

Here D is the closed unit disk in the plane R^2 with center at the origin O . The quotient is obtained by identifying certain pairs of points on ∂D . This boundary ∂D is traced out by the path

$$[0, 1] \rightarrow \partial D : t \mapsto x_t = (-\sin(2\pi t), \cos(2\pi t)).$$

We divide ∂D into consecutive arcs $\tilde{K}_1, \dots, \tilde{K}_N$, for a certain integer $N > 1$, where $\tilde{K}_j$ is the curve traced by x_t for $t \in [t_{j-1}, t_j]$, with $t_j = j/N$. The map q identifies certain pairs of arcs $\tilde{K}_j, \tilde{K}_{j'}$, i.e. $q(x_t) = q(x_{t'})$ for all $t \in [t_{j-1}, t_j]$ and $[t_{j-1}, t_j] \rightarrow [t_{j'-1}, t_{j'}] : t \mapsto t'$ is a 'linear' bijection. The point $q(O)$ will often be denoted $*$. $\square$

Remark. A detailed description of the quotient map q appears in the Appendix. $\square$

2.2. *Notation* (K_i, K'_i, S_j, W_Σ : See Figure 1). We will assume, as we may, that by a suitable naming of the arcs $q(\tilde{K}_i)$ by the letters a , a_i , b_i , B_i , and c_i , the 'word' $q(\tilde{K}_N) \dots q(\tilde{K}_1)$ reads

- (1) $a^{-1}ac_k^{-1}B_kc_k \dots c_1^{-1}B_1c_1$ (where, as always, a^{-1} or $\bar{a}$ denotes the arc obtained by reversing a) if Σ is the sphere with k holes,
- (2) $[a_n, b_n] \dots [a_1, b_1]c_k^{-1}B_kc_k \dots c_1^{-1}B_1c_1$, where $[x, y] = xyx^{-1}y^{-1}$, if Σ is the connected sum of n tori with k holes,
- (3) $a_na_n \dots a_1a_1c_k^{-1}B_kc_k \dots c_1^{-1}B_1c_1$ if Σ is the connected sum of n projective planes with k holes.

The case (1) will always require special treatment in our discussions because of the presence of the loop $\bar{a}a$. Note that $a_i, b_i, c_i^{-1}B_i c_i$ are closed loops in Σ based at $q(x_0)$. To specify these in terms of the $\tilde{K}_i$'s we define arcs $K_1, \dots, K_n$ by the requirements that (a) $K_n \dots K_1 = \partial D$, (b) each K_i is the composite of a number of consecutive $\tilde{K}_i$ and (c) each $q(K_i)$ is a simple closed curve based at $q(x_0)$. Note that in the case (1), where Σ is a sphere with $k(\geq 0)$ holes, $q(K_n)$ corresponds to $a^{-1}a$.

We write Ox_t to denote a (suitably parametrized) radial path from O to x_t , and by x_tO the reversed path. Thus $q(x_{t_0}O)q(K_j)q(Ox_{t_0})$ is a closed curve in Σ based at o . We write :

$$K'_i = q(x_{t_0}O)q(K_i)q(Ox_{t_0}).$$

The loops K'_i generate $\pi_1(\Sigma, o)$ subject to the relation $K'_n \dots K'_1 = 1$ in homotopy and, in case Σ is the sphere with or without holes, then also $K'_n = 1$ in homotopy. We choose a minimal collection $\{S_1, \dots, S_p\}$ such that each K'_i is either S_j or $\overline{S}_j$ for some j ; in case Σ is the sphere with $k(\geq 0)$ holes, we take S_p to be the “last” loop K'_n . Then $K'_n \dots K'_1$ is a word W_Σ in the letters $S_1, \overline{S}_1, \dots, S_p, \overline{S}_p$. If $(g_1, \dots, g_p) \in G^p$ then we shall write $W_\Sigma(g_1, \dots, g_p)$ for the element of G obtained by evaluating the word W_Σ as a product with g_i substituted for S_i and g_i^{-1} for $\overline{S}_i$. For example, if $p = 2$ and $W_\Sigma = \overline{S}_2 \overline{S}_1 S_2 S_1$ (i.e. Σ is a torus) then $W_\Sigma(g_1, g_2) = g_2^{-1} g_1^{-1} g_2 g_1$. In case Σ is the sphere with one hole, we have $\partial D = \tilde{K}_5 \tilde{K}_4 \tilde{K}_3 \tilde{K}_2 \tilde{K}_1 = K_2 K_1$, where $\tilde{K}_3$ is identified with $\overline{\tilde{K}}_1$, and $\overline{\tilde{K}}_5$ with $\tilde{K}_4$, $q(\tilde{K}_5) \dots q(\tilde{K}_1)$ is of the form $a^{-1} a c_1 B_1 c_1^{-1}$, $W_\Sigma = S_2 S_1$ and $W_\Sigma(g_1, g_2) = g_2 g_1$. $\square$

3. The Structure of Bundles and Connections over Compact Surfaces

Consider a compact surface (that is, a compact connected two-dimensional manifold possibly with boundary) Σ and a path-connected topological group G . Two principal G -bundles $P \rightarrow \Sigma$ and $Q \rightarrow \Sigma$ are *equivalent* if there is a homeomorphism $f : P \rightarrow Q$ such that (i) $f(pg) = f(p)g$ for every $p \in P$ and $g \in G$, and (ii) f maps the fiber in P over any point of Σ onto the fiber in Q over the same point. We denote by $\pi_1(G)$ the first homotopy group of G consisting of homotopy classes of loops in G based at the identity $e \in G$, with multiplication defined by composition of representative loops. It is a basic fact that $\pi_1(G)$ is a commutative group (see [St]). In this section we shall describe a specific method of classifying principal G -bundles over Σ . For oriented Σ without boundary we shall construct a method of associating an element of $\pi_1(G)$ to each principal G -bundle over Σ . This element will identify the bundle up to equivalence. For non-orientable Σ (without boundary) each equivalence class of principal G -bundles will be described by an element of the quotient group $\pi_1(G)/\pi_1(G)^2$ where $\pi_1(G)^2 = \{z^2 : z \in \pi_1(G)\}$. If Σ has boundary then any principal G -bundle over Σ is equivalent to the product bundle $\Sigma \times G \rightarrow \Sigma : (x, g) \mapsto x$.

In the following result we use the simple fact that any principal bundle, with path-connected structure group, over a 1-complex admits a global section (see [St; section 29.1]). As a consequence we have that if $\pi : P \rightarrow \Sigma$ is a principal G -bundle and $q : D \rightarrow \Sigma$ is the usual covering (as in Section 2) of Σ by the disk D then there is a section of P over the set $q(\partial D)$. We also use the fact that any principal bundle over the disk has a global section (because the disk is contractible).

3.1. Theorem. *Assume that the surface Σ has no boundary. Let G be a path-connected topological group, and $\pi : P \rightarrow \Sigma$ a principal G -bundle over the connected compact 2-manifold Σ . We display Σ as the quotient $q : D \rightarrow \Sigma$ by pasting together arcs on ∂D as described in Section 2. Let $\sigma : q(\partial D) \rightarrow P$ be a continuous section of P over $q(\partial D)$. Let $\Theta : D \rightarrow P$ be a continuous map such that $\pi\Theta(x) = q(x)$ for every $x \in D$. The existence of such Θ is a consequence of the fact that, since D is contractible, the pullback bundle $q^*P \rightarrow D$ of P by $q : D \rightarrow \Sigma$, admits a global section. Moreover, we will take, as we may, $\Theta(n) = \sigma q(n)$, where n is the point $(0, 1) \in \mathbb{R}^2$ at the 'top' of ∂D . Define $\gamma_\sigma^\Theta : \partial D \rightarrow G$ by :*

$$\sigma q(x) = \Theta(x)\gamma_\sigma^\Theta(x)$$

for all $x \in \partial D$. Note that $\gamma_\sigma^\Theta(n) = e$ (the identity element of G). The loop

$$\gamma_\sigma^\Theta : (\partial D, n) \rightarrow (G, e)$$

(wherein ∂D is traced anticlockwise) defines a homotopy class $[\gamma_\sigma^\Theta] \in \pi_1(G)$. Then :

(i) If Σ is orientable then $[\gamma_\sigma^\Theta]$ is independent of the choices of Θ and σ . We shall write $[P]$ for $[\gamma_\sigma^\Theta]$ if Σ is orientable. If Σ is not orientable then $[\gamma_\sigma^\Theta]\pi_1(G)^2 \in \pi_1(G)/\pi_1(G)^2$ is independent of Θ and σ and in this case we shall write $[P] = [\gamma_\sigma^\Theta]\pi_1(G)^2$;

(ii) If P and Q are principal G -bundles over Σ then P and Q are equivalent if and only if $[P] = [Q]$;

(iii) If Σ is orientable and $h \in \pi_1(G)$, or if Σ is unorientable and $h \in \pi_1(G)/\pi_1(G)^2$, then there is a principal G -bundle P over Σ with $[P] = h$; if G is a Lie group then P is a smooth principal G -bundle.

Proof: (i) Let σ and Θ be as described above and let σ' and Θ' be another set of choices for σ and Θ , respectively. The definition of γ_σ^Θ shows that if g is any fixed element of G and $\Theta(x)$ is replaced by $\Theta(x)g$ and $\sigma q(x)$ is replaced by $\sigma q(x)g$ then $\gamma_\sigma^\Theta(x)$ is transformed into $g^{-1}\gamma_\sigma^\Theta(x)g$; for :

$$\sigma q(x)g = \Theta(x)\gamma_\sigma^\Theta(x)g = \Theta(x)g \cdot g^{-1}\gamma_\sigma^\Theta(x)g.$$

Since G is path-connected, this procedure does not alter the homotopy class $[\gamma_\sigma^\Theta]$. Therefore we may and will assume that $\Theta(n) = \Theta'(n)$. (For σ' and Θ' may be replaced by their right translates by that element $h \in G$ for which $\Theta'(n)h = \Theta(n)$, and this will not alter $[\gamma_{\sigma'}^{\Theta'}]$). Define $\tau : q(\partial D) \rightarrow G$ by requiring that $\sigma'(q(x)) = \sigma(q(x))\tau(q(x))$ for every $x \in \partial D$. Since σ and σ' are continuous it follows by considering local sections that τ is also continuous. We have for every $x \in \partial D$:

$$\Theta'(x)\gamma_{\sigma'}^{\Theta'}(x) = \sigma'q(x) = \sigma q(x)\tau q(x) = \Theta(x)\gamma_\sigma^\Theta(x)\tau q(x).$$

Define $\xi : D \rightarrow G$ by requiring that for every $x \in D$

$$\Theta'(x) = \Theta(x)\xi(x).$$

By considering trivializations we see that ξ is continuous. The loop $\xi : (\partial D, n) \rightarrow (G, e)$ is null-homotopic; a homotopy is given by $[0, 1] \times \partial D \rightarrow G : (s, x_t) \mapsto \xi(sx_t)\xi(sx_0)^{-1}$, wherein $x_t = (-\sin(2\pi t), \cos(2\pi t))$.

Since :

$$\gamma_{\sigma'}^{\Theta'}(x) = \xi(x)^{-1}\gamma_\sigma^\Theta(x)\tau(q(x))$$

for every $x \in \partial D$, it follows that $\gamma_{\sigma'}^{\Theta'}$ and $x \mapsto \gamma_\sigma^\Theta(x)\tau(q(x))$ are homotopic as maps $(\partial D, n) \rightarrow (G, e)$. Therefore it will suffice to prove that if Σ is orientable then $\tau \circ q$ is null-homotopic as a map $(\partial D, n) \rightarrow (G, e)$, while if Σ is unorientable then we should prove that the homotopy class $[\tau \circ q]$ belongs to $\pi_1(G)^2$.

If Σ is orientable (and $\partial\Sigma = \emptyset$) then q maps ∂D into a loop in Σ which is a composite of loops of the form $ab\bar{a}\bar{b}$ (or a loop of the form $c\bar{c}$ in case Σ is a sphere) where a and b are loops in Σ . Since $\pi_1(G)$ is abelian we conclude that $[\tau q]$ is the identity element in $\pi_1(G)$. Therefore $[\gamma_\sigma^\Theta]$ is independent of the choices of Θ and σ , when Σ is orientable.

If Σ is not orientable (and $\partial\Sigma = \emptyset$) then $q|\partial D$ is a composite of loops of the form cc , where c is a loop in Σ . In this case $[\tau \circ q] \in \pi_1(G)$ is therefore in $\pi_1(G)^2$ and so the element $[\gamma_\sigma^\Theta] \in \pi_1(G)/\pi_1(G)^2$ is independent of the choices of Θ and σ .

Note that if Σ has boundary then $q|\partial D$ contains a loop in Σ which is not paired off with any loop occurring elsewhere in $q|\partial D$ and so the homotopy class of γ_σ^Θ depends on Θ and σ .

Proof of (ii) : Suppose P and Q are equivalent principal bundles over Σ . Let Θ and σ be as in the definition of $[P]$ and let $f : P \rightarrow Q$ be a bundle equivalence. Then $f \circ \Theta$ and $f \circ \sigma$ are analogs of Θ and σ , respectively, for Q . From $\sigma q(x) = \Theta(x)\gamma_\sigma^\Theta(x)$, for $x \in \partial D$, we have,

$$f\sigma q(x) = f\Theta(x)\gamma_\sigma^\Theta(x) \quad \text{for every } x \in \partial D.$$

Thus :

$$\gamma_{f\sigma}^{f\Theta} = \gamma_\sigma^\Theta$$

and so $[P] = [Q]$.

Now suppose, conversely, that $[P] = [Q]$. Denote the σ and Θ associated to P by σ_P and Θ_P , respectively, and those to Q by the corresponding symbols with P replaced by Q in the subscripts. Then $\gamma_{\sigma_P}^{\Theta_P}$ is defined by :

$$\sigma_P q(x) = \Theta_P(x)\gamma_{\sigma_P}^{\Theta_P}(x)$$

for every $x \in \partial D$. We will write γ_P for $\gamma_{\sigma_P}^{\Theta_P}$. We define γ_Q analogously for Q . If Σ is orientable then the hypothesis $[P] = [Q]$ says that $[\gamma_P] = [\gamma_Q]$ as elements of $\pi_1(G)$; if Σ is unorientable then $[\gamma_P] = [\gamma_Q]$ as elements of $\pi_1(G)/\pi_1(G)^2$. Define $\gamma : \partial D \rightarrow G$ by :

$$\gamma_Q(x) = \gamma(x)\gamma_P(x)$$

for every $x \in \partial D$. Note that $\gamma(n) = e$ since both $\gamma_Q(n)$ and $\gamma_P(n)$ equal e . We show next that it may be assumed that γ extends to a continuous map $D \rightarrow G$.

If Σ is orientable then the loop $\gamma : (\partial D, n) \rightarrow (G, e)$ is homotopically trivial since $[\gamma_P] = [\gamma_Q]$, because the operation of multiplying two loops in G pointwise coincides with composition in homotopy (see [St; 16.7]). Therefore γ extends to a continuous map $\gamma : D \rightarrow G$.

If Σ is non-orientable then $[P] = [Q]$ means that $[\gamma_P]\pi_1(G)^2 = [\gamma_Q]\pi_1(G)^2$ as elements of $\pi_1(G)/\pi_1(G)^2$. Recall that γ_P is defined by :

$$\sigma_P q(x) = \Theta_P(x)\gamma_P(x).$$

If $\sigma_P(y)$ is replaced by $\sigma_P(y)\tau(y)$, where $\tau : q(\partial D) \rightarrow G$ is continuous, then $\gamma_P(x)$ is replaced by $\gamma_P(x)\tau(q(x))$. Because Σ is unorientable, $q|\partial D$ contains a loop of the form cc where c is also a loop in Σ . Set $\tau(y) = e$ for y not on c . Then, since $[\gamma_Q][\gamma_P]^{-1} \in \pi_1(G)^2$, it is possible to choose τ on c in such a way that $(\partial D, n) \rightarrow (G, e) : x \mapsto \gamma_P(x)\tau(q(x))$ is homotopic to γ_Q . So we may and will assume that $[\gamma_P] = [\gamma_Q]$ in $\pi_1(G)$. Thus $\gamma : \partial D \rightarrow G$ extends to a continuous map $\gamma : D \rightarrow G$.

Now define a G -equivariant map $\phi : P \rightarrow Q$ by :

$$\phi(\Theta_P(x)g) = \Theta_Q(x)\gamma(x)g$$

for all $x \in \partial D$ and $g \in G$. We verify that ϕ is well-defined. The only possible ambiguity occurs when $x \in \partial D$. But, for $x \in \partial D$, the above equation for ϕ is equivalent to $\phi(\sigma_P q(x)g) = \sigma_Q q(x)g$ and here there is no ambiguity.

Finally, we need to prove continuity of ϕ . Continuity on $P|q(D \setminus \partial D)$ follows directly from the fact that $q : (D \setminus \partial D) \rightarrow q(D \setminus \partial D)$ is a homeomorphism. So we prove continuity at points on $P|q(\partial D)$. If $x \in \partial D$, then there is a neighborhood U of $q(x)$ in Σ such that $q^{-1}(\overline{U})$ is the union of a finite collection of closed (topological) half-disks C such that q maps each such half-disk homeomorphically onto its image. (In case Σ is a sphere and x is either n or the diametrically opposite point s then, pictorially speaking, these half disks are ‘quarter disks’). The map $C \times G \rightarrow P|q(C) : (x, g) \rightarrow \Theta_P(x)g$ is a continuous bijection and is G -equivariant (G acts on $C \times G$ by $(c, h)g = (c, hg)$). Therefore (see the argument below for ϕ), it is a homeomorphism, since $q|C : C \rightarrow q(C)$ is a homeomorphism. So on the closed set $P|q(C) \subset P|\overline{U}$ the map ϕ is continuous. Since $\overline{U}$ is a finite union of closed sets of the form $q(C)$ and $P|\overline{U}$ is a finite union of the closed sets $P|q(C)$ (this set is closed because it equals $\pi^{-1}q(C)$ and $q(C)$ being compact is closed) it follows that ϕ is continuous on $P|\overline{U}$ and hence also on $P|U$.

Therefore $\phi : P \rightarrow Q$ is a well-defined G -equivariant continuous map which takes $\pi_P^{-1}(x)$ into $\pi_Q^{-1}(x)$ (here $\pi_Q : Q \rightarrow \Sigma$ is the bundle projection), for every $x \in \Sigma$.

It follows that ϕ is a homeomorphism. For, in local trivializations, $P|U$ is equivalent to $U \times G$, and $Q|U$ is equivalent to $U \times G$, and then the map $\phi : P|U \rightarrow Q|U$ has the form $(x, g) \mapsto (x, g\Phi(x))$ where $\Phi : U \rightarrow G$ is continuous ($x \mapsto (x, e) \mapsto (x, \Phi(x)) \mapsto \Phi(x)$). It is then clear that ϕ is surjective and $\phi^{-1} : Q|U \rightarrow P|U$ has the form $(x, h) \mapsto (x, h\Phi(x)^{-1})$ which is also continuous.

Thus we have proven that $\phi : P \rightarrow Q$ is a bundle equivalence.

Proof of (iii) : Consider $h \in \pi_1(G)$. Note that the following arguments are valid whether Σ is orientable or not. Choose continuous $s : (\partial D, n) \rightarrow (G, e)$ such that $[s] = h$ in $\pi_1(G)$. Extend s to a continuous map $D \setminus \{0\} \rightarrow G$. Let $U_1 = q(D \setminus \partial D)$ and $U_2 = q(D \setminus \{O\})$. These are open subsets of Σ . The function $U_1 \cap U_2 \rightarrow G : q(x) \mapsto sq(x)$ is continuous (since s is continuous and q maps $(D \setminus \{O\}) \cap (D \setminus \partial D)$ homeomorphically onto $U_1 \cap U_2$). Therefore, using the standard prescription of constructing bundles (see [St; 2.4, 8.1]) there is a principal G -bundle $\pi : P \rightarrow \Sigma$ such that there are trivializations $\psi_i : U_i \times G \rightarrow P|U_i$ ($i = 1, 2$) satisfying

$$\psi_2(q(x), g) = \psi_1(q(x), s(x)g)$$

for every $(q(x), g) \in (U_1 \cap U_2) \times G$. By saying that ψ_i is a ‘trivialization’ we mean that it is a homeomorphism of $U_i \times G$ onto $P|U_i$ for which $\psi_i(x, g)h = \psi_i(x, gh)$ and $\pi\psi(x, g) = x$ for every $x \in U_i$ and $g, h \in G$.

Define $\Theta : D \rightarrow P$ by :

$$\Theta(x) = \begin{cases} \psi_1(q(x), e) & \text{if } x \in D \setminus \partial D; \\ \psi_2(q(x), s(x)^{-1}) & \text{if } x \in D \setminus \{O\}. \end{cases}$$

Then $\Theta(x)$ is well-defined for $x \in (D \setminus \partial D) \cap (D \setminus \{O\})$, and $\pi\Theta(x) = q(x)$ for every $x \in D$. Since Θ is continuous over $D \setminus \partial D$ and over $D \setminus \{O\}$, it is continuous over D .

Define

$$\sigma : q(\partial D) \rightarrow P : y \mapsto \psi_2(y, e).$$

Then σ is a continuous section of P over $q(\partial D)$.

Now for $x \in \partial D$ we have :

$$\Theta(x)s(x) = \psi_2(q(x), s(x)^{-1})s(x) = \psi_2(q(x), e) = \sigma q(x)$$

and this shows that (using our previous notation for γ_σ^Θ) :

$$\gamma_\sigma^\Theta = s|_{\partial D}.$$

Thus $[\gamma_\sigma^\Theta] = [s] = h$, the specified element of $\pi_1(G)$ we started with.

If G is a Lie group then we take, as we may, s to be smooth and we obtain a smooth principal bundle P . $\square$

3.2. Remark. If Σ has boundary then every principal G -bundle over Σ is equivalent to the product bundle $\Sigma \times G \rightarrow \Sigma : (x, g) \mapsto x$. To see this in the context of Theorem 3.1, let Θ , σ , γ_σ^Θ be as before. Now $q(\partial D)$ contains a loop c which is not paired off with any loop elsewhere in $q(\partial D)$. Define $\tau : q(\partial D) \rightarrow G$ by setting $\tau(y) = e$ for y not on c and choose τ on c so that $[\tau q|c] = [\gamma_\sigma^\Theta] \in \pi_1(G)$. Then $[\tau q] = [\tau q|c] = [\gamma_\sigma^\Theta]$. Now let $\sigma' : q(\partial D) \rightarrow G : y \mapsto \sigma(y)\tau(y)$. Then, for $x \in \partial D$,

$$\Theta(x) = \sigma q(x)\gamma_\sigma^\Theta(x) = \sigma' q(x)\{\tau q(x)\}^{-1}\gamma_\sigma^\Theta(x)$$

and so $\gamma_{\sigma'}^\Theta(x) = \tau(q(x))^{-1}\gamma_\sigma^\Theta(x)$ which implies that $[\gamma_{\sigma'}^\Theta] = [\tau q]^{-1}[\gamma_\sigma^\Theta] = e \in \pi_1(G)$. Then the arguments for Theorem 3.1(ii) above prove that P is equivalent to the product bundle $\Sigma \times G \rightarrow \Sigma$, since $[\gamma_{\tilde{\sigma}}^\Theta] = e$ where $\tilde{\Theta}$ and $\tilde{\sigma}$ are the analogs of Θ and σ for the product bundle. $\square$

3.3. Hypotheses. From now on we shall assume that G is a connected Lie group, with Lie algebra $\underline{g}$, and P is a smooth principal G -bundle over Σ . We equip Σ with a Riemannian metric. The projection $q : D \rightarrow \Sigma$ will be assumed to have the properties described in the Appendix. $\square$

We turn now to the problem of classifying connections on P up to equivalence.

3.4. Terminology (Parallel transport $\tau_\omega(c)$). Let ω be a smooth connection on P . Recall that ∂D is divided into arcs $\bar{K}_1, \dots, \bar{K}_N$. Denote by D' the set D minus the set of endpoints of the arcs $\bar{K}_j$. Let $c : [0, 1] \rightarrow D$ be a continuous map such that $q \circ c$ (which we shall write as qc) is a piecewise smooth path in Σ . Let u be a point on the fiber $\pi^{-1}(qc(0))$ in P over the initial point $qc(0)$. There is a unique piecewise smooth path $\tilde{q}c : [0, 1] \rightarrow P$ satisfying : (i) $\tilde{q}c(0) = u$, (ii) $\pi[\tilde{q}c(t)] = qc(t)$ for every $t \in [0, 1]$ and (iii) $\omega(\tilde{q}c'(t)) = 0$ whenever the derivative exists. This path is called the ω -horizontal lift of qc . The point $\tilde{q}c(1)$ is called the parallel translate of u along c by ω , and we denote it by $\tau_\omega(qc)u$ or more simply by $\tau_\omega(c)u$. $\square$

3.5. Notation (Θ_ω). Fix $u \in \pi^{-1}(o)$, where $o = q(O)$. Define $\Theta_\omega : D \rightarrow P$ by setting $\Theta_\omega(x)$ to be the point in P obtained by parallel-translating u along the radial segment Ox . Then, by Lemma A8, Θ_ω is continuous on D , smooth on D' , and $\pi\Theta_\omega = q$. $\square$

3.6. *Notation* (Area element $d\sigma'$). Let $d\sigma'$ be the smooth positively oriented 2-form on D' such that $|d\sigma'|$ is the pull-back of the Riemannian area element $d\sigma$ on Σ ; i.e., if $v, w \in T_x D$ then $|d\sigma'_x(v, w)| = d\sigma_{q(x)}(q_*v, q_*w)$. $\square$

3.7. *Definition of $f_\omega : D \rightarrow \underline{g}$* . Let Ω^ω be the curvature of ω . Let $x \in D$ and equip a neighborhood B around $q(x) \in \Sigma$ with the orientation which makes q orientation preserving as a map from a neighborhood of x (having the usual orientation from that of D) into (possibly part of) B . Choose a continuously varying pair (e_1, e_2) of vector fields on $\pi^{-1}(B)$ whose projections, by π_* , onto B are positively oriented. Then $f_\omega(x) = \Omega^\omega|_{\Theta_\omega(x)}(e_1, e_2)$. Since Ω^ω is a 2-form which vanishes on vertical vectors, $f_\omega(x)$ is independent of the specific choice of (e_1, e_2) . Since Θ_ω , Ω^ω and (e_1, e_2) are continuous, so is $f_\omega : D \rightarrow \underline{g}$. On D' we have $\Theta_\omega^*(\Omega^\omega) = f_\omega d\sigma'$. $\square$

3.8. *Lemma*. Recall that ∂D is divided into arcs K_j , $j \in \{1, \dots, n\}$, defined by $[s_{j-1}, s_j] \rightarrow \partial D : t \mapsto x_t$; moreover, the points $q(x_{s_j})$ all coincide. Let $t \in [s_{i-1}, s_i]$ and $t' \in [s_{j-1}, s_j]$, and $i' \in \{j-1, j\}$, be such that q identifies the arc $x_{s_{i-1}}x_t$ with the arc $x_{s_{i'}}x_{t'}$ (these are parts of, or reversed part of, the path $s \mapsto x_s$). Let Ox_s denote the radial path from O , the center of D , to x_s , parametrized so that its image under q is piecewise smooth; denote by x_sO the reversed path. Let L_s (with $s \in [0, 1]$) be the loop $x_sO \cdot x_0x_s \cdot x_0x_0$. Then, for any connection ω on P and $u \in \pi^{-1}(q(O))$, :

$$g_u(q(x_{t'}O) \cdot q(Ox_t); \omega) = \gamma(t')g_u(K'_{i'} \dots K'_1; \omega)^{-1}g_u(K'_{i-1} \dots K'_1; \omega)\gamma(t)^{-1}$$

where

$$\gamma(s) = g_u(q(L_s); \omega),$$

and

$$K'_r = q(x_{t_0}O) \cdot q(x_{t_{r-1}}x_{t_r}) \cdot q(Ox_{t_0}).$$

Consequently,

$$\Theta_\omega(x_t) = \Theta_\omega(x_{t'})\gamma(t')g_u(K'_{i'} \dots K'_1; \omega)^{-1}g_u(K'_{i-1} \dots K'_1; \omega)\gamma(t)^{-1}.$$

(Note that if Σ is a sphere then the K'_r terms may be dropped.)

Proof: If C is a loop in Σ , based at $q(O)$, we write $g(C)$ to mean $g_u(C; \omega)$. If $A_1, \dots, A_r$ are curves in D such that the composite $q(A_r) \dots q(A_1)$ is a closed curve in Σ based at $q(O)$, then we shall write $g(A_r \dots A_1)$ to denote $g_u(q(A_r) \dots q(A_1); \omega)$.

Since $q(x_{s_{i-1}}x_t) = q(x_{s_{i'}}x_{t'})$, we have :

$$g(x_{t'}O \cdot Ox_t) = g(x_{t'}O \cdot x_{s_{i'}}x_{t'} \cdot Ox_{s_{i'}})g(x_{s_{i'}}O \cdot Ox_{s_{i-1}})g(x_{s_{i-1}}O \cdot x_t x_{s_{i-1}} \cdot Ox_t).$$

Writing this in terms of $\gamma(\cdot)$, we have :

$$g(x_{t'}O \cdot Ox_t) = \gamma(t')\gamma(s_{i'})^{-1}g(x_{s_{i'}}O \cdot Ox_{s_{i-1}})\gamma(s_{i-1})\gamma(t)^{-1} \quad (*)$$

From

$$g(x_{s_r}O \cdot x_{s_0}x_{s_r} \cdot Ox_{s_0}) = g(x_{s_r}O \cdot Ox_{s_0})g(x_{s_0}O \cdot x_{s_0}x_{s_r} \cdot Ox_{s_0}) = g(x_{s_r}O \cdot Ox_{s_0})g(K'_r \dots K'_1)$$

(the K'_i terms are absent if $r = 0$), we obtain :

$$g(x_{s_r}, O.Ox_{s_0}) = \gamma(s_r)g(K'_r \dots K'_1)^{-1}.$$

This implies :

$$g(x_{s_{i'}} O.Ox_{s_{i-1}}) = \gamma(s_{i'})g(K'_{i'} \dots K'_1)^{-1}g(K'_{i-1} \dots K'_1)\gamma(s_{i-1})^{-1}$$

(the terms involving the K'_r are absent in case Σ is a sphere). Using this in (*) we conclude that :

$$g(x_{t'} O.Ox_t) = \gamma(t')g_u(K'_{i'} \dots K'_1; \omega)^{-1}g_u(K'_{i-1} \dots K'_1; \omega)\gamma(t)^{-1}.$$

The result for Θ_ω follows from :

$$\Theta_\omega(x_t) = \tau_\omega(q(Ox_t))u = \tau_\omega(q(Ox_{t'}))ug_u(q(x_{t'}O) \cdot q(Ox_t); \omega) = \Theta_\omega(x_{t'})g(x_{t'}O.Ox_t). \quad \square$$

The following results (3.9-3.12), excluding the parts relating to smoothness, will be used in Section 6 as guiding ideas in defining the Yang-Mills measure. However, the results in Section 6 do not have any logical dependence on the following.

3.9. Theorem. *Let G be a connected Lie group, Σ a connected two-dimensional smooth manifold without boundary and $\pi : P \rightarrow \Sigma$ a smooth principal G -bundle. As usual, $q : D \rightarrow \Sigma$ is the projection with $q(O) = o$, and u is a fixed point on the fiber in P over o . For $t \in [0, 1]$ let L_t be the loop $x_t O.x_{t_0} x_t.Ox_{t_0}$, as in Lemma 3.8. Let ω be a smooth connection on P . Define*

$$\gamma : [0, 1] \rightarrow G : t \mapsto g_u(q(L_t); \omega)$$

(this is well-defined since $q(L_t)$ is smoothly parametrizable by Lemma A5); i.e. $\gamma(t)$ is the holonomy of ω around the loop $q(L_t)$ with initial point u .

Recall the homotopy basis loops $S_1, \dots, S_p$ in Σ based at o . Lift the curve $\gamma : [0, 1] \rightarrow G$ to the universal cover $\tilde{G}$ of G to obtain a path $\tilde{\gamma} : [0, 1] \rightarrow \tilde{G}$ with $\tilde{\gamma}(0) = \tilde{e}$ (the identity element of $\tilde{G}$) with $\tilde{\gamma}(t)$ covering $\gamma(t)$. For each $i = 1, \dots, p$, pick an element $\tilde{g}_u(S_i; \omega) \in \tilde{G}$ which projects to the holonomy element $g_u(S_i; \omega) \in G$. Then

$$z \stackrel{\text{def}}{=} \tilde{\gamma}(1)W_\Sigma(\tilde{g}_u(S_1; \omega), \dots, \tilde{g}_u(S_p; \omega))^{-1}$$

lies in the kernel of the projection $\tilde{G} \rightarrow G$ (and therefore in the center of $\tilde{G}$), and any path in $\tilde{G}$ from e to z projects to a loop in G , based at $\tilde{e}$, whose homotopy class $[z]$ is equal to $[P]$ if Σ is orientable, while $[z]\pi_1(G)^2 = [P]$ if Σ is unorientable.

3.10. Theorem. *Two connections ω_1 and ω_2 on the principal G -bundle P are gauge-equivalent under $\mathcal{G}_o$ (in the sense that there is a gauge transformation $\phi \in \mathcal{G}_o$ with $\phi^*\omega_2 = \omega_1$) if and only if :*

- (i) *they have the same holonomies around the loops S_i*

(ii) $f_{\omega_1} = f_{\omega_2}$.

3.11. Let $\alpha_1, \dots, \alpha_p \in G$, and f a continuous $\underline{g}$ -valued function on D which satisfies suitable smoothness/compatibility conditions (two such conditions are stated in Proposition A14) on ∂D . Assume that $\tilde{\gamma}(1)^{-1}W_{\Sigma}(\tilde{\alpha}_1, \dots, \tilde{\alpha}_p) \in \ker(\tilde{G} \rightarrow G)$, where each $\tilde{\alpha}_i \in \tilde{G}$ is any element covering $\alpha_i \in G$, and $\tilde{\gamma} : [0, 1] \rightarrow \tilde{G}$ is the continuous map, with $\tilde{\gamma}(0) = \tilde{e}$ (the identity in $\tilde{G}$), which satisfies (see equation (A6))

$$\frac{d\tilde{\gamma}_t}{dt} = -\frac{d}{dt} \left(\int_{D_t} f d\sigma' \right) \tilde{\gamma}_t$$

for $t \in [0, 1] \setminus \{t_0, \dots, t_{N-1}\}$, where $D_t = \{rx_s : r \in [0, 1], s \in [0, t]\}$ and $d\sigma'$ denotes the area form on D' obtained by pulling back the Riemannian area measure on Σ . There is then a smooth connection ω on a principal G -bundle $P \rightarrow \Sigma$, unique up to gauge equivalence, with $g_u(S_i; \omega) = \alpha_i$, for every $i \in \{1, \dots, p\}$, and with $f_{\omega} = f$ on D .

We shall not formulate the “suitable smoothness/compatibility” conditions precisely, and will thus leave 3.11 as a plausible but imprecise and unproven statement. A precise formulation, and proof, is possible but is very complicated because of the unwieldy behavior of q at the points x_{t_i} . Proposition A14 and Remarks A13 explain, in part, how 3.11 can be made precise. None of our results will have logical dependence on 3.11.

Proof of Theorem 3.9. Let Θ_{ω} be as in Notation 3.5 and choose σ , a section of $P|_{q(\partial D)}$, such that

$$\Theta_{\omega}(x_{t_0}) = \sigma q(x_{t_0}).$$

Then $\gamma_{\sigma}^{\Theta_{\omega}}$ is defined by :

$$\sigma q(x_t) = \Theta_{\omega}(x_t) \gamma_{\sigma}^{\Theta_{\omega}}(x_t).$$

So $[P]$ is determined by the homotopy class of the loop $[0, 1] \rightarrow G : t \mapsto \gamma_{\sigma}^{\Theta_{\omega}}(t)$. Define $\xi(t)$ by :

$$\gamma_{\sigma}^{\Theta_{\omega}}(x_t) = g_u(qL_t; \omega) \xi(t)^{-1} = \gamma(t) \xi(t)^{-1}.$$

It will suffice to prove that if $\tilde{\xi} : [0, 1] \rightarrow \tilde{G}$ is the lift of ξ to $\tilde{G}$ with $\tilde{\xi}(0) = \tilde{e}$, then :

$$\tilde{\xi}(1) = W_{\Sigma}(\tilde{g}_u(S_1), \dots, \tilde{g}_u(S_p)) \quad (*)$$

Recall that ∂D is divided into arcs $K_1, \dots, K_N$, with K_j being $[s_{j-1}, s_j] \rightarrow \partial D : t \mapsto x_t$; q identifies certain pairs of arcs K_j . Recall also that the loops $K'_j = q(x_{t_0} O) q(K_j) q(O x_{t_0})$ are the loops $S_i, \bar{S}_i$ in such an order that $K'_N \dots K'_1 = W_{\Sigma}(S_1, \dots, S_p)$.

To prove $(*)$ we shall verify that : (1) $\xi(t) \xi(s_{i-1})^{-1} = \xi(t') \xi(s_j)^{-1}$ if $t \in [s_{i-1}, s_i]$, $t' \in [s_{j'-1}, s_{j'}]$, and q identifies the arc $x_{s_{i-1}} x_t$ with $x_{s_j} x_{t'}$ (j being either i' or $i' - 1$); and (2) $\xi(s_r) = g_u(K'_r \dots K'_1; \omega)$ for every $r \in \{1, \dots, N\}$.

Then Lemma 3.13 below, applied to (1) and (2) will imply $(*)$.

(1) : Let $i, i' \in \{1, \dots, N\}$, $t \in [s_{i-1}, s_i]$, $t' \in [s_{i'-1}, s_{i'}]$, and q identify the arc $x_{s_{i-1}}x_t$ with $x_{s_{i'-1}}x_{t'}$. Then :

$$\Theta_\omega(x_t)\gamma(t)\xi(t)^{-1} = \sigma q(x_t) = \sigma q(x_{t'}) = \Theta_\omega(x_{t'})\gamma(t')\xi(t')^{-1}.$$

By Lemma 3.8, this implies :

$$\gamma(t')g_u(K'_j \dots K'_1; \omega)^{-1}g_u(K'_{i-1} \dots K'_1; \omega)\gamma(t)^{-1} \cdot \gamma(t)\xi(t)^{-1} = \gamma(t')\xi(t')^{-1}.$$

So :

$$\xi(t')^{-1}\xi(t) = g_u(K'_j \dots K'_1; \omega)^{-1}g_u(K'_{i-1} \dots K'_1; \omega) \quad (**)$$

does not depend on t as long as $t \in [s_{i-1}, s_i]$.

(2) : This follows from (**) by taking $t' = t_j$ and $i = 1$. $\square$

3.12. *Remark* (the case $\partial\Sigma \neq \emptyset$). If Σ has boundary then every principal bundle over Σ , with path connected structure group G , is equivalent to the product bundle. The condition in this case would just be :

$$\gamma(1) = W_\Sigma(g_u(S_1), \dots, g_u(S_p)). \quad \square$$

3.13. *Lemma*. Suppose $\xi : [0, 1] \rightarrow G$ is continuous, with $\xi(0) = e$, and $0 < t_1 < t_2 < \dots < t_{N-1} < t_N = 1$. Suppose that for some $i, i' \in \{1, \dots, N\}$, there is a homeomorphism $\phi : [t_{i-1}, t_i] \rightarrow [t_{i'-1}, t_{i'}] : t \mapsto t'$, such that

$$\xi(t)\xi(t_{i-1})^{-1} = \xi(t')\xi(t'_{i-1})^{-1} \quad (***)$$

Let $\alpha_1, \dots, \alpha_N \in G$ be defined by:

$$\xi(t_r) = \alpha_r \alpha_{r-1} \dots \alpha_1, \quad r \in \{1, \dots, N\}.$$

Then there exist $\tilde{\alpha}_1, \dots, \tilde{\alpha}_N \in \tilde{G}$, with each $\tilde{\alpha}_r$ covering α_r , and with

$$\tilde{\alpha}_{i'} = \begin{cases} \tilde{\alpha}_i & \text{if } t'_{i-1} = t_{i'-1}; \\ \tilde{\alpha}_i^{-1} & \text{if } t'_{i-1} = t'_i. \end{cases}$$

Moreover,

$$\tilde{\xi}(1) = \tilde{\alpha}_N \tilde{\alpha}_{N-1} \dots \tilde{\alpha}_1$$

where $\tilde{\xi} : [0, 1] \rightarrow \tilde{G}$ is the lift of $\xi : [0, 1] \rightarrow G$ with $\tilde{\xi}(0) = \tilde{e}$ (the identity of $\tilde{G}$). A similar conclusion holds if there are other pairs like i and i' (the relationship between $\tilde{\alpha}_i$ and $\tilde{\alpha}_{i'}$ described above holds for each such pair i, i').

Proof: Define $\tilde{\alpha}_r = \tilde{\xi}(t_r)\tilde{\xi}(t_{r-1})^{-1}$. Then it is clear that $\tilde{\alpha}_r$ covers $\alpha_r \in G$, and

$$\tilde{\xi}(1) = \tilde{\alpha}_N \tilde{\alpha}_{N-1} \dots \tilde{\alpha}_1.$$

Now

$$[t_{i-1}, t_i] \rightarrow \tilde{G} : t \mapsto \tilde{\xi}(t)\tilde{\xi}(t_{i-1})^{-1}$$

and

$$[t_{i-1}, t_i] \rightarrow \tilde{G} : t \mapsto \tilde{\xi}(t')\tilde{\xi}(t'_{i-1})^{-1}$$

are the same because both start at $\bar{e}$ and, by hypothesis (**), both cover the same path on G . Setting $t = t_i$, we then obtain :

$$\tilde{\xi}(t_i)\tilde{\xi}(t_{i-1})^{-1} = \tilde{\xi}(t'_i)\tilde{\xi}(t'_{i-1})^{-1}.$$

Since $t \mapsto t'$ is a homeomorphism, t'_{i-1} is either $t_{i'-1}$ or $t_{i'}$. If $t'_{i-1} = t_{i'-1}$, then the right side of the above equation is $\tilde{\alpha}_{i'}$. If, on the other hand, $t'_{i-1} = t_{i'}$ then $t'_i = t_{i'-1}$, and in this case the right side is $\tilde{\xi}(t_{i'-1})\tilde{\xi}(t_{i'})^{-1} = \tilde{\alpha}_{i'}^{-1}$. This proves (**). $\square$

Proof of Theorem 3.10:

The basic idea is quite simple. The connections ω_1 and ω_2 are gauge equivalent under $\mathcal{G}_o$ if and only if $g_u(C; \omega_1) = g_u(C; \omega_2)$ for every piecewise smooth closed loop C in Σ based at o . From this, and the fact that f_ω is a measure of ‘infinitesimal holonomy’, it follows that if ω_1 and ω_2 are gauge equivalent under $\mathcal{G}_o$ then $g_u(S_i; \omega_1) = g_u(S_i; \omega_2)$ for each S_i , and that $f_{\omega_1} = f_{\omega_2}$. For the converse we note that for a nice enough loop C , the holonomy $g_u(C; \omega)$ is a product of certain $g_u(S_i; \omega)$ (or their inverses) and some $g_u(q \circ C'; \omega)$ where C' are loops, based at O , in D . Since the holonomies $g_u(q \circ C'; \omega)$ are determined by f_ω , we see that the $g_u(S_i; \omega)$ and f_ω determine ω up to gauge equivalence under $\mathcal{G}_o$. The technical details of this argument are contained in Lemmas A6 and A7. $\square$

3.14. Remark (Existence of bundles with given h). Assume that G is a path connected topological group with a universal cover $\tilde{G}$. Identify $\pi_1(G)$ with $\ker(\tilde{G} \rightarrow G)$, and write $\pi_1(G)^2 \stackrel{\text{def}}{=} \{x^2 : x \in \pi_1(G)\}$. Let Σ be our compact surface. Define the group G' as follows. If Σ is orientable and without boundary then $G' = \tilde{G}$; if Σ is unorientable and without boundary then $G' = \tilde{G}/\pi_1(G)^2$; if Σ has boundary then $G' = G = \tilde{G}/\pi_1(G)$. In all cases, there is a natural quotient homomorphism $G' \rightarrow G$. If $G' = \tilde{G}/\pi_1(G)^2$, then $\ker(G' \rightarrow G) = \pi_1(G)/\pi_1(G)^2$. Then Theorem 3.1 implies that a principal G -bundle P over Σ is classified by an element $h \in \ker(G' \rightarrow G)$. $\square$

3.15. Remark (the function $W_\Sigma : G^p \rightarrow G'$). Let Σ , G and G' be as in the preceding remark. Recall the function W_Σ of p variables associated to Σ . If $(a_1, \dots, a_p) \in G^p$ and $a'_i \in G'$ covers $a_i \in G$, for each $i \in \{1, \dots, p\}$, then $W_\Sigma(a'_1, \dots, a'_p)$ depends only on $(a_1, \dots, a_p)$ and not on the choices of a'_i . Moreover, $(a_1, \dots, a_p) \rightarrow W_\Sigma(a'_1, \dots, a'_p)$ is continuous since $G' \rightarrow G$ is a local homeomorphism. $\square$

3.16. Theorem. Let $\phi : \Sigma \rightarrow \Sigma$ be a homeomorphism and $\pi : P \rightarrow \Sigma$ a principal G -bundle, where G is a path-connected topological group. Denote by ϕ^*P the pullback bundle $\cup_{x \in \Sigma} \{x\} \times \pi^{-1}\phi(x) \rightarrow \Sigma : (x, y) \mapsto x$. Then :

$$[\phi^*P] = [P]^{\text{sgn } \phi}$$

wherein $\text{sgn } \phi = 1$ if Σ is orientable and ϕ is orientation-preserving, $\text{sgn } \phi = -1$ if Σ is orientable and ϕ is orientation-reversing, and $\text{sgn } \phi = \pm 1$ if Σ is not orientable. $\square$

In order to prove this, we will define an invariant $[P]'$ for which the theorem is easily provable. Then we will demonstrate (Proposition 3.18) that in fact $[P]' = [P]$. Since Theorem 3.16 is trivial in case Σ has boundary, we shall assume that Σ has no boundary.

3.17. Definition of $[P]'$. Let Σ be a connected compact two dimensional manifold without boundary. If Σ is orientable then we equip it with a fixed orientation. Let S be a simplicial complex triangulating Σ . Let s be a section of P over the 1-skeleton of S . For each oriented 2-simplex Δ of S , let s_Δ be a section of P over Δ . Then $\partial\Delta \rightarrow G : x \mapsto s_\Delta(x)^{-1}s(x)$ defines an element γ_Δ of $\pi_1(G)$; here we are writing $s_\Delta(x)^{-1}s(x)$ to denote the element $g(x) \in G$, depending continuously on x , for which $s(x) = s_\Delta(x)g(x)$. Note also that here $\partial\Delta$ is oriented as the boundary of Δ . If Σ is oriented then we define $[P]'$ to be the product of the γ_Δ as Δ runs over all the positively oriented 2-simplices of S . If Σ is not orientable, let $\bar{\gamma}_\Delta = \gamma_\Delta \pi_1(G)^2$, an element of $\pi_1(G)/\pi_1(G)^2$; note that $\bar{\gamma}_\Delta$ has the same value irrespective of the orientation of Δ . If Σ is not orientable we define $[P]'$ to be the product of all the $\bar{\gamma}_\Delta$ (i.e. if $\Delta_1, \dots, \Delta_n$ are all the unoriented 2-simplices of S then $[P]' = \bar{\gamma}_{\Delta_1} \dots \bar{\gamma}_{\Delta_n}$). $\square$

Fact 1 : $[P]'$ does not depend on the choices of s_Δ and s .

Proof : Changing s_Δ does not change γ_Δ and hence does not change $[P]'$. Next consider sections s and s' of P over the 1-skeleton of S . By a homotopy of s' we can make s and s' agree on the 0-skeleton of S ; this procedure does not alter $[P]'$. So we assume that s and s' agree on the zero skeleton of S . In this case the $\gamma_\Delta^{-1}\gamma'_\Delta$ (wherein γ'_Δ is defined with respect to s' as γ_Δ is with respect to s) specify a $\pi_1(G)$ -valued 1-cochain z on S ; i.e. for each oriented 1-simplex b of S there is a $z_b \in \pi_1(G)$ such that $z_{\bar{b}} = z_b^{-1}$ and such that $\gamma_\Delta^{-1}\gamma'_\Delta = z_{b_3}z_{b_2}z_{b_1}$ if $\partial\Delta = b_3b_2b_1$ (this makes sense since $\pi_1(G)$ is abelian). In case Σ is oriented it is now clear that $\prod_\Delta \gamma_\Delta^{-1}\gamma'_\Delta = e$, where the product is over positively oriented 2-simplices Δ and wherein e is the identity in $\pi_1(G)$. Thus if Σ is oriented then $[P]'$ has the same value whether s is used or whether s' is used in its definition. If Σ is not orientable then the product of the $\gamma_\Delta^{-1}\gamma'_\Delta$ belongs to $\pi_1(G)^2$ and therefore, again, $[P]'$ has the same value if s is replaced by s' . $\square$

Fact 2 : If Σ is subdivided then $[P]'$ is unaltered.

Proof. Let S' be a subdivision of S . Extend s to a section of P over the 1-skeleton of S' . We use the sections s_Δ , with Δ running over 2-simplices of S , to define, by restriction, the sections $s_{\Delta'}$ for 2-simplices Δ' of S' . Let Δ be a 2-simplex of S divided into 2-simplices $\Delta_1, \dots, \Delta_m$ of S' . Then $\gamma_{\Delta_1} \dots \gamma_{\Delta_m} = \gamma_\Delta$, assuming that each Δ_i is oriented coherently with Δ and γ_{Δ_i} is defined with respect to S' and the s_{Δ_i} . It follows that $[P]'$ has the same value whether S or S' is used in its definition. $\square$

Proof of 3.16 : If P and Q are equivalent bundles then it follows directly that $[P]' = [Q]'$, wherein $[P]'$ and $[Q]'$ are both defined with respect to the same (arbitrary) triangulation of Σ . Now let $\phi : \Sigma \rightarrow \Sigma$ be a homeomorphism and let S and S' be triangulations of Σ . Then by the Hauptvermutung (Theorem 4.6 in [Br]), there exist subdivisions of S and S' , which we denote again by S and S' , respectively, and there is a homeomorphism $\psi : \Sigma \rightarrow \Sigma$ which is isotopic to ϕ and which maps the subdivided complex S isomorphically onto the subdivided complex S' . Since ϕ and ψ are homotopic, ϕ^*P and ψ^*P are equivalent and so $[\phi^*P]' = [\psi^*P]'$, with respect to any triangulation of Σ . Since ψ takes S isomorphically onto S' , the definition of $[\psi^*P]'$ with respect to S , using appropriate

choices for s and s_Δ coincides with the definition of $[P]'^{\pm 1}$ with respect to S' , where -1 is used only if Σ is orientable and ψ (equivalently, ϕ) is orientation reversing. Taking ϕ to be the identity map on Σ we see that $[P]'$ is independent of the specific triangulation used in its definition. We conclude that $[\phi^*P]' = [P]^{\text{sgn } \phi}$. $\square$

3.18 Proposition. *Let $q : D \rightarrow \Sigma$ be the quotient map as in Notation 2.1. If Σ is orientable we equip it with the orientation which makes q orientation preserving (from the usual orientation of the closed unit disk $D \subset \mathbb{R}^2$). Let $\pi : P \rightarrow \Sigma$ be a principal G -bundle, where G is a path connected topological group. Then $[P]' = [P]$.*

Proof: We triangulate D in such a way that its image under q provides a triangulation of Σ . From the definition of $[P]$ in Theorem 3.1 we see, in the notation used in the statement of Theorem 3.1, that the restrictions of Θ to the various 2-simplexes T (triangulating D or, by projection, Σ) provide the sections s_Δ needed to define $[P]'$; moreover, $\sigma|_{q(\partial D)}$ can be extended to a section s over the 1-skeleton of the triangulation of Σ . The product of the corresponding γ_Δ equals γ_σ^Θ , and therefore $[P]' = [P]$. $\square$

3.19. Remark. The γ_Δ used in defining $[P]'$ specify an element of $H^2(\Sigma; \pi_1(G))$, and $[P]'$ is obtained by ‘evaluating’ it on a generator of $H_2(\Sigma; \mathbb{Z})$ or, in case Σ is unorientable, of $H_2(\Sigma; \mathbb{Z}/2\mathbb{Z})$. This interpretation of $[P]$ provides the link between our approach and that of the more general machinery developed in [St : sections 31-37]. $\square$

4. Quantum Gauge Theory on the Disk

We review several facts about planar quantum gauge theory. The results and techniques in this section are drawn largely from [Se1] where they are proved for the case of compact groups. These results were also proved in [GKS] and [Dr] by different methods (see [Se1] for other references). We do provide details not explained in [Se1].

Let D be the unit disk in the plane, centered at the origin O . We equip D with a Riemannian metric. (We could develop the theory with the metric defined on D minus a finite set of points on the boundary ∂D but the theory would be essentially the same and therefore we will just work on D itself.)

4.1. Hypotheses concerning G . Let G be a connected Lie group which is a compact group times R^d (d some non-negative integer). The Lie algebra $\underline{g}$ of G is equipped with an Ad -invariant metric $\langle \cdot, \cdot \rangle_{\underline{g}}$. In particular, G is isomorphic to a matrix group. Our notation will often assume that G is a matrix group. $\square$

For this section only, we denote by P the product bundle $\pi : D \times G \rightarrow D$. Let u be the point (O, e) on $\pi^{-1}(O)$. For ω a connection on P , define the section $s_\omega : D \rightarrow P$ by setting $s_\omega(p)$ to be the parallel translate, by ω , of u along the radial path Op . Define $F^\omega : D \rightarrow \underline{g}$ by requiring that :

$$F^\omega d\sigma = s_\omega^* \Omega^\omega$$

where Ω^ω is the curvature of ω and $d\sigma$ is the area 2-form on D (positively oriented); i.e. for every $p \in D$, :

$$F^\omega(p) = \Omega^\omega(e_1, e_2)$$

wherein e_1, e_2 are a pair of vectors in $T_{s_\omega(p)}P$ for which $(\pi_*(e_1), \pi_*(e_2))$ is a positively oriented orthonormal basis of $T_p D$.

The association $\omega \mapsto F^\omega$ sets up a bijection of $\mathcal{A}/\mathcal{G}_O$ with a linear space of $\underline{g}$ -valued functions on D . Some heuristic arguments now let us identify the Yang-Mills measure $d\mu_{YM, disk} = Z^{-1} e^{-S(\omega)} [D\omega]$, for this situation, as a Gaussian measure 'on' $L^2(D; \underline{g})$ (the Hilbert space of all measurable functions $F : D \rightarrow \underline{g}$ with $\|F\|^2 \stackrel{\text{def}}{=} \int_D \|F\|_{\underline{g}}^2 d\sigma < \infty$). First we observe that $S(\omega) = \frac{1}{2} \|F^\omega\|^2$. Let $s : D \rightarrow P$ be a fixed section with $s(O) = u$; say $s(\cdot) = (\cdot, e)$. The association $\omega \mapsto \omega' = s^* \omega$ sets up an affine (isometric) isomorphism of the space of all connections on P with the space of all smooth $\underline{g}$ -valued 1-forms on D . Each such 1-form ω' is 'gauge equivalent' to a 1-form $\omega_\theta d\theta$ which vanishes on radial vectors; in fact, $\omega_\theta d\theta = s_\omega^* \omega$ if $\omega' = s^* \omega$. Thus $\mathcal{A}/\mathcal{G}_O$ may be identified with the space $\mathcal{A}_\theta$ of $\underline{g}$ -valued 1-forms ω' on D for which $\omega'(\frac{\partial}{\partial r}) = 0$. As argued in the Remarks 4.2 below, $[D\omega]$ corresponds to 'Lebesgue measure' on $\mathcal{A}_\theta$ (this linear space has an obvious inner-product structure). Finally, $\|F^\omega\|^2 = \|d(\omega_\theta d\theta)/d\sigma\|^2$, and the map $\omega_\theta d\theta \mapsto F^\omega$, being linear, should take Lebesgue measure to a multiple of Lebesgue measure. Thus it is reasonable to view $[D\omega]$ as Lebesgue measure on the space $L^2(D; \underline{g})$ of F 's. Combining

all these informal observations we see that the Yang-Mills measure for gauge fields on the disk D should be taken as Gaussian measure (with ‘density’ proportional to $e^{-\frac{1}{2}\|F\|^2}$) on $L^2(D; \underline{g})$. As is well known (see [Ku]) such a measure does make sense but actually lives on a larger Hilbert space $\overline{L^2(D; \underline{g})}$, which is a Hilbert-Schmidt closure of $L^2(D; \underline{g})$. We will usually write Ω to denote this Hilbert space (which is our probability space). In this stochastic setting, the ‘curvature’ F is a $\underline{g}$ -valued white-noise; this means that for each Borel set $E \subset D$, one has a $\underline{g}$ -valued random variable $F(E) : \omega \mapsto F^\omega(E)$, specified almost-everywhere on $\overline{L^2(D; \underline{g})}$, such that $F(E)$ is Gaussian with mean zero and, for every Borel $A, B \subset D$ and $v, w \in \underline{g}$, the expectation value of $\langle F(A), v \rangle_{\underline{g}} \langle w, F(B) \rangle_{\underline{g}}$ equals $|A \cap B| \langle v, w \rangle_{\underline{g}}$ (wherein $|A \cap B|$ is the area of $A \cap B$, measured by $d\sigma$).

4.2. Remarks on $[D\omega]$. We show that $[D\omega]$ can be taken to be ‘Lebesgue measure’ on A_θ . The argument here (which is a Faddeev-Popov type procedure) is, of course, heuristic but a rigorous analog can be formulated in finite dimensions. To this end, we let, as we may, $\mathcal{A}$ be the space of all $\underline{g}$ -valued 1-forms on D , and we denote by $\mathcal{G}_O$ the set of all smooth maps $D \rightarrow G$ taking the value e at the center O . If $v, w \in \mathcal{A}$ then we have the inner-product $\langle v, w \rangle = \int_D \langle v(e_1), w(e_1) \rangle d\sigma + \int_D \langle v(e_2), w(e_2) \rangle d\sigma$, where (e_1, e_2) are orthonormal vectors in R^2 (the tangent space at any point of D). The group $\mathcal{G}_O$ acts isometrically on $\mathcal{A}$ by : $(g, A) \mapsto \psi_g(A) \stackrel{\text{def}}{=} gAg^{-1} + gd(g^{-1})$. Let $\psi : \mathcal{G}_O \times A_\theta \rightarrow \mathcal{A} : (g, A_\theta d\theta) \mapsto g(A_\theta d\theta)g^{-1} + gd(g^{-1})$. For the illustrative purposes of this formal argument we will work as if ψ were a diffeomorphism. Let f be a $\mathcal{G}_O$ -invariant function on $\mathcal{A}$. Then the ‘Lebesgue integral’ $\int_{\mathcal{A}} f(A) \mathcal{D}A$ equals $\int_{\mathcal{G}} \mathcal{D}g \int_{A_\theta} \mathcal{D}A_\theta f(A_\theta d\theta) J(g, A_\theta)$, wherein $J(g, A_\theta)$ is the Jacobian of ψ at $(g, A_\theta d\theta)$. Since $(g, A) \mapsto \psi_g(A)$ is an action of $\mathcal{G}_O$ on $\mathcal{A}$, the derivative $d\psi_{(g, A_\theta d\theta)}$ equals the composite $d\psi_g|_{A_\theta d\theta} \circ d\psi_{(e, A_\theta d\theta)}$. Since $d\psi_g|_{A_\theta d\theta}$ is an isometry, it follows that the determinant $J(g, A_\theta)$ equals $J(e, A_\theta)$. Thus $\int_{\mathcal{A}} f(A) \mathcal{D}A$ equals the ‘volume’ of $\mathcal{G}_O$ times $\int_{A_\theta} \mathcal{D}A_\theta f(A_\theta d\theta) J(e, A_\theta)$. Now $d\psi_{(e, A_\theta d\theta)} : T_e \mathcal{G}_O \oplus T_{A_\theta d\theta} A_\theta \rightarrow T_{A_\theta d\theta} \mathcal{A}$. Splitting $T_{A_\theta d\theta} \mathcal{A}$ into A_θ and its orthogonal complement $A_\theta^\perp$, and examining a matrix representation of $d\psi_{(e, A_\theta d\theta)}$ relative to orthonormal bases, we see that $J(e, A_\theta)$ equals $|\det(P_{A_\theta^\perp} d\psi_e|_{A_\theta d\theta} : T_e \mathcal{G}_O \rightarrow A_\theta^\perp)|$; here $P_{A_\theta^\perp}$ is the orthogonal projection onto $A_\theta^\perp$ in $\mathcal{A}$, and the determinant, calculated by using orthonormal bases, is ‘well-defined’ up to sign. Now the Lie algebra $T_e \mathcal{G}_O$ is identifiable as the space of functions $D \rightarrow \underline{g}$ vanishing at O , and $d\psi_e|_{A_\theta d\theta}(H) = -(dH + [A_\theta, H]d\theta)$; so $P_{A_\theta^\perp} d\psi_e|_{A_\theta d\theta}$ is independent of $A_\theta d\theta$. Thus $J(e, A_\theta d\theta)$ is a constant. Therefore, $\int_{\mathcal{A}} f(A) \mathcal{D}A$ is a constant times $\int_{A_\theta} f(A_\theta d\theta) \mathcal{D}A_\theta$. This shows that $[D\omega]$ may be taken to be ‘Lebesgue measure’ on A_θ . $\square$

We shall use polar coordinates (r, θ) on D , with θ measured anticlockwise from the positive y -axis.

4.3. Terminology. A horizontal curve C is given by a continuous map $C : [a, b] \rightarrow D : \theta \mapsto (\theta, \sigma_C(\theta))$, in polar coordinates, where $0 \leq a < b \leq 2\pi$, and $\sigma_C(\theta) > 0$ for $\theta \in (a, b)$. We will not distinguish between reparametrizations of the same curve. If $\theta \in [a, b]$, we denote by C_θ the loop $C(\theta)O \cdot C[[a, \theta] \cdot OC(a)$; i.e. proceed along the radius from O to $C(a)$, then along C upto θ , and then back along the radius to O . $\square$

Let ω be a smooth connection on P , and C , as above, a horizontal curve. Assume that C is piecewise smooth. Denote by E_θ the subset of D whose boundary is C_θ . Then

the holonomy $g_\theta = g_u(C_\theta; \omega)$, is the solution of :

$$dg_\theta = -dF_\theta^\omega g_\theta \quad (4.1)$$

with initial condition

$$g_a = e,$$

where

$$F_\theta^\omega = \int_{E_\theta} F^\omega d\sigma.$$

Passing over to the case where $\omega \in \overline{L^2(D; \underline{g})}$ it is necessary to reinterpret (4.1) as a stochastic differential equation (see [McK] or [Em] for details on such equations). The following definition is based on [GKS] (the use of stochastic differential equations in the context of parallel transport appeared earlier in [AHH]).

4.4. *Definition (Stochastic Parallel Transport).* Let $C : [a, b] \rightarrow D$ be a horizontal curve in D . *Stochastic parallel transport* along C is described by the solution of the Stratonovich stochastic differential equation :

$$dg_\theta = (-dF_\theta) \circ g_\theta \quad (4.2)$$

(or, as an Ito stochastic differential equation,

$$dg_\theta = -dF_\theta g_\theta + \frac{1}{2}(dF_\theta)^2 g_\theta \quad)$$

with initial condition

$$g_a = e,$$

where $F_\theta = F(E_\theta)$, and E_θ is the region enclosed by the loop C_θ . The *stochastic holonomy* around the loop C_θ is the G -valued random variable g_θ . We will also write $g(C_\theta; \omega)$ for $g_\theta(\omega)$, and $g(C; \omega)$ for $g(C_b; \omega)$. We will often denote the random variable $\omega \mapsto g(C; \omega)$ by $g(C)$.

If $\overline{C}$ is the curve obtained by traversing C in reverse then we define :

$$g(\overline{C}; \omega) = g(C; \omega)^{-1}.$$

If M is a radial segment or a constant curve then we define the random variable $g(M)$ to be the constant e . If K is a curve which is a composite $K_m \dots K_1$, where each K_i is radial or horizontal or reversed horizontal, we define $g(K)$ to be $g(K_m) \dots g(K_1)$. It is clear that two representations of K as a composite of such segments yield, almost surely, the same random variable $g(K)$. $\square$

4.5. *Remarks* (Details on the solution of (4.2)). Recall that our probability space is Ω . The process $(\theta, \omega) \mapsto F_\theta^\omega = F^\omega(E_\theta)$ has the same finite dimensional distributions as Brownian motion on $\underline{g}$, if time is clocked by the areas $|E_\theta|$ (instead of by θ). Therefore, by Kolmogoroff's Lemma, there is a modified process $(\theta, \omega) \mapsto F'_\theta(\omega)$ such that $\theta \mapsto F'_\theta(\omega)$ is continuous for all $\omega \in \Omega$, and such that, for each θ , $F_\theta = F'_\theta$ almost everywhere. To

understand g_θ , split up $[a, b]$ into intervals of length $\frac{1}{2^n}$ (fixed integer $n \geq 1$). Suppose θ lies in the k -th interval $[a_k, a_{k+1}]$. Define $[a, b] \times \Omega \rightarrow G : \theta \mapsto g^{(n)}(\theta; \omega)$ inductively by setting (we suppress ω) $g^{(n)}(a) = e$ and $g^{(n)}(\theta) = \exp[-(F'_\theta - F'_{a_k})]g^{(n)}(a_k)$. Then (as shown in [McK]), for almost every $\omega \in \Omega$, $g^{(n)}(\theta)$ converges uniformly to a continuous function $[a, b] \rightarrow G$. This limit is $\theta \mapsto g_\theta$, the solution of (4.2). Thus the solution of (4.2) is a map $\Omega \times [a, b] \rightarrow G : (\omega, \theta) \mapsto g_\theta(\omega)$, and there is a set $\Omega' \subset \Omega$, of measure 1, such that $\theta \mapsto g_\theta(\omega)$ is continuous for all $\omega \in \Omega'$. It is well-defined to the extent that if g'_θ is another solution then there is a set $N \subset \Omega$ of measure 0 such that $g_\theta(\omega) = g'_\theta(\omega)$ for every $\theta \in [a, b]$ and for every $\omega \in \Omega \setminus N$. The process $\theta \mapsto g_\theta$ describes Brownian motion on G with time clocked by the areas $|E_\theta|$ instead of by θ . $\square$

4.6. *Terminology (Grid, Lasso* : See Figure 2). Consider a set of curves in D consisting of radii $M_0, \dots, M_n$ (from O to points on ∂D), in anticlockwise order starting with M_0 the radius along the positive y -axis, and horizontal curves $\theta \mapsto (\theta, \sigma_{ij}(\theta))$ (in polar coordinates) running from the M_i to M_{i+1} . Here, and always, we shall write $M_{n+1} = M_0$. We label the horizontal curves running from M_i to M_{i+1} by $\sigma_{i1}, \dots, \sigma_{ik_i}$, ordered so that $\sigma_{i1} \leq \dots \leq \sigma_{ik_i}$. We assume that : (i) σ_{ik_i} runs along ∂D (i.e. $\sigma_{ik_i} = 1$), and (2) if $j \neq l$ then σ_{ij} and σ_{il} do not have any interior points in common. Such a system of curves will be called a *grid*. (Note : For later sections, we will adopt certain further conditions stated below in Convention 4.13). We write σ_{i0} to denote the constant curve at O . Denote by L_{ij} the closed loop based at O formed by following the radial segment along M_i to the initial point of σ_{ij} , followed by σ_{ij} , followed by the radial segment along $\overline{M}_{i+1}$ to the final point of σ_{ij} , followed by $\overline{\sigma}_{i+1,j}$, followed by the radial segment along $\overline{M}_i$ back to O . Such a loop will be called a *lasso*. The lassos in a grid can be ordered in a natural sequence $L_1, \dots, L_m$ such that $\overline{M}_0 \partial D M_0$ is a reduction of the composite loop $L_m \dots L_1$ in the sense that the loop $\overline{M}_0 \partial D M_0$ is obtained by dropping successively all curve segments in $L_m \dots L_1$ which are traversed in opposite directions consecutively. $\square$

4.7. *Notation (dx , $Q_t(x)$)*. Equip G with a fixed bi-invariant Haar measure dx . Let $Q_t(x)$ denote the probability density, relative to dx , at time t of the standard Brownian motion $b_t(x)$ (governed by the metric $\langle \cdot, \cdot \rangle_g$) on G . Thus $Q_t(x)$ is a multiple of the heat kernel on G , satisfying $\frac{\partial Q_t(x)}{\partial t} = \frac{1}{2} \Delta Q_t(x)$, where Δ is the Laplacian on G associated to the metric $\langle \cdot, \cdot \rangle_g$. $\square$

The following result will be a fundamental tool for our ultimate goal of calculating loop expectation values for surfaces, in Section 8.

4.8. *Theorem. Let $L_1, \dots, L_m$ be the sequence of lassos in a grid as described above. Let A_i denote the area of the bounded open subset of the plane enclosed by (part of) L_i . Then for every bounded measurable function f on G^m ,*

$$\begin{aligned} \int f(g(L_1; \omega), \dots, g(L_m; \omega)) \, d\mu_{\text{YM, disk}}(\omega) \\ = \int_{G^m} f(x_1, \dots, x_m) Q_{A_1}(x_1) \dots Q_{A_m}(x_m) \, dx_1 \dots dx_m. \end{aligned}$$

That is, the $g(L_i)$'s are independent random variables with density function $Q_{A_i}(x)$. $\square$

This result is from [Sel], in the case G is compact. We provide details here which were not explained in [Sel].

Theorem 4.8 is direct consequence of the following Proposition :

4.9. Proposition. Let $c_i : [a, b] \rightarrow D : \theta \mapsto (c_i(\theta), \theta)$ (in polar coordinates (r, θ) as usual), for $i = 1, 2$, be continuous curves, where $0 \leq a \leq b \leq 2\pi$. Suppose $c_1 \leq c_2$. Then the G -valued random variable $g(c_1)^{-1}g(c_2)$, on $(\Omega, \mu_{YM, disk})$, has probability density $Q_A(x)$ with respect to the Haar measure dx on G , where A is the $d\sigma$ -area of the region $\{(r, \theta) : a \leq \theta \leq b, c_1(\theta) \leq r \leq c_2(\theta)\}$. Moreover, $g(c_1)^{-1}g(c_2)$ is measurable with respect to the σ -algebra generated by the random variables $F(E)$ as E runs over $\{(r, \theta) : r \in [0, c_2(\theta)], \theta \in [a, b]\} \subset D$, but is independent of the σ -algebra generated by all the random variables $F(E)$ as E runs over Borel subsets of $D \setminus \{(r, \theta) : a \leq \theta \leq b, c_1(\theta) \leq r \leq c_2(\theta)\}$. $\square$

The proof of Proposition 4.9 is obtained from Lemma 4.10. In the application to Proposition 4.9, (i) $\mathcal{F}_t^i$ is the (completed) σ -algebra generated by all random variables $F(E)$ as E runs over Borel subsets of $\{(r, \theta) : 0 \leq r \leq c_1(\theta), \theta \in [a, a+t]\}$ (as before, we are abusing notation by describing points by their polar coordinates), (ii) $\mathcal{F}_t^2$ is defined analogously with respect to c_2 , and (iii) $\mathcal{F}_t^3$ is the completed σ -algebra generated by $F(E)$ with E running over all Borel subsets of $\{(r, \theta) : \theta \in [a, a+t], c_1(\theta) \leq r \leq c_2(\theta)\}$. Moreover, we take $F_1(t) = F(\{(r, \theta) : \theta \in [a, a+t], 0 \leq r \leq c_1(t)\})$, define F_2 analogously with respect to c_2 , and $F_3(t) = F(\{(r, \theta) : r \in [c_1(\theta), c_2(\theta)], \theta \in [a, a+t]\})$.

4.10 Lemma. Let $(\Omega, \mathcal{F}, P)$ be a probability space, $T > 0$, and $(\mathcal{F}_t^1)_{0 \leq t \leq T}, (\mathcal{F}_t^2)_{0 \leq t \leq T}, (\mathcal{F}_t^3)_{0 \leq t \leq T}$ increasing families of σ -subalgebras of $\mathcal{F}$, such that :

- (1) each $\mathcal{F}_t^i$ contains all sets A in $\mathcal{F}$ which have measure $P(A) = 0$
- (2) $\mathcal{F}_t^2 = \sigma(\mathcal{F}_t^1 \cup \mathcal{F}_t^3)$
- (3) the σ -algebras $\mathcal{F}_T^3$ and $\mathcal{F}_T^1$ are independent.

For $i \in \{1, 2, 3\}$ and $t \in [0, T]$, let $(F_i(t))$ be a continuous square-integrable $\underline{g}$ -valued $(\mathcal{F}_t^i)$ -adapted process, with mean 0, such that $F_i(t) - F_i(s)$ is independent of $\mathcal{F}_s^i$ whenever $0 \leq s \leq t$, and with quadratic variation denoted $(dF_i(t))^2$. Assume also that for each $i \in \{1, 2, 3\}$, $t \geq 0$, and $x \in G$, the random variable $x F_i(t) x^{-1}$ has the same distribution as $F_i(t)$ (for instance, $F_i(t)$ is Gaussian with density proportional to $e^{-C_i^i \|x\|_2^2}$, for some constant $C_i^i > 0$). For $i \in \{1, 2, 3\}$, let $(h_i(t))$ denote the continuous $(\mathcal{F}_t^i)$ -adapted G -valued process which satisfies the stochastic differential equation (for $0 \leq t \leq T$)

$$dh_i(t) = \left(-dF_i(t) + \frac{1}{2} \{dF_i(t)\}^2 \right) h_i(t)$$

with initial condition $h_i(0) = e$. Let

$$h(s) \stackrel{\text{def}}{=} h_1(s)^{-1} h_2(s).$$

Then $t \mapsto h(t)$ is a continuous process having the same finite dimensional distributions as $h_3(t)$. If $\mathcal{A}$ is any σ -subalgebra of $\mathcal{F}$ for which the σ -algebra $\sigma(\mathcal{A} \cup \mathcal{F}_t^1)$ is independent of $\sigma\{F_u^3 - F_t^3 : t \leq u \leq T\}$, then $h(t)$ is independent of $\mathcal{A}$.

Proof: By Ito's Lemma :

$$dh(s) = dh_1(s)^{-1} h_2(s) + h_1(s)^{-1} dh_2(s) + dh_1(s)^{-1} dh_2(s).$$

Again, by Ito's Lemma,

$$d(h_1(s)^{-1}) = -h_1(s)^{-1}dh_1(s)h_1(s)^{-1} + \frac{1}{2}2h_1(s)^{-1}\left(dh_1(s)\right)h_1(s)^{-1}\left(dh_1(s)\right)h_1(s)^{-1}.$$

So, using

$$dh_1(s) = -dF_1(s)h_1(s) + \frac{1}{2}\left(dF_1(s)\right)^2h_1(s),$$

we obtain

$$dh_1(s)^{-1} = h_1(s)^{-1}dF_1(s) + \frac{1}{2}h_1(s)^{-1}\left(dF_1(s)\right)^2.$$

Then

$$\begin{aligned} dh(s) &= d\left(h_1(s)^{-1}\right)h_2(s) + h_1(s)^{-1}dh_2(s) + d\left(h_1(s)^{-1}\right)dh_2(s) \\ &= h_1(s)^{-1}dF_1(s)h_2(s) + \frac{1}{2}h_1(s)^{-1}\left(dF_1(s)\right)^2h_2(s) \\ &\quad - h_1(s)^{-1}dF_2(s)h_2(s) + \frac{1}{2}h_1(s)^{-1}\left(dF_2(s)\right)^2h_2(s) - h_1(s)^{-1}dF_1(s)dF_2(s)h_2(s). \end{aligned}$$

So

$$\begin{aligned} dh(s) &= h_1(s)^{-1}dF_1(s)h_2(s) + \frac{1}{2}h_1(s)^{-1}\left(dF_1(s)\right)^2h_2(s) - h_1(s)^{-1}dF_2(s)h_2(s) \\ &\quad + \frac{1}{2}h_1(s)^{-1}\left(dF_2(s)\right)^2h_2(s) - h_1(s)^{-1}dF_1(s)dF_2(s)h_2(s) \\ &= h_1(s)^{-1}\left[dF_1(s) - dF_2(s) + \frac{1}{2}\left\{\left(dF_1(s)\right)^2 + \left(dF_2(s)\right)^2 - 2dF_1(s)dF_2(s)\right\}\right]h_2(s). \end{aligned}$$

Now F_2 is the sum of the independent, mean zero, processes F_1 and F_3 . So :

$$\begin{aligned} \left(dF_1(s)\right)^2 + \left(dF_2(s)\right)^2 - 2dF_1(s)dF_2(s) &= \left(dF_1(s)\right)^2 + \left(dF_3(s)\right)^2 + \left(dF_1(s)\right)^2 - \\ &\quad 2dF_1(s)\left(dF_3(s) + dF_1(s)\right) \\ &= \left(dF_3(s)\right)^2 + 2\left(dF_1(s)\right)^2 - 0 - 2\left(dF_1(s)\right)^2 \\ &= \left(dF_3(s)\right)^2 \end{aligned}$$

Thus :

$$dh(s) = h_1(s)^{-1}\left[-dF_3(s) + \frac{1}{2}\left(dF_3(s)\right)^2\right]h_1(s)h(s).$$

Let

$$j(s) = \int_0^s \left[-dF_3(u) + \frac{1}{2}\left(dF_3(u)\right)^2\right]$$

and

$$\beta(s) = \int_0^s h_1(u)^{-1} dj(u) h_1(u).$$

Since $\underline{g} \rightarrow \underline{g} : X \mapsto gXg^{-1}$ is an isometry for each $g \in G$, it follows, upon writing out the definition of $\beta(s)$ in terms of an orthonormal basis of $\underline{g}$, that the integral defining $\beta(s)$ exists. Thus :

$$dh(s) = d\beta(s)h(s).$$

By Lemma 4.11 below, for any $0 \leq s_1 \leq \dots \leq s_m \leq T$, $(\beta(s_1), \dots, \beta(s_m))$ has the same distribution as $(j(s_1), \dots, j(s_m))$, and is independent of $\mathcal{A}$.

Now $h(s) = \lim_{n \rightarrow \infty} h^{(n)}(s)$, almost surely, where $h^{(n)}$ is defined ‘inductively’ by :

$$h^{(n)}(s) \stackrel{\text{def}}{=} \exp \left(\beta(s) - \beta \left(\frac{l}{2^n} \right) \right) h^{(n)} \left(\frac{l}{2^n} \right),$$

with $h^{(n)}(0) = e$, and $l = [2^n s]$. Since $(\beta(s_1), \beta(s_2) - \beta(s_1), \dots, \beta(s_m) - \beta(s_{m-1}))$ is independent of $\mathcal{A}$, it follows that $(h^{(n)}(s_1), \dots, h^{(n)}(s_m))$ is independent of $\mathcal{A}$. Then by the reasoning used in the proof of Lemma 4.11, $(h(s_1), \dots, h(s_m))$ is independent of $\mathcal{A}$. Since, by Lemma 4.11, $\{\beta(s)\}$ has the same finite dimensional distributions as $\{j(s)\}$, it follows that $s \mapsto h^{(n)}(s)$ has the same finite dimensional distributions as $s \mapsto h_3^{(n)}(s)$, where the latter is defined in the same way as $h^{(n)}(s)$ except with β replaced by j . Since $h_3^{(n)} \rightarrow h_3$, almost surely, it follows, as in the proof of Lemma 4.11, that $s \mapsto h(s)$ has the same finite dimensional distributions as $s \mapsto h_3(s)$. $\square$

4.11. Lemma. With notation and hypotheses as in Lemma 4.10, define for every $t \in [0, T]$,

$$\beta(t) \stackrel{\text{def}}{=} \int_0^t h_1(s)^{-1} [-dF_3(s) + \frac{1}{2} (dF_3(s))^2] h_1(s).$$

Let $\mathcal{A}$ be a σ -algebra of subsets of Ω such that $\sigma(\mathcal{F}_s^1 \cup \mathcal{A})$ is independent of $\sigma\{F_3(s') - F_3(s) : s' \in [s, T]\}$, for every $s \in [0, T]$. Then

- (i) for $0 \leq s_1 \leq \dots \leq s_n \leq T$, $(\beta(s_1), \beta(s_2), \dots, \beta(s_n))$ is independent of $\mathcal{A}$;
- (ii) the process $s \mapsto \beta(s)$ has the same finite dimensional distributions as the process $s \mapsto j(s) = \int_0^s [-dF_3(u) + \frac{1}{2} (dF_3(u))^2]$.

Proof. Let

$$X^{(n)}(s) = \int_0^s h_1^{(n)}(u)^{-1} dj(u) h_1^{(n)}(u)$$

where $h_1^{(n)}(u) = h_1(2^{-n}[2^nu])$ and j is as above. By continuity of h_1 , we have $h_1^{(n)} \rightarrow h_1$ uniformly on $[0, 1]$, and so, by the Borel-Cantelli Lemma, every subsequence of $(h_1^{(n)})$ has a further subsequence $(h_1^{(n_k)})$ such that

$$P \left[\int_0^1 \|h_1(u) - h_1^{(n_k)}(u)\|_{\underline{g}}^2 du \leq 2^{-k}, \text{ as } k \uparrow \infty \right] = 1.$$

A similar statement holds for $\text{Ad} \left(h_1^{(n)}(u)^{-1} \right)$. This implies, by the definition of the stochastic integral (see [McK]), that

$$\beta(s) = \lim_{k \rightarrow \infty} X^{(n_k)}(s) \quad \text{almost surely.}$$

Now

$$X^{(n)}(s) = \sum_{1 \leq k \leq l} h_1 \left(\frac{k-1}{2^n} \right)^{-1} (\Delta_k j) h_1 \left(\frac{k-1}{2^n} \right) + h_1 \left(\frac{l}{2^n} \right)^{-1} \left[j(s) - j \left(\frac{l}{2^n} \right) \right] h_1 \left(\frac{l}{2^n} \right)$$

where $l = [2^n s]$, and

$$\Delta_k j \stackrel{\text{def}}{=} j \left(\frac{k}{2^n} \right) - j \left(\frac{k-1}{2^n} \right).$$

Since $\langle \cdot, \cdot \rangle_g$ is Ad-invariant it follows that for every $x \in G$, $x^{-1} \Delta_k j x$ has the same distribution as $\Delta_k j$. So, since $\Delta_k j$ is independent of the $\sigma(\mathcal{A} \cup \mathcal{F}_t^2)$, for $t \in [0, \frac{k-1}{2^n}]$, and $h_1 \left(\frac{k-1}{2^n} \right)$ is $\mathcal{F}_{(k-1)2^{-n}}^1$ -measurable, it follows by Lemma 4.12 that $\text{Ad} \left(h_1 \left(\frac{k-1}{2^n} \right)^{-1} \right) \Delta_k j$ has the same distribution as $\Delta_k j$, and is independent of $\sigma(\mathcal{A} \cup \mathcal{F}_t^2)$, for all $t \in [0, (k-1)2^{-n}]$. The same holds for the last term in the sum for $X^{(n)}(s)$. Thus each term in the sum for $X^{(n)}(s)$ is independent of all the preceding terms, and $X^{(n)}(s)$ is independent of $\mathcal{A}$ (being the sum of terms *jointly* independent of $\mathcal{A}$). Moreover, the variables $j(\frac{k}{2^n}) - j(\frac{k-1}{2^n})$ (and $j(s) - j(l/2^n)$) are also mutually independent.

Thus $X^{(n)}(s)$ has the same distribution as $j(s)$, and is independent of $\mathcal{A}$. Applying this argument to $X^{(n)}(s') - X^{(n)}(s)$, with $s \leq s'$, we see that $X^{(n)}(s') - X^{(n)}(s)$ is independent of $\sigma(\mathcal{A} \cup \mathcal{F}_s^2)$, and has the same distribution as $j(s') - j(s)$. So if $0 = s_0 \leq s_1 \leq \dots \leq s_k$, then the r -th term in $(X^{(n)}(s_1), X^{(n)}(s_2) - X^{(n)}(s_1), \dots, X^{(n)}(s_k) - X^{(n)}(s_{k-1}))$ is independent of $\sigma(\mathcal{A} \cup \mathcal{F}_{s_{r-1}}^2)$, and the r -th term is distributed as $j(s_r) - j(s_{r-1})$.

Therefore, the variables $X^{(n)}(s_r) - X^{(n)}(s_{r-1})$ are mutually independent and

$$\begin{aligned} (X^{(n)}(s_1), \dots, X^{(n)}(s_k)) &= \\ (X^{(n)}(s_1), X^{(n)}(s_1) + X^{(n)}(s_2) - X^{(n)}(s_1), \dots, X^{(n)}(s_1) + \dots + X^{(n)}(s_k) - X^{(n)}(s_{k-1})) \end{aligned}$$

has the same distribution as does

$$\begin{aligned} (j(s_1), \dots, j(s_k)) \\ = (j(s_1), j(s_1) + j(s_2) - j(s_1), \dots, j(s_1) + j(s_2) - j(s_1) + \dots + j(s_k) - j(s_{k-1})). \end{aligned}$$

That is, the finite dimensional distributions of $s \mapsto X^{(n)}(s)$ are the same as those of $s \mapsto j(s)$. Moreover, each $X^{(n)}(s)$ is independent of $\mathcal{A}$.

Now recall that $\beta(s) = \lim_{k \rightarrow \infty} X^{(n_k)}(s)$, almost surely. For any continuous bounded function f on g and $B \in \mathcal{A}$ we have :

$$\begin{aligned} \lim_{k \rightarrow \infty} \int_{\Omega} f \left(X^{(n_k)}(s) \right) 1_B dP &= \int_{\Omega} \lim_{k \rightarrow \infty} f \left(X^{(n_k)}(s) \right) 1_B dP \quad \text{by dominated convergence} \\ &= \int_{\Omega} f \left(\lim_{k \rightarrow \infty} X^{(n_k)}(s) \right) 1_B dP \quad \text{because } f \text{ is continuous} \\ &= \int_{\Omega} f(\beta(s)) 1_B dP. \end{aligned}$$

But, on the other hand, since $X^{(n_k)}(s)$ is independent of $\mathcal{A}$,

$$\int_{\Omega} f(X^{(n_k)}(s)) 1_B dP = \int_{\Omega} f(X^{(n_k)}(s)) dP \cdot P(B) \rightarrow \int_{\Omega} f(\beta(s)) dP \cdot P(B).$$

Thus,

$$\int_{\Omega} f(\beta(s)) 1_B dP = \int_{\Omega} f(\beta(s)) dP \cdot P(B).$$

This being true for all continuous bounded functions f on $\underline{g}$, it is also true for $f = 1_C$ for any Borel $C \subset \underline{g}$. (For if C is closed then this follows by using dominated convergence applied to the functions $f_n(\cdot) = \max\{1 - n d(\cdot, C), 0\}$, with d denoting distance; since the class of all C for which the claim holds is closed under complements and countable disjoint unions, it contains all Borel $C \subset \underline{g}$). That is, $\beta(s)$ is independent of $\mathcal{A}$. Similarly, for $s_1 \leq s_2 \leq \dots \leq s_m$,

$$(\beta(s_1), \beta(s_2), \dots, \beta(s_m))$$

is independent of $\mathcal{A}$.

We also know that for any bounded continuous function f on $\underline{g}^m$

$$\int_{\Omega} f(X^{(n)}(s_1), X^{(n)}(s_2), \dots, X^{(n)}(s_m)) dP = \int_{\Omega} f(j(s_1), j(s_2), \dots, j(s_m)) dP.$$

Then by, for instance, the limiting argument used above (now applied to the function f on $\underline{g}^m$) we conclude that the $\underline{g}$ -valued random variable $(\beta(s_1), \beta(s_2), \dots, \beta(s_m))$ has the same distribution as $(j(s_1), j(s_2), \dots, j(s_m))$. $\square$

4.12 Lemma. *Let $(\Omega, \mathcal{F}, P)$ be a probability space, $M : \Omega \rightarrow \underline{g}$ a random variable, Σ a σ -algebra of measurable subsets of Ω , h a G -valued Σ -measurable function on Ω . Assume that M is independent of Σ , and that M has the same distribution as $x^{-1}Mx$ for every $x \in G$. Then the $\underline{g}$ -valued random variable $h^{-1}Mh$ is independent of Σ and has the same distribution as $\bar{M}$.*

Proof: A simple application of Fubini's theorem. See [Se 1]. $\square$

4.13. Convention for grids. We will always assume that grids have the following properties (in the notation of Terminology 4.6 and 2.2): (1) every σ_{ik_i} lies on some $\bar{K}_j(\subset \partial D)$; (2) if $q : D \rightarrow \Sigma$ is as in Section 3, and if $q(\sigma_{ik_i})$ and $q(\sigma_{jk_j})$ have any point, other than endpoints, in common, then σ_{ik_i} is identified exactly with σ_{jk_j} or with $\bar{\sigma}_{jk_j}$, by q . It is easily verifiable that any grid, as in 4.6, can be enlarged by addition of some radial lines to satisfy these conditions. $\square$

4.14. Remark. For the following section (specifically, in its application to the construction of the Yang-Mills measure), we will think of the Yang-Mills probability space on the disk as a product $(\Omega_1, \mathcal{F}_1, P_1) \times (\Omega_2, \mathcal{F}_2, P_2)$, where $(\Omega_1, \mathcal{F}_1, P_1)$ is the Yang-Mills space for the “right-half” of the disk D and $(\Omega_2, \mathcal{F}_2, P_2)$ is the space for the “left-half” of D . This is a technical device which helps in the construction of the conditional probability measure in Section 5. $\square$

5. A Conditional Probability Measure

In this section we construct a conditional probability measure which will be used in Section 8 to define the Yang-Mills measure.

5.1. Theorem. *Suppose X and Y are complete separable metric spaces with Borel σ -fields denoted $\mathcal{B}(X)$ and $\mathcal{B}(Y)$, respectively. Suppose $\mu : \mathcal{B}(X) \times \mathcal{B}(Y) \rightarrow [0, \infty)$ is a function with the property that $A \mapsto \mu(A, B)$ is a finite measure on $\mathcal{B}(X)$, for each fixed $B \in \mathcal{B}(Y)$, and $B \mapsto \mu(A, B)$ is a finite measure on $\mathcal{B}(Y)$ for each fixed $A \in \mathcal{B}(X)$. Then there is a unique measure μ on the product σ -algebra $\mathcal{B}(X) \otimes \mathcal{B}(Y)$ such that $\mu(A \times B) = \mu(A, B)$.*

Proof: If either X or Y is countable then the result is easily verifiable. For instance, if X is countable then $\mu(E) = \sum_{x \in X} \mu(\{x\}, E_x)$, where $E_x = \{y \in Y : (x, y) \in E\}$, is the required measure.

We assume that X and Y are both uncountable. Then by a classical result (see [KRP] for an exposition) X and Y are Borel isomorphic to $[0, 1]$ with its usual Borel σ -algebra.

Step 1 : Let $A_1, \dots, A_l$ be pairwise disjoint sets in $\mathcal{B}(X)$, and $B_1, \dots, B_m$ pairwise disjoint sets in $\mathcal{B}(Y)$. Then :

$$\mu(A, B) = \sum_{1 \leq i \leq l, 1 \leq j \leq m} \mu(A_i, B_j)$$

where $A = \cup_{i=1}^l A_i$ and $B = \cup_{j=1}^m B_j$. This follows from :

$$\mu(A, B) = \sum_{1 \leq i \leq l} \mu(A_i, B) = \sum_{1 \leq i \leq l} \sum_{1 \leq j \leq m} \mu(A_i, B_j).$$

Step 2 : Next suppose $A_1 \times B_1, \dots, A_n \times B_n$ are disjoint sets with each A_i in $\mathcal{B}(X)$ and each B_i in $\mathcal{B}(Y)$, and suppose $\{C_i \times D_i\}_{i=1, \dots, m}$ is a family of sets with each $C_i \in \mathcal{B}(X)$ and each $D_i \in \mathcal{B}(Y)$, such that :

$$\cup_{i=1}^n (A_i \times B_i) \subset \cup_{j=1}^m (C_j \times D_j).$$

Then :

$$\sum_{i=1}^n \mu(A_i, B_i) \leq \sum_{j=1}^m \mu(C_j, D_j) \quad (*)$$

In particular, equality holds if the $C_i \times D_i$ are also mutually disjoint and if $\cup_{i=1}^n (A_i \times B_i) = \cup_{j=1}^m (C_j \times D_j)$.

To prove $(*)$ we choose, as we may, $\{A_\alpha\}_{\alpha \in J}$, $\{B_\beta\}_{\beta \in K}$, finite families of disjoint sets in $\mathcal{B}(X)$ and $\mathcal{B}(Y)$, respectively, such that each A_i and each C_j is a union of some of the sets A_α , and each B_i and each D_j is a union of some of the sets B_β . Then, by Step (1),

$\sum_{i=1}^n \mu(A_i, B_i)$ equals the sum of all the $\mu(A_\alpha, B_\beta)$ for which $A_\alpha \times B_\beta$ is contained in some $A_i \times B_i$. On the other hand, $\sum_{j=1}^m \mu(C_j, D_j)$ equals the sum of all these $\mu(A_\alpha, B_\beta)$, possibly with repetitions, plus possibly some other $\mu(A_\alpha, B_\beta)$ as well. Consequently, (*) holds.

Step 3. Definition : If $E \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$ is a finite union of disjoint sets $A_i \times B_i$, $i = 1, \dots, m$, with each $A_i \in \mathcal{B}(X)$ and each $B_i \in \mathcal{B}(Y)$, then we write : $\mu(E) = \sum_{i=1}^m \mu(A_i, B_i)$. By Step (2) it follows that this is independent of the choice of the sets $A_i \times B_i$. The set $\mathcal{F}$ of all finite unions of sets of the type $A \times B$ with $A \in \mathcal{B}(X)$ and $B \in \mathcal{B}(Y)$, is a field; i.e., it is closed under finite unions and complements. Every element of $\mathcal{F}$ can be expressed as a finite disjoint union of sets of the type $A \times B \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$. Thus μ is defined on $\mathcal{F}$.

Step 2 implies that μ is finitely additive on $\mathcal{F}$.

Step 4 : μ is countably additive on $\mathcal{F}$.

proof : Since every set in $\mathcal{F}$ is a finite disjoint union of sets $A \times B \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$, it will suffice to show that for every sequence of disjoint sets $A_i \times B_i \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$, $i \geq 1$, with $\cup_{i \geq 1} A_i \times B_i = A \times B \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$,

$$\sum_{i \geq 1} \mu(A_i \times B_i) = \mu(A \times B).$$

Fact 2 implies that ' $\leq$ ' holds. We now prove that ' $\geq$ ' also holds. Recall that X and Y are Borel isomorphic to $[0, 1]$ with its usual Borel structure; we identify X and Y with $[0, 1]$ in this way. Let $\delta > 0$. Since $\mu(\cdot, [0, 1])$ is a finite Borel measure on $[0, 1]$, there is a closed set $A' \subset A$, and there are open sets $A'_i \supset A_i$ such that :

$$\mu(A, [0, 1]) - \mu(A', [0, 1]) < \delta/4$$

and

$$\mu(A'_i, [0, 1]) - \mu(A_i, [0, 1]) < \frac{\delta}{2^{i+2}}.$$

Choose B' and B_i analogously for B and B_i .

Then :

$$\begin{aligned} \mu(A, B) - \mu(A', B') &= \mu(A, B) - \mu(A', B) + \mu(A', B) - \mu(A', B') \\ &= \mu(A - A', B) + \mu(A', B - B') \\ &\leq \mu(A - A', [0, 1]) + \mu([0, 1], B - B') \\ &< \delta/2. \end{aligned}$$

Similarly,

$$\mu(A'_i, B'_i) - \mu(A_i, B_i) < \delta/2^{i+1}.$$

We can choose finite $m (\geq 1)$ such that the sets $A'_i \times B'_i$, with $1 \leq i \leq m$, cover the compact set $A' \times B' \subset [0, 1] \times [0, 1]$. Recall that μ , as defined on the field $\mathcal{F}$, is non-negative and finitely-additive. Consequently, μ is finitely-subadditive on $\mathcal{F}$. Then :

$$\begin{aligned}
\mu(A \times B) &< \mu(A' \times B') + \delta/2 \\
&\leq \mu(\cup_{i=1}^m A'_i \times B'_i) + \delta/2 \\
&\leq \sum_{i=1}^m \mu(A'_i \times B'_i) + \delta/2 \\
&< \sum_{i=1}^{\infty} \mu(A_i \times B_i) + \delta/2 + \delta/2.
\end{aligned}$$

Letting $\delta \rightarrow 0$, we obtain

$$\mu(A \times B) \leq \sum_{i=1}^{\infty} \mu(A_i \times B_i),$$

which was our goal.

Thus μ is non-negative and countably additive on the field $\mathcal{F}$, and $\mu(\emptyset) = 0$. Hence, by standard measure theory, μ extends to a measure on the σ -algebra generated by $\mathcal{F}$; i.e. to $\mathcal{B}(X) \otimes \mathcal{B}(Y)$.

Since the sets $A \times B \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$ generate the σ -algebra $\mathcal{B}(X) \otimes \mathcal{B}(Y)$, and since $\mu(X, Y)$ is finite, μ is the only measure on $\mathcal{B}(X) \otimes \mathcal{B}(Y)$ with the property that $\mu(A \times B) = \mu(A, B)$ for every $A \times B \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$. $\square$

5.2. Theorem. *Let $(X, \mathcal{B}(X))$ and $(Y, \mathcal{B}(Y))$ be complete separable metric spaces and let $g_X : X \rightarrow G$, $g_Y : Y \rightarrow G$ be Borel measurable functions, where G is a locally compact σ -compact topological group with a fixed bi-invariant Haar measure (usually to be denoted dx). Let P_X be a probability measure on $\mathcal{B}(X)$, and P_Y a probability measure on $\mathcal{B}(Y)$.*

- (H1) *Assume that there exist bounded measurable functions $\rho_X : G \rightarrow [0, \infty)$ and $\rho_Y : G \rightarrow [0, \infty)$, such that $P_X(g_X^{-1}(E)) = \int_E \rho_X(x) dx$, and $P_Y(g_Y^{-1}(E)) = \int_E \rho_Y(x) dx$, for every Borel set $E \subset G$.*
- (H2) *Assume that $\rho_X > 0$ and $\rho_Y > 0$ everywhere (i.e. that it is possible to choose the densities ρ_X and ρ_Y with this property; this implies that G has at most countably many components and hence that it is σ -compact).*

Then there is a unique assignment $(E, z) \mapsto P(E|g_X g_Y = z)$, where $E \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$ and $z \in G$, such that :

- (1) *for every $z \in G$, the assignment $E \mapsto P(E|g_X g_Y = z)$ is a probability measure on $\mathcal{B}(X) \otimes \mathcal{B}(Y)$;*
- (2) *for every $E \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$, the function $z \mapsto P(E|g_X g_Y = z)$ is measurable, and for every Borel set $V \subset G$,*

$$\int_V P(E|g_X g_Y = z) dp_{12}(z) = P_X \otimes P_Y(E \cap \{g_X g_Y \in V\})$$

where p_{12} is the probability measure on G describing the distribution of $g_X g_Y$ (i.e. $p_{12}(\cdot) = P_X \otimes P_Y(\{(\omega_1, \omega_2) : g_X(\omega_1)g_Y(\omega_2) \in \cdot\})$) and we use the standard notation

$$\{g_X g_Y \in V\} = \{(\omega_1, \omega_2) : g_X(\omega_1)g_Y(\omega_2) \in V\};$$

(3) for every $A \in \mathcal{B}(X)$ and $B \in \mathcal{B}(Y)$, the function $z \mapsto P(A \times B | g_X g_Y = z)$ is continuous.

Moreover, $P(\cdot | g_X g_Y = z)$ satisfies :

$$(4) \quad P(\{g_X g_Y = z\} | g_X g_Y = z) = 1,$$

and, for every $A \times B \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$, the function $z \mapsto P(A \times B | g_X g_Y = z)$ is uniformly continuous on G .

Proof : If E is a Borel subset of G , we write $|E|$ to denote the Haar measure of G . If $A \in \mathcal{B}(X)$ then

$$\mathcal{B}(X) \rightarrow [0, 1] : B \mapsto P_X(A \cap \{g_X \in E\})$$

is a measure satisfying

$$P_X(A \cap g_X^{-1}(E)) \leq P_X(g_X \in E) = 0 \quad \text{if } |E| = 0.$$

Therefore there is a measurable function $x \mapsto \rho_X(A, x)$ on G , with $0 \leq \rho_X(A, x) \leq \rho_X(x)$ for every $x \in G$, satisfying

$$\int_E \rho_X(A, x) dx = P_X(A \cap \{g_X \in E\}) \text{ for every Borel } E \subset G.$$

Define $\rho_Y(B, y)$ similarly, for $B \in \mathcal{B}(Y)$ and $y \in G$.

Moreover, we take, as we may,

$$\rho_X(X, x) = \rho_X(x) \text{ and } \rho_Y(Y, y) = \rho_Y(y).$$

Define :

$$\mu_z(A, B) = \frac{\int_G \rho_X(A, x) \rho_Y(B, x^{-1}z) dx}{\int_G \rho_X(x) \rho_Y(x^{-1}z) dx}.$$

Step 1 : We show that $\mu_z(A, B)$ is well-defined, finite and non-negative, and that $\mu_z(\cdot, \cdot)$ is a measure in each variable when the other variable is fixed. We prove also that $z \mapsto \mu_z(A, B)$ is continuous on G , for every $A \times B \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$.

Our hypothesis that ρ_X and ρ_Y are everywhere positive guarantees that the denominator in μ_z is not zero. The integrands in the numerator and denominator are non-negative measurable functions.

Recall that $x \mapsto \rho_X(A, x)$ is a Borel measurable function on G with $0 \leq \rho_X(A, x) \leq \rho_X(x)$, and that ρ_X is bounded. Thus $\rho_X(A, \cdot)$ and $\rho_Y(B, \cdot)$ are bounded functions in $L^1(G, dx)$, and therefore the convolution $\int_G \rho_X(A, x) \rho_Y(B, x^{-1}z) dx$ is finite, non-negative and uniformly continuous in z (recall that G is equipped with a bi-invariant Haar measure). This proves that $\mu_z(A, B)$ is finite, non-negative, and uniformly continuous in z .

Step 2 : We show that, for fixed $B \in \mathcal{B}(Y)$, the assignment

$$A \mapsto p(A) = \int_G \rho_X(A, x) \rho_Y(B, x^{-1}z) dx$$

is a finite measure on $\mathcal{B}(X)$. Replacing x with zx^{-1} , and using the fact that bi-invariance of dx implies $d(x^{-1}) = dx$, and by interchanging A and B in the arguments below, it will follow that $\int_G \rho_X(A, x) \rho_Y(B, x^{-1}z) dx$ is a measure in $B \in \mathcal{B}(Y)$, for every fixed $A \in \mathcal{B}(X)$.

For any $A \in \mathcal{B}(X)$, we have :

$$p(A) \leq \|\rho_Y\|_{\sup} \int_G \rho_X(x) dx = \|\rho_Y\|_{\sup}.$$

Now we claim that for every $A \in \mathcal{B}(X)$:

$$p(A) = \int_A dP_X(\omega_1) \rho_Y(B, g_X(\omega_1)^{-1}z) \quad (*)$$

This would directly imply that $A \mapsto p(A)$ is a measure on $\mathcal{B}(X)$.

Now we prove (*). For any Borel $C \subset G$, we have from the definition of $\rho_X(A, \cdot)$:

$$\int_A dP_X(\omega_1) 1_C(g_X(\omega_1)) = \int_G dx \rho_X(A, x) 1_C(x).$$

Taking linear combinations and monotone limits we obtain, for every non-negative Borel function f on G :

$$\int_A dP_X(\omega_1) f(g_X(\omega_1)) = \int_G dx \rho_X(A, x) f(x) \quad (**)$$

Setting $f(x) = \rho_Y(B, x^{-1}z)$, which we know to be measurable in x and non-negative, we obtain (*).

Step 3 : Thus by Theorem 5.1, there is a unique measure $P(\cdot | g_X g_Y = z)$ on $\mathcal{B}(X) \otimes \mathcal{B}(Y)$ such that $P(A \times B | g_X g_Y = z) = \mu_z(A, B)$ for every $A \in \mathcal{B}(X)$ and $B \in \mathcal{B}(Y)$. This completes the proof of (1). In view of Step 1, we have also proven (3).

Step 4 : We prove the first statement in (2); i.e. that $z \mapsto P(E | g_X g_Y = z)$ is a Borel function on G , for every $E \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$. For this we observe that the class of all sets $E \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$ for which $z \mapsto P(E | g_X g_Y = z)$ is Borel, contains all sets $A \times B \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$, and is closed under complements and countable disjoint unions. Therefore, by Dynkin's $\pi - \lambda$ theorem, this class contains all $E \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$.

Step 5 : We prove the second part of (2); i.e. that for every Borel $V \subset G$ and every $E \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$:

$$\int_V P(E | g_X g_Y = z) dp_{12}(z) = P_X \otimes P_Y(E \cap \{g_X g_Y \in V\}) \quad (***)$$

where $p_{12}(\cdot) = P_X \otimes P_Y(g_X g_Y \in \cdot)$. First we observe that both sides of (***) are finite measures in E . Therefore it will suffice to prove (***) for E of the form $A \times B$, where $A \in \mathcal{B}(X)$ and $B \in \mathcal{B}(Y)$. Then we have (using **) in the third step):

$$\begin{aligned}
P_X \otimes P_Y((A \times B) \cap \{g_X g_Y \in V\}) &= \int_A dP_X(\omega_1) P_Y(B \cap \{g_Y \in g_X(\omega_1)^{-1}V\}) \\
&= \int_A dP_X(\omega_1) \int_V dz \rho_Y(B, g_X(\omega_1)^{-1}z) \\
&= \int_V dz \int_G dx \rho_X(A, x) \rho_Y(B, x^{-1}z) \\
&= \int_V dz \rho_{12}(z) P(A \times B | g_X g_Y = z)
\end{aligned}$$

wherein

$$\rho_{12}(z) = \int_G dx \rho_X(x) \rho_Y(x^{-1}z).$$

Put $A = X$, and $B = Y$, in this to obtain :

$$P_X \otimes P_Y(\{g_X g_Y \in V\}) = \int_V dz \rho_{12}(z).$$

So

$$dp_{12}(z) = \rho_{12}(z) dz.$$

This completes the proof of (***) for $E = A \times B$, and hence for all $E \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$. This completes the proof of (2).

Step 6 : We prove that $(E, z) \mapsto P(E | g_X g_Y = z)$ is the unique assignment which satisfies (1), (2), and (3).

From (2), we see that for fixed $E \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$, $z \mapsto P(E | g_X g_Y = z)$ is uniquely defined up to sets of p_{12} -measure zero. Now, if V is a Borel subset of G of positive Haar measure, then

$$p_{12}(V) = \int_V \rho_X(x) \rho_Y(x^{-1}z) dx > 0$$

since $|V| > 0$, and $\rho_X > 0$ and $\rho_Y > 0$. Therefore, in particular, $p_{12}(V) > 0$ if V is a non-empty open subset of G . Thus (2) and (3) specify $P(A \times B | g_X g_Y = z)$ uniquely for all $z \in G$, $A \times B \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$. Since these $A \times B$ generate $\mathcal{B}(X) \otimes \mathcal{B}(Y)$, it follows now from (1) that $P(E | g_X g_Y = z)$ is uniquely specified for every $E \in \mathcal{B}(X) \otimes \mathcal{B}(Y)$.

Step 7: We prove that for every $z \in G$:

$$P(\{g_X g_Y = z\} | g_X g_Y = z) = 1.$$

Recall that

$$P(A \times B | g_X g_Y = z) = \frac{\int_G \rho_X(A, x) \rho_Y(B, x^{-1}z) dx}{\int_G \rho_X(x) \rho_Y(x^{-1}z) dx}.$$

Define a measure m on G by :

$$dm(x) = \rho_X(x) \rho_Y(x^{-1}z) dx.$$

Since ρ_X and ρ_Y are positive, bounded, and in $L^1(G, dx)$, the measure m is (positive and) finite. Set :

$$A = \{\omega_1 \in X : g_X(\omega_1) \in A'\}$$

and

$$B = \{\omega_2 \in Y : g_Y(\omega_2)^{-1} \in B'\}$$

where A' and B' are Borel subsets of G .

Then we can take :

$$\rho_X(A, x) = 1_{A'}(x) \rho_X(x)$$

because, for every Borel $V \subset G$:

$$\begin{aligned} \int_V \rho_X(A, x) dx &= P_X(A \cap \{g_X \in V\}) \\ &= P_X(g_X \in V \cap A') \\ &= \int_G 1_V(x) 1_{A'}(x) \rho_X(x) dx \\ &= \int_V 1_{A'}(x) \rho_X(x) dx, \end{aligned}$$

and because $0 \leq 1_{A'} \rho_X \leq \rho_X$.

Similarly,

$$\rho_Y(B, y) = 1_{B'}(y^{-1}) \rho_Y(y).$$

Therefore,

$$\begin{aligned} P(g_X \in A', g_Y^{-1} \in B' | g_X g_Y = z) &= \frac{\int_G \rho_X(x) \rho_Y(x^{-1}z) 1_{A'}(x) 1_{B'}(z^{-1}x) dx}{\int_G \rho_X(x) \rho_Y(x^{-1}z) dx} \\ &= \frac{\int_G dm(x) 1_{A' \cap zB'}(x)}{m(G)} \\ &= \frac{m(A' \cap zB')}{m(G)}. \end{aligned}$$

Notice that $m(G) > 0$.

So :

$$P(g_X \in A', g_Y \in B' | g_X g_Y = z) = \frac{m(A' \cap z(B')^{-1})}{m(G)} \quad (5.1)$$

Let $p_1 : G \times G \rightarrow G : (x, y) \mapsto x$ and

$$\Delta_z = \{(x, x^{-1}z) : x \in G\} \subset G \times G.$$

Then :

$$x \in A' \cap z(B')^{-1} \text{ if and only if } (x, x^{-1}z) \in \Delta_z \cap (A' \times B').$$

Therefore,

$$p_1[\Delta_z \cap (A' \times B')] = A' \cap z(B')^{-1} \quad (5.2)$$

Now

$$\mu : E \mapsto m(p_1(E \cap \Delta_z))/m(G) \quad (5.3)$$

is a Borel measure on $G \times G$. This is so because : (i) p_1 maps Δ_z homeomorphically onto G ; (ii) $E \cap \Delta_z = (\text{incl} : \Delta_z \rightarrow G \times G)^{-1}(E)$ is Borel in Δ_z (hence Borel in $G \times G$, since Δ_z is closed) whenever E is Borel in $G \times G$; (iii) $p_1(E_1 \cap \Delta_z)$ and $p_1(E_2 \cap \Delta_z)$ are disjoint if $E_1 \cap E_2 = \emptyset$; (iv) p_1 maps countable disjoint unions into countable disjoint unions.

By equations (5.1), (5.2) and (5.3), :

$$\mu(A' \times B') = P((g_X, g_Y) \in A' \times B' | g_X g_Y = z).$$

Thus μ and $P((g_X, g_Y) \in \cdot | g_X g_Y = z)$ are finite Borel measures on $G \times G$ which agree on all Borel sets of the form $A \times B$. Therefore :

$$\mu(V) = P((g_X, g_Y) \in V | g_X g_Y = z)$$

for every Borel $V \subset G \times G$.

In particular, taking $V = \Delta_z$, we have :

$$\begin{aligned} P(\{g_X g_Y = z\} | g_X g_Y = z) &= \mu(V) \\ &= \frac{m(p_1(V \cap \Delta_z))}{m(G)} \\ &= \frac{m(p_1(\Delta_z))}{m(G)} \\ &= \frac{m(G)}{m(G)} \\ &= 1. \quad \square \end{aligned}$$

5.3 Remark. We have seen that :

$$P(A \times B | g_X g_Y = z) = \frac{1}{\rho_{12}(z)} \int_A dP_X(\omega_1) \rho_Y(B, g_X(\omega_1)^{-1} z).$$

This shows that $P(A \times B | g_X g_Y = z) = 0$ if $P_X(A) = 0$. Interchanging the roles of A and B we obtain the same conclusion under the assumption that $P_Y(B) = 0$. Thus $P(A \times B | g_X g_Y = z) = 0$ if $P_X(A)P_Y(B) = 0$. This has the following significance. We will, in later work, use G -valued random variables on $(\Omega, \mathcal{F})$, where Ω will have the form $\Omega_1 \times \Omega_2 \times \Omega_3$, which are obtained as products of random variables X_{ij} defined on the component spaces Ω_i , $i \in \{1, 2, 3\}$. If each X_{ij} is redefined on a set of P_i -measure 0 then Lemma 5.9 (which is a consequence of the observation above) implies that the product of the X_{ij} is changed only on a set of $P(\cdot | g_1 g_2 = z g_3)$ -measure zero. Thus the product of the X_{ij} is a well-defined random variable on $(\Omega, \mathcal{F}, P(\cdot | g_1 g_2 = z g_3))$. $\square$

5.4. *Definition.* Let $(\Omega_1, \mathcal{F}_1, P_1), (\Omega_2, \mathcal{F}_2, P_2), (\Omega_3, \mathcal{F}_3, P_3)$ be probability spaces. Write $\Omega = \Omega_1 \times \Omega_2 \times \Omega_3$, and $\mathcal{F} = \mathcal{F}_1 \otimes \mathcal{F}_2 \otimes \mathcal{F}_3$. Assume that for $i \in \{1, 2\}$, Ω_i is a complete separable metric space and $\mathcal{F}_i$ is its Borel σ -algebra. Let G be a locally compact σ -compact topological group with a fixed bi-invariant Haar measure. Let $g_i : \Omega_i \rightarrow G$ be $\mathcal{F}_i$ -measurable, for $i \in \{1, 2, 3\}$. Assume that g_1 and g_2 have bounded positive densities ρ_1 and ρ_2 , respectively, with respect to the Haar measure on G . For $z \in G$ and $E \in \mathcal{F} = \mathcal{F}_1 \otimes \mathcal{F}_2 \otimes \mathcal{F}_3$, we set :

$$P(E|g_1g_2 = zg_3) \stackrel{\text{def}}{=} \frac{1}{Z(z)} \int_{\Omega_3} dP_3(\omega_3) P(E^{\omega_3}|g_1g_2 = zg_3(\omega_3)) \rho_{12}(zg_3(\omega_3))$$

wherein :

$$E^{\omega_3} = \{(\omega_1, \omega_2) \in \Omega_1 \times \Omega_2 : (\omega_1, \omega_2, \omega_3) \in E\},$$

$$\rho_{12}(x) = \int_G \rho_1(y) \rho_2(y^{-1}x) dy,$$

and

$$Z(z) = \int_{\Omega_3} dP_3(\omega_3) \rho_{12}(zg_3(\omega_3)).$$

($Z(z) > 0$ because $\rho_{12} > 0$.) $\square$

5.5. *Remark.* We verify that $P(\cdot|g_1g_2 = zg_3)$ is well-defined. The class $\mathcal{B}$ of all $E \in \mathcal{F}$ for which $\xi : \omega_3 \mapsto P(E^{\omega_3}|g_1g_2 = zg_3(\omega_3))$ is defined and measurable contains all sets of the form $E = E_1 \times H$, where $E_1 \in \mathcal{F}_1 \otimes \mathcal{F}_2$ and $H \in \mathcal{F}_3$, because then $\xi(\omega_3) = 1_H(\omega_3) P(E_1|g_1g_2 = zg_3(\omega_3))$ and this is $\mathcal{F}_3$ -measurable since $x \mapsto P(E|g_1g_2 = x)$ is Borel measurable on G . Moreover, $\mathcal{B}$ is closed under complements and countable disjoint unions. Therefore $\mathcal{B} = \mathcal{F}$, by Dynkin's $\pi - \lambda$ theorem. $\square$

5.6. *Lemma.* For every $A \in \mathcal{F}_1$, $B \in \mathcal{F}_2$, $C \in \mathcal{F}_3$, the function $G \rightarrow [0, 1] : z \mapsto P(A \times B \times C|g_1g_2 = zg_3)$ is continuous.

Proof. The definition of $P(\cdot|g_1g_2 = zg_3)$ gives :

$$\begin{aligned} P(A \times B \times C|g_1g_2 = zg_3) \\ = \frac{1}{Z(z)} \int_{\Omega_3} dP_3(\omega_3) 1_C(\omega_3) P(A \times B|g_1g_2 = zg_3(\omega_3)) \rho_{12}(zg_3(\omega_3)). \end{aligned}$$

Since ρ_{12} and $z \mapsto P(A \times B|g_1g_2 = zg_3(\omega_3))$ are bounded uniformly continuous functions on G , the function

$$G \rightarrow [0, 1] : z \mapsto \int_{\Omega_3} dP_3(\omega_3) 1_C(\omega_3) P(A \times B|g_1g_2 = zg_3(\omega_3)) \rho_{12}(zg_3(\omega_3))$$

is continuous.

Taking $A \times B \times C = \Omega_1 \times \Omega_2 \times \Omega_3$, we see that $z \mapsto Z(z)$ is continuous. $\square$

5.7. Lemma. For every Borel set $V \subset G$ and $E \in \mathcal{F}$,

$$\int_V P(E|g_1g_2 = zg_3) dp_{123}(z) = P_1 \otimes P_2 \otimes P_3(E \cap \{g_1g_2g_3^{-1} \in V\})$$

where $p_{123}(V) = P_1 \otimes P_2 \otimes P_3(\{g_1g_2g_3^{-1} \in V\})$.

Proof: We have :

$$dp_{123}(z) = \left[\int_{\Omega_3} \rho_{12}(zg_3(\omega_3)) dP_3(\omega_3) \right] dz = Z(z)dz.$$

Then :

$$\begin{aligned} \int_V P(E|g_1g_2 = zg_3) dp_{123}(z) &= \int_V P(E|g_1g_2 = zg_3) Z(z) dz \\ &= \int_V \left[\frac{1}{Z(z)} \int_{\Omega_3} dP_3(\omega_3) P(E^{\omega_3}|g_1g_2 = zg_3(\omega_3)) \cdot \right. \\ &\quad \left. \cdot \rho_{12}(zg_3(\omega_3)) \right] Z(z) dz \\ &= \int_V dz \int_{\Omega_3} dP_3(\omega_3) P(E^{\omega_3}|g_1g_2 = zg_3(\omega_3)) \rho_{12}(zg_3(\omega_3)) \\ &= \int_{\Omega_3} dP_3(\omega_3) \int_{Vg_3(\omega_3)} dz P(E^{\omega_3}|g_1g_2 = z) \rho_{12}(z) \\ &= \int_{\Omega_3} dP_3(\omega_3) P_1 \otimes P_2(E^{\omega_3} \cap \{g_1g_2 \in Vg_3(\omega_3)\}) \\ &= P_1 \otimes P_2 \otimes P_3(E \cap \{g_1g_2 \in Vg_3\}) \\ &= P_1 \otimes P_2 \otimes P_3(E \cap \{g_1g_2g_3^{-1} \in V\}) \quad \square. \end{aligned}$$

5.8 Lemma. For every $z \in G$:

$$P(\{g_1g_2 = zg_3\}|g_1g_2 = zg_3) = 1.$$

Proof:

$$\begin{aligned} P(\{g_1g_2 = zg_3\}|g_1g_2 = zg_3) &= \frac{1}{Z(z)} \int_{\Omega_3} P(\{g_1g_2 = zg_3(\omega_3)\}|g_1g_2 = zg_3(\omega_3)) \cdot \\ &\quad \cdot \rho_{12}(zg_3(\omega_3)) dP_3(\omega_3) \\ &= \frac{1}{Z(z)} \int_{\Omega_3} dP_3(\omega_3) 1 \cdot \rho_{12}(zg_3(\omega_3)) \\ &= \frac{1}{Z(z)} Z(z) \\ &= 1. \quad \square \end{aligned}$$

5.9. Lemma. If $A \in \mathcal{F}_1$, $B \in \mathcal{F}_2$, and $C \in \mathcal{F}_3$ are such that $P_1(A)P_2(B)P_3(C) = 0$, then $P(A \times B \times C|g_1g_2 = zg_3) = 0$ for every $z \in G$.

Proof.

$$P(A \times B \times C | g_1 g_2 = z g_3) = \frac{1}{Z(z)} \int_C dP_3(\omega_3) P(A \times B | g_1 g_2 = z g_3(\omega_3)) \rho_{12}(z g_3(\omega_3)).$$

By Remark 5.3, the integrand equals zero if $P_1(A)P_2(B) = 0$. On the other hand, if $P_3(C) = 0$ then it is clear that the integral is zero. $\square$

5.10. Proposition. We use the notation and hypotheses introduced above concerning the space $(\Omega_i, \mathcal{F}_i, P_i)$, the locally compact σ -compact group G with a fixed bi-invariant Haar measure, and the functions $g_i : \Omega_i \rightarrow G$. Let $\phi = (\phi_1, \dots, \phi_m) : \Omega_1 \rightarrow G^m$, $\psi = (\psi_1, \dots, \psi_n) : \Omega_2 \rightarrow G^n$, and $g_3 : \Omega_3 \rightarrow G$ be measurable functions. Suppose $\phi_1 \dots \phi_k = g_1$, and $\psi_1 \dots \psi_l = g_2$, for some $k \in \{1, \dots, m\}$ and $l \in \{1, \dots, n\}$. Suppose that ϕ has a bounded density ρ_ϕ on G^m and ψ has bounded density ρ_ψ on G^n (with respect to the product Haar measures on G^m and G^n). Then for every bounded or non-negative measurable function f on $G^m \times G^n \times \Omega_3$, we have the expectation values formula (wherein $\omega = (\omega_1, \omega_2, \omega_3)$) :

$$\begin{aligned} & \int_{\Omega} f(\phi(\omega_1), \psi(\omega_2), \omega_3) dP(\omega | g_1 g_2 = z g_3) \\ &= \frac{1}{Z(z)} \int_{\Omega_3} dP_3(\omega_3) \int_{G^{m+n}} dx dy f(x, y, \omega_3) \rho_\phi(x) \rho_\psi(y) \delta(x_1 \dots x_k y_1 \dots y_l g_3(\omega_3)^{-1} z^{-1}), \end{aligned}$$

wherein $dx = dx_1 \dots dx_m$, $dy = dy_1 \dots dy_n$, $\delta(\cdot)$ means that we can drop dx_j (or dy_j) and the $\delta(\cdot)$ in the integration and in the rest of the integrand give x_j (y_j , respectively) the value for which $x_1 \dots x_k y_1 \dots y_l = z g_3(\omega_3)$. The notation $Z(z)$ is as before.

Proof: It suffices to prove the case where f is of the form $1_{A \times B \times C}$. We use notation as in Theorem 5.2 except we use the subscript 1 instead of X , 2 for Y and 3 for Z . Then :

$$\begin{aligned} P(\phi \in A, \psi \in B, C) &= \frac{1}{Z(z)} \int dP_3(\omega_3) 1_C(\omega_3) P(\phi \in A, \psi \in B | g_1 g_2 = z g_3(\omega_3)) \cdot \rho_{12}(z g_3(\omega_3)) \\ &= \frac{1}{Z(z)} \int dP_3(\omega_3) 1_C(\omega_3) \int_G dt \rho_1(\phi \in A, t) \rho_2(\psi \in B, t^{-1} z g_3(\omega_3)). \end{aligned}$$

Thus it suffices to show that $\rho_1(\phi \in A, t)$ equals $\int \rho_\phi(x) \delta(x_1 \dots x_k t^{-1}) dx_1 \dots dx_k$ and that $\int_G dt \delta(x_1 \dots x_k t^{-1}) \delta(y_1 \dots y_l (t^{-1} z g_3(\omega_3))^{-1})$ can be replaced by $\delta(x_1 \dots x_k y_1 \dots y_l z g_3(\omega_3)^{-1})$. These facts can be verified readily. A detailed proof is in [Se2] (Lemmas 4.2 and 4.3 and Proposition 4.5 of [Se 2]; note that compactness of G is not used in the proof). $\square$

6. The Yang-Mills Measure

In this section we construct (Definition 6.2) a probability measure $\mu_{YM}^{\Sigma, h}$ which we shall take to be the Yang-Mills measure for gauge fields over Σ belonging to a specified topological class. Throughout this section G will be a compact connected Lie group with an Ad-invariant metric $\langle \cdot, \cdot \rangle_{\underline{g}}$ on its Lie algebra $\underline{g}$. Recall that Σ is obtained as a quotient $q : D \rightarrow \Sigma$ from the disk D ; the surface Σ has an area measure $d\sigma$, induced by its Riemannian metric, and pulling this up to D we obtain a measure on D also to be denoted $d\sigma$.

Motivated by the conclusions of Section 3 (in particular, 3.9-3.12) and of Section 4 we seek to define the Yang-Mills measure as the product of a Gaussian measure (with ‘density’ proportional to $e^{-\frac{1}{2} \int_D \|F\|_{\underline{g}}^2 d\sigma}$) on the space of all ‘ F ’, with Haar measure on G^p (one copy of G for each homotopy basis loop S_i in Σ) *conditioned* to satisfy the constraint imposed by Theorem 3.9.

6.1. *Notation.* We shall use the group G' defined by :

$$G' \stackrel{\text{def}}{=} \begin{cases} \tilde{G} & \text{if } \partial\Sigma = \emptyset \text{ and } \Sigma \text{ is orientable} \\ \tilde{G} \stackrel{\text{def}}{=} \tilde{G}/\pi_1(G)^2 & \text{if } \partial\Sigma = \emptyset \text{ and } \Sigma \text{ is unorientable} \\ G \stackrel{\text{def}}{=} \tilde{G}/\pi_1(G) & \text{if } \partial\Sigma \neq \emptyset \end{cases}$$

where we have identified $\pi_1(G)$ with $\ker(\tilde{G} \rightarrow G)$, and $\pi_1(G)^2 \stackrel{\text{def}}{=} \{k^2 : k \in \pi_1(G)\}$. $\square$

For technical reasons, we shall take the probability space for F to be the product space $(\Omega_1, \mathcal{F}_1, P_1) \times (\Omega_2, \mathcal{F}_2, P_2)$, where Ω_1 is the space of F restricted to the right half D_r of the disk D , and Ω_2 is the corresponding space for the left half D_l .

Throughout this section we shall work with three probability spaces $(\Omega_i, \mathcal{F}_i, P_i)$, $i = 1, 2, 3$; Ω_1 will be a Hilbert-Schmidt closure of the real Hilbert space $\underline{g} \otimes L^2(D_r; d\sigma)$, where D_r is the ‘right-half’ of the disk D . Then Ω_1 is a complete separable metric space and, as mentioned in Section 4, there is a probability measure P_1 , with formal ‘density’ proportional to $e^{-\frac{1}{2} \int_{D_r} \|F\|_{\underline{g}}^2 d\sigma}$, such that for any Borel sets $E_1, E_2 \subset D_r$, there are P_1 -almost-everywhere defined $\underline{g}$ -valued random variables $F(E_1)$ and $F(E_2)$ on Ω_1 such that $F(E_i)$ is Gaussian with mean 0 and, for every $v, w \in \underline{g}$:

$$\int_{\Omega_1} \langle v, F(E_1)(\omega) \rangle_{\underline{g}} \langle F(E_2)(\omega), w \rangle_{\underline{g}} dP_1(\omega) = \langle v, w \rangle_{\underline{g}} |E_1 \cap E_2|$$

wherein $|E_1 \cap E_2|$ is the σ -measure of $E_1 \cap E_2$. The space Ω_2 is the exact analog, with D_r replaced by D_l .

The definition of $(\Omega_3, \mathcal{F}_3, P_3)$ needs to be described carefully so as to include the case where Σ is the sphere with or without holes. We define $(\Omega_3, \mathcal{F}_3, P_3)$ to be G^p equipped with its Borel σ -algebra and :

- (1) in case Σ is the sphere with $k \geq 0$ holes, P_3 is the product of Haar measure (of unit total mass) on the first G^{p-1} and the unit mass δ_e at e in the last copy of G ;
- (2) in all other cases P_3 is the Haar measure on G^p of unit total mass.

We shall take g_3 to be specified by

$$g_3 : G^p \rightarrow G' : (\alpha_1, \dots, \alpha_p) \mapsto W'_\Sigma(\alpha_1, \dots, \alpha_p) = W_\Sigma(\alpha'_1, \dots, \alpha'_p)$$

where W_Σ is the word (see Notation 2.2) associated to the surface Σ , and for each $i \in \{1, \dots, p\}$, $\alpha'_i \in G'$ covers $\alpha_i \in G$ under the covering map $G' \rightarrow G$. As we have seen before in 3.15, g_3 is well-defined and it is continuous because the covering map is a local homeomorphism.

The reason why P_3 is defined in this special way when Σ is a sphere with $k(\geq 0)$ holes is as follows. In our application, $(\alpha_1, \dots, \alpha_p) \in G^p$ corresponds to the basic holonomies $(g_u(S_1; \omega), \dots, g_u(S_p; \omega))$, and for a smooth ω , when Σ is the sphere with $k(\geq 0)$ holes the term $g_u(S_p; \omega)$ is identically equal to e since S_p is the loop of the form $\bar{a}a$.

Recall the quantum gauge theory on the disk developed in Section 4. We shall use that theory for the group G' (i.e. apply the theory of Section 4 in the case the structure group is G'). Let g_1 be the G' -valued random variable on $(\Omega_1, \mathcal{F}_1, P_1)$ given by :

$$g_1(\omega_1) = g'(\partial D_r; \omega_1)$$

where ∂D_r is the loop based at O going anticlockwise around the right half of the disk D and

$$g_2(\omega_2) = g'(\partial D_l; \omega_2)$$

where ∂D_l is the loop based at O going around the anticlockwise boundary of D_l . The notation $g'(C; \omega)$ is for stochastic parallel-transport around C as in Definition 4.4, with G' as the gauge group. Recall that $* = q(O)$ is a fixed base point on Σ .

Motivation for Definition 6.2. Let P be a principal G -bundle over Σ , of topological class $[P] = h \in G'$ (see Remark 3.14). In view of the results 3.9-3.12, the space of (gauge equivalence classes of) connections on P should correspond to points in $\Omega_1 \times \Omega_2 \times G^p$

where $g_1 g_2 = h g_3$. The formal expression $e^{-\frac{1}{2} \|F\|_{L^2(D; \mathfrak{g})}^2}$ for the density of the Yang-Mills measure does not contain the holonomy variables $(g_u(S_1; \omega), \dots, g_u(S_p; \omega))$ explicitly, and is Gaussian in F . Recall that, for technical convenience, we have taken the probability space of F as the product $\Omega_1 \times \Omega_2$, with the product Gaussian measure. Thus it seems natural to take the Yang-Mills measure to be the product of the Gaussian measure for the F with Haar measure for the S_i -holonomy variables *conditioned* to satisfy $g_1 g_2 = h g_3$. We do not require any pointwise properties, such as smoothness or continuity, of ω (these form zero-measure subsets under the Gaussian measure on F even before conditioning) and so we ignore the parts of 3.11 which relate to such properties of ω . This is the motivation for Definition 6.2 below. $\blacksquare$

From Theorem 4.8 on stochastic parallel transport it follows that the distribution of g_1 on G' is $Q'_{|D_r|}(x) dx$ where dx is a bi-invariant Haar measure on G' , $|D_r|$ is the σ -area of D_r and Q'_t is the density, with respect to dx , at time t of the standard (i.e. governed by

the metric $\langle \cdot, \cdot \rangle_g$ Brownian motion $b_t(x)$ on G' . For the same reason g_2 has the density $Q'_{|D_l|}(x) dx$ on G' . Therefore, $g_1 g_2$ has the density $\rho_{12}(x) = Q'_{|D_r|}(y) Q'_{|D_l|}(y^{-1}x) dy$ (dy being the Haar measure), which on using the convolution semigroup property of Q'_t equals $Q'_{|\Sigma|}(x)$ (we have used $|D| = |\Sigma|$).

6.2. *Definition of the Yang-Mills measure* $\mu_{\Sigma}^{\Sigma, h}$. With the notation just introduced, we shall take $P(\cdot | g_1 g_2 = h g_3)$, to be denoted by $\mu_{\Sigma}^{\Sigma, h}$, as the Yang-Mills measure for gauge fields over Σ on the principal G -bundle P of topological class $[P] = h \in G'$. We shall write $\overline{\mathcal{C}}_{\Sigma}^{\Sigma}$ for Ω , the space on which $\mu_{\Sigma}^{\Sigma, h}$ is defined. Thus the probability space $(\overline{\mathcal{C}}_{\Sigma}^{\Sigma}, \mathcal{F}, \mu_{\Sigma}^{\Sigma, h})$ describes quantum gauge theory on the bundle $P \rightarrow \Sigma$ of class $[P] = h$. According to Lemma 5.8, the measure $\mu_{\Sigma}^{\Sigma, h}$ actually lives on the set where $g_1 g_2 = h g_3$ holds; i.e., $g'(\partial D_r; \omega) g'(\partial D_l; \omega) = h W'_{\Sigma}(\alpha_1, \dots, \alpha_p)$ for $\mu_{\Sigma}^{\Sigma, h}$ -almost-every ω , wherein $(\alpha_1, \dots, \alpha_p) = g_3(\omega)$. This agrees with our description of connections in Theorem 3.9. $\square$

From the remarks preceding Definition 6.2, the expression for $Z(h)$ in Definition 5.4 is seen to be equal to Z_{Σ}^h , defined by ($d\alpha_i$ being Haar measure on G with total mass 1)

$$Z_{\Sigma}^h = \int_{G^p} Q'_{|\Sigma|}(h W'_{\Sigma}(\alpha_1, \dots, \alpha_p)) d\alpha_1 \dots d\alpha_{p-1}$$

unless Σ is the sphere with $k(\geq 0)$ holes, in which case Z_{Σ}^h is :

$$Z_{\Sigma}^h = \int_{G^p} Q'_{|\Sigma|}(h W_{\Sigma}(\alpha'_1, \dots, \alpha'_{p-1}, e)) d\alpha_1 \dots d\alpha_{p-1}.$$

6.3. *Notation.* To save space we adopt the following convention : if Σ is the sphere with $k(\geq 0)$ holes we write $d'\alpha$ to denote $d\alpha_1 \dots d\alpha_{p-1} \delta(\alpha_p) d\alpha_p$, where $\delta_e(\alpha_p) d\alpha_p$ is the unit mass at e ; in all other cases, $d'\alpha$ will stand for $d\alpha_1 \dots d\alpha_p$.

Thus,

$$Z_{\Sigma}^h = \int_{G^p} Q'_{|\Sigma|}(h W'_{\Sigma}(\alpha_1, \dots, \alpha_p)) d'\alpha.$$

Observe that if G' is compact and is equipped with Haar measure of total mass 1, then

$$Z_{\Sigma}^h = \int_{G^p} Q'_{|\Sigma|}(h W_{\Sigma}(\alpha'_1, \dots, \alpha'_p)) d'\alpha'. \quad \square$$

6.4. *Remark on the role of the metric on G .* Notice that $d'\alpha$, which is an invariant measure of unit total mass on G^p , plays the role of P_3 ; on the other hand, the (unnormalized) Haar measure G' enters the discussion through Q'_t which is relevant to g_1 and g_2 . Note that Q'_t depends on the choice of the Haar measure on G' and on the Ad-invariant metric on $\underline{g}$ (used to define 'standard' Brownian motion on G'). $\square$

6.5. *Remark.* Suppose Σ is connected, compact, oriented and of genus $g \geq 1$. Suppose also that G is semisimple (and, of course, compact and connected). Equip $\tilde{G}$ with Haar measure of unit total mass. Then Z_{Σ}^h may be calculated more explicitly by expanding $\tilde{Q}_{|\Sigma|}$ in terms of the characters of $\tilde{G}$. Let χ_{α} denote the character associated to an irreducible representation α of $\tilde{G}$. If $X_1, \dots, X_d$ is an orthonormal basis of the Lie algebra of G , then

$\sum_{i=1}^d \alpha_*(X_i)^2$ is of the form $-C_\alpha I$, wherein C_α is a scalar ('Casimir') and I is the identity operator on the representation space of α . Then $\tilde{Q}_t(\tilde{x}) = \sum_\alpha (\dim \alpha) e^{-C_\alpha t/2} \chi_\alpha(\tilde{x})$, where the convergence is pointwise (see explanation below). Using this and the fact (see Ex 4.17.3 in [Brtd]) that $\int_{\tilde{G}} \chi_\alpha(aba^{-1}c) da = (\dim \alpha)^{-1} \chi_\alpha(b) \chi_\alpha(c)$, we find, after commuting the integral with the series sum (this is justified since $\sum_\alpha e^{-C_\alpha t/2} (\dim \alpha) \int |\chi_\alpha(\cdots)| d\cdots \leq \sum_\alpha e^{-C_\alpha t/2} (\dim \alpha)^2 = \tilde{Q}_t(e) < \infty$) :

$$Z_\Sigma^h = \sum_\alpha \frac{e^{-C_\alpha |\Sigma|/2} \chi_\alpha(h)}{(\dim \alpha)^{2g-1}},$$

where the sum is over all inequivalent irreducible representations α of $\tilde{G}$. The expansion of $\tilde{Q}_t(x)$ is usually justified as follows. If Δ is the Laplacian on $\tilde{G}$ associated to the Ad-invariant inner-product on $\mathfrak{g}$, then $(e^{t\Delta/2})f(y) = \int_{\tilde{G}} \tilde{Q}_t(yx^{-1})f(x) dx$, for $f \in L^2(\tilde{G}, dx)$. Moreover, this is a Hilbert-Schmidt operator on $L^2(\tilde{G}, dx)$ whose eigenvalues are the C_α , with the corresponding eigenfunctions $\phi_{\alpha,j}$ being suitably normalized matrix-elements of the representation α (with respect to any orthonormal basis). From this, one may deduce that $\tilde{Q}_t(xy^{-1}) = \sum_{\alpha,j} e^{-tC_\alpha/2} \phi_{\alpha,j}(x) \overline{\phi_{\alpha,j}(y)}$, where the convergence is in $L^2(\tilde{G} \times \tilde{G})$. That this convergence is actually pointwise (uniform and absolute) is now deduced by Mercer's theorem (see Section 98 in [RN]) using the fact that $\tilde{Q}_t$ is continuous and that $\tilde{Q}_t(x) \geq 0$. Finally, taking y to be the identity in the series expansion of $\tilde{Q}_t(xy^{-1})$, and adding up the $\phi_{\alpha,j}$ for each fixed α , we obtain $\tilde{Q}_t(\tilde{x}) = \sum_\alpha (\dim \alpha) e^{-C_\alpha t/2} \chi_\alpha(\tilde{x})$, with pointwise (absolute and uniform) convergence. $\square$

7. Invariants of Systems of Curves

We introduce certain topological invariants associated to configurations of curves on a surface. These invariants will be used in Section 8 to express loop expectation values.

7.1. Terminology (graphs, chains, H -fields, cycles) : Let $\mathcal{G}$ be a graph with bonds running between vertices. Each bond $[b]$ is equipped with two opposite orientations b and $\bar{b}$. Each oriented bond b runs from its *initial point* b^i to its final point b^f , and its reverse $\bar{b}$ runs from b^f to b^i . We denote by $E(\mathcal{G})$ the set of all oriented bonds of $\mathcal{G}$ and by $V(\mathcal{G})$ the set of all vertices of $\mathcal{G}$. If H is a group then an H -field over $\mathcal{G}$ (or a 1-chain in $\mathcal{G}$ with coefficients in H) is a map $x : E(\mathcal{G}) \rightarrow H : b \mapsto x(b) = x_b$ having the property that $x(\bar{b}) = x(b)^{-1}$. A *chain* in $\mathcal{G}$ is a sequence $b_k \dots b_1$ of oriented bonds in $\mathcal{G}$ with $b_j^f = b_{j+1}^i$. This chain is a *cycle* or *loop* if $b_k^f = b_1^i$. If $K = b_j \dots b_1$ is a sequence of oriented bonds in $\mathcal{G}$ and x is an H -field then we write $x(K) = x(b_j) \dots x(b_1)$. Let $[b_1], \dots, [b_n]$ be all the distinct bonds of a graph $\mathcal{G}$ and let F be a function of H -fields x over $\mathcal{G}$ defined by :

$$F(x) = F'(x_{b_1}, x_{\bar{b}_1}, \dots, x_{b_n}, x_{\bar{b}_n})$$

for some function F' on H^{2n} . Then we write $\int F(x) dx$ for $\int_{H^n} F'(x_1, x_1^{-1}, \dots, x_n, x_n^{-1}) dx_1 \dots dx_n$ whenever the latter makes sense. $\square$

7.2. Assumptions about (S, Γ) : Let S be a connected two dimensional finite simplicial complex, and $\Gamma = \{c_1, \dots, c_k\}$ a set of paths where each c_i is of the form $b_{i n_1} \dots b_{i 1}$ with b_{ij} being directed bonds (i.e. oriented 1-simplices) in S and $[b_{ij}] \neq [b_{i' j'}]$ unless $i = i'$ and $j = j'$. Recall that we do not distinguish between reparametrizations of a curve. We write $|S|$ for the underlying topological space of a complex S (warning : in other contexts we have used and will use $|S|$ to denote the 'area' of S). We will write $|\Gamma|$ for the set of all points of $|S|$ covered by the curves in Γ . If S is orientable then we equip it with some fixed orientation. We will assume that Γ is such that if Δ and Δ' are any two-simplices of S whose interiors lie in the same connected component of $|S| \setminus |\Gamma|$, then there is a sequence $\Delta_0 = \Delta, \Delta_1, \dots, \Delta_n = \Delta'$ of 2-simplices, all with interiors lying in the same component of $|S| \setminus |\Gamma|$, such that every consecutive pair Δ_i and Δ_{i+1} share an edge (a "face") which does not lie on any curve in Γ . Moreover, we assume the same with Γ replaced by $\emptyset$. $\square$

7.3. Notation. We denote by $\mathcal{R}(\Gamma)$ the (finite) set of components of $|S| \setminus |\Gamma|$. $\square$

If each $R \in \mathcal{R}(\Gamma)$ is a topological 2-manifold, possibly with boundary, (for example, if S is a topological 2-manifold) then Lemma A5 of [Se2] says that the above assumptions are valid.

7.4. Assumptions about H , h , and $Q_t(x)$: Let H be a compact connected Lie group, equipped with an Ad-invariant metric. Fix h in the center of H . We equip H with Haar measure of unit total mass, and we denote by $Q_t(x)$ the corresponding density at time t of the standard Brownian motion $b_t(x)$ on H (governed by the metric on H). Thus $Q_t(x)$ is a multiple of the heat kernel on H . $\square$

7.5. Lemma. Let (S, Γ) and $\mathcal{R}(\Gamma)$ be as above. For each $R \in \mathcal{R}(\Gamma)$, fix a number $a(R) > 0$. Choose any positive numbers $a(\Delta)$, one for each 2-simplex Δ of S , such that the sum of all the $a(\Delta)$, with $|\Delta| \subset |R|$, is $a(R)$. Let H , h and $Q_t(x)$ be as in section (7.4) above.

Assume: either (i) that S is orientable, with a chosen orientation, or (ii) that $h = h^{-1}$. Let f be a bounded Borel function on H . Assume that, for every $(y_1, \dots, y_k) \in H^k$, the value $f(y_1, \dots, y_k)$ remains invariant when any y_i is replaced by $h^{\pm 1}y_i$ (for arbitrary $y_1, \dots, y_k \in H$). Then the number

$$I(f) = \int_{H^k} f(x(c_1), \dots, x(c_k)) Q_{a(\Delta^*)}(hx(\partial\Delta^*)) \cdot \prod_{\Delta \in S_2, \Delta \neq \Delta^*} Q_{a(\Delta)}(x(\partial\Delta)) \delta(x(c_1)y_1^{-1}) \dots \delta(x(c_k)y_k^{-1}) dx dy_1 \dots dy_k$$

is independent of the choice of Δ^* . (The notation in $I(f)$: The integration dx is over all H -fields x over the graph specified by the 1-simplices of S and the interpretation of the δ 's is as before (see Proposition 5.10). By S_2 we have denoted the set of all 2-simplices in S . Moreover, if S is orientable, with a chosen orientation, then $\partial\Delta^*$ and the $\partial\Delta$ are oriented according to the orientation chosen for S . In the case that S is not orientable the hypothesis $h = h^{-1}$, combined with the properties $Q_t(x) = Q_t(x^{-1})$ and $Q_t(hxh^{-1}) = Q_t(x)$, ensures that the orientation of each $\partial\Delta$ can be chosen arbitrarily without affecting the value of $I(f)$.)

Proof: Let Δ_1 share a bond b with Δ^* , where b appears in $\partial\Delta^*$ and $\bar{b}$ appears in $\partial\Delta_1$. We make the change of variables $x_b \mapsto h^{-1}x_b$. Then the term $Q_{a(\Delta^*)}(hx(\partial\Delta^*))$ in $I(f)$ becomes $Q_{a(\Delta^*)}(x(\partial\Delta^*))$, while the term $Q_{a(\Delta_1)}(x(\partial\Delta_1))$ becomes $Q_{a(\Delta_1)}(hx(\partial\Delta_1))$. If b or $\bar{b}$ appears in some (unique) c_j then $x(c_j)$ is changed to $h^{-1}x(c_j)$ or $hx(c_j)$. Then by the property assumed of f , the integrand in $I(f)$ retains the same value. So Δ^* may be replaced by Δ_1 . Since for any $\Delta' \in S_2$ there is a sequence $\Delta^* = \Delta_0, \Delta_1, \dots, \Delta_m = \Delta'$ with Δ_i and Δ_{i+1} sharing a common bond, it follows that Δ^* in $I(f)$ can be replaced by any $\Delta' \in S_2$ without changing the value of $I(f)$. $\square$

7.6. *Definition ($Z_{S,\Gamma}^h$):* Assume that $|S| \setminus |\Gamma|$ has only one component. Define, for fixed $a(S) > 0$, and any choices $a(\Delta) > 0$ adding up to $a(S)$,

$$Z_{S,\Gamma}^h(y_1, \dots, y_k) = \int Q_{a(\Delta^*)}(hx(\partial\Delta^*)) \prod_{\Delta \in S_2, \Delta \neq \Delta^*} Q_{a(\Delta)}(x(\partial\Delta)) \delta(x(c_1)y_1^{-1}) \dots \delta(x(c_k)y_k^{-1}) dx,$$

where $\partial\Delta^*$ and each $\partial\Delta$ is chosen coherently with the orientation of S if S is oriented, otherwise each $\partial\Delta$ is oriented arbitrarily (but recall that then $h = h^{-1}$). Recall that h lies in the center of G . The meaning of the presence of the $\delta(\cdot)$ -terms is as in Proposition 5.10. If $h = e$, the identity in H , then we will usually write $Z_{S,\Gamma}$ instead of $Z_{S,\Gamma}^h$. $\square$

Fact 0. $Z_{S,\Gamma}^h(y_1, \dots, y_k)$ is independent of the choice of Δ^* .

Proof: Let $\Delta_1 \in S_2$ share a bond b with Δ^* where $[b]$ does not lie on any $c_j \in \Gamma$. We may assume that $\partial\Delta^*$ contains b while $\partial\Delta_1$ contains $\bar{b}$. In the integration defining $Z_{S,\Gamma}^h(y_1, \dots, y_k)$ we can change variables: $x_b \mapsto h^{-1}x_b$. Since $[b]$ does not lie on any c_j , the $\delta(\cdot)$ -terms remain as they are. The only change in the integrand is that Δ^* is replaced by Δ_1 . Since any $\Delta' \in S_2$ can be connected to Δ^* by a chain of 2-simplices

$\Delta_0 = \Delta^*, \Delta_1, \dots, \Delta_m = \Delta'$ with consecutive 2-simplices sharing a common bond not appearing in any curve of Γ , we see that Δ^* may be replaced by Δ' without changing the value of $Z_{S,\Gamma}^h(y_1, \dots, y_k)$. $\square$

Fact 1. $Z_{S,\Gamma}^h(y_1, \dots, y_k)$ is independent of the choice of the individual numbers $a(\Delta)$ as long as their sum $a(S)$ is fixed.

Fact 2. If S' is a subdivision of S then $Z_{S,\Gamma}^h = Z_{S',\Gamma}^h$ (assuming $a(S) = a(S')$).

Fact 3. Suppose (S', Γ') is another pair like (S, Γ) and that there is a homeomorphism $\phi : |S| \rightarrow |S'|$ carrying the curves of $\Gamma = \{c_1, \dots, c_k\}$ into the curves of $\Gamma' = \{c'_1, \dots, c'_k\}$ with the indexing done so that c'_i is $\phi \circ c_i$ (possibly reparametrized). Assume also that $a(S) = a(S')$. Then

$$Z_{S,\Gamma}^h = Z_{S',\Gamma'}^{h^{\pm 1}}$$

where ' $\pm$ ' is '+' if S is oriented and ϕ is orientation-preserving and '-' otherwise (recall that in case Σ is not orientable $h^{-1} = h$).

Proof of Fact 1. If $\Delta, \Delta' \in S_2$ share a common edge not appearing in any curve in Γ then the convolution property $Q_{a(\Delta)} * Q_{a(\Delta')} = Q_{a(\Delta)+a(\Delta')}$ shows that $Z_{S,\Gamma}^h(y_1, \dots, y_k)$ depends on $a(\Delta) + a(\Delta')$ but not on the individual values $a(\Delta)$ and $a(\Delta')$. Since any $\Delta_1, \Delta_2 \in S_2$ can be connected by a chain of elements of S_2 , any consecutive pair of which have the property described above for Δ and Δ' , it follows by Lemma A17 that $Z_{S,\Gamma}^h(y_1, \dots, y_k)$ depends on the sum $\sum_{\Delta \in S_2} a(\Delta) = a(S)$ but not on the individual values $a(\Delta)$. $\square$

Proof of Fact 2 : We make some preliminary observations. First we observe that the value of the integral in the definition is unchanged if x is taken to be a G -field over a (finite) graph obtained by subdividing the 1-simplices of S . This is seen easily by a change-of-variables argument together with the fact that G has total mass 1. Next we observe that if P is a polygon, equipped with a fixed orientation, divided into 2-simplices Δ , then for any G -field z on the bonds of these 2-simplices, we have :

$$\int Q_{a(\Delta^*)}(h^j z(\partial \Delta^*)) \prod_{\Delta \neq \Delta^*} Q_{a(\Delta)}(z(\partial \Delta)) dz = Q_{\Sigma_{\Delta} a(\Delta)}(h^j z(\partial P)) \quad (*)$$

where the integration is over all z_b with b running over the bonds not lying on ∂P , j any integer, the $a(\Delta)$ are positive numbers, Δ^* is some positively oriented 2-simplex appearing in P , and $\prod_{\Delta}$ is over positively oriented 2-simplices Δ . Equation (*) may be proven by induction on the number m of 2-simplices Δ making up P . It is clearly true if there is only one Δ (in which case there is no integration). Assuming the result for polygons divided into less than m two-simplices, we can consider P as being obtained by adjoining one 'outer simplex', i.e. a simplex with one or two bonds lying on ∂P , to a polygon P' divided into $m-1$ two-simplices. Since (*) holds for P' , the convolution property of Q_t implies that (*) holds for P as well.

We turn now to the main argument. If we compare the integrals defining $Z_{S,\Gamma}^h(y_1, \dots, y_k)$ and $Z_{S',\Gamma}^h(y_1, \dots, y_k)$, we see that for each term $Q_{a(\Delta)}(x(\partial \Delta))$ appearing in the integrand for $Z_{S,\Gamma}^h(y_1, \dots, y_k)$ there is, in the integrand for $Z_{S',\Gamma}^h(y_1, \dots, y_k)$, a product of terms $Q_{a(\Delta')}(x'(\partial \Delta'))$, with Δ' running over the 2-simplices of S' subdividing Δ , and correspondingly there are integrations over x'_b for 'inner bonds', i.e. bonds b of S' not lying on

$\partial\Delta$. An analogous situation holds for the simplex Δ^* of S which appears with the h -term. Using (*) we see that carrying out the integration over the dx'_b with b running over 'inner bonds' transforms $Z_{S',\Gamma}^h(y_1, \dots, y_k)$ into an integral which differs from the defining integral of $Z_{S,\Gamma}^h(y_1, \dots, y_k)$ only in that x runs over G -fields x' on a graph obtained by subdividing the bonds of S . However, as we noted in our first observation above, this is equal to $Z_{S,\Gamma}^h(y_1, \dots, y_k)$. $\square$

Proof of Fact 3: By subdividing S and S' if necessary, we may and will assume that ϕ maps the 1-simplices appearing in the curves c_i into 1-simplices of S' (though perhaps not linearly). By deforming ϕ on the 2-simplices 'bordering on' the c_i we can isotopy ϕ so that it becomes linear on the simplices appearing in the curves c_i (see (4.4) of [Br] for a more detailed argument). Then by the *Hauptvermutung* (Theorem 4.6 in [Br]), there is a subdivision $\tilde{S}$ of S and a subdivision $\tilde{S}'$ of S' , and a simplicial isomorphism $\phi: \tilde{S} \rightarrow \tilde{S}'$ such that $\tilde{\phi} \circ c_i = c'_i$ for each $i \in \{1, \dots, k\}$.

Suppose first that S is oriented (in which case so is S') and that ϕ is orientation preserving. Then $\phi(\Delta)$ is positively oriented if Δ is. Therefore, with $a(\tilde{S}) = a(\tilde{S}') = a(S)$, the integral defining $Z_{S,\Gamma}^h(y_1, \dots, y_k)$ and the integral defining $Z_{\tilde{S}',\Gamma'}^h(y_1, \dots, y_k)$ are manifestly equal. By Fact 2, it follows then that $Z_{S,\Gamma}^h$ equals $Z_{S',\Gamma'}^h$. Next suppose S and S' are oriented and suppose ϕ is orientation reversing. Then $\phi(\Delta)$ is negatively oriented if Δ is positively oriented. We observe that since $Q_t(y) = Q_t(y^{-1})$ and since $Q_t(aba^{-1}) = Q_t(b)$, we have :

$$Q_t(x(\partial\Delta)) = Q_t(x(\partial\Delta)^{-1}),$$

and

$$Q_t(hx(\partial\Delta)) = Q_t(x(\partial\Delta)^{-1}h^{-1}) = Q_t(h^{-1}x(\partial\Delta)^{-1}).$$

Since all the 2-simplices Δ , including Δ^* , appearing in the defining integral for $Z_{S',\Gamma'}^h$ are positively oriented, it follows from the observations just made that $Z_{S',\Gamma'}^h = Z_{S,\Gamma}^h$. If S is unorientable then our hypothesis $h = h^{-1}$ implies, again by the observations made above, that the integrals defining $Z_{S',\Gamma'}^h$ and $Z_{S,\Gamma}^h$ are equal. $\square$

7.7. Remark/Notation. In view of Fact 3, it makes sense to write $Z_{S,\Gamma}^h$ as $Z_{|S|,\Gamma}^h$; thus $|S|$ here is a topological space which can be triangulated in such a way that each curve in Γ is made up of a number of oriented 1-simplices, satisfying the hypotheses in 7.2 and 7.6. $\square$

7.8. Remark. Triangulating a surface using a grid: Let Σ be a compact surface displayed as a quotient of the closed unit disk D by $q: D \rightarrow \Sigma$, as in Notation 2.1. Let Γ be a grid in D . Then a finite number of extra segments and vertices can be added to Γ to obtain a larger grid Γ_1 such that each component R of $D \setminus |\Gamma_1|$ is bounded by three bonds b_1, b_2, b_3 of Γ_1 and there is a homeomorphism of a standard 2-simplex onto $\bar{R}$ such that the edges of the 2-simplex are mapped into b_1, b_2 and b_3 , respectively. By an appropriate choice of these homeomorphisms we can paste copies of the standard 2-simplex 'linearly' together so as to form a simplicial complex $\bar{D}$, and there is a homeomorphism $|\bar{D}| \rightarrow D$ carrying each bond of $\bar{D}$ onto a bond of Γ_1 , and every bond of Γ_1 is of this form. This also yields a triangulation of Σ such that the bonds of $q(\Gamma_1)$ consist of 1-simplices of the triangulation. $\square$

8. Loop Expectation Values I

In this section we define the stochastic analog of the holonomy $g_u(C; \omega)$ (notation as in Section 2; we will drop the subscript u) as a random variable on $(\overline{\mathcal{C}_o^{\Sigma, h}}, \mathcal{F}, \mu_{YM}^{\Sigma, h})$. We then compute the joint distribution of random variables $g(C_1; \omega), \dots, g(C_m; \omega)$ for collections of closed loops C_i on Σ based at o . The strategy is to lift up all the curves to the disk D , work on D and then project back to Σ .

We shall use the covering $q : D \rightarrow \Sigma$ described in Section 3. Recall also that a grid (Definition 4.6 and Convention 4.13) is a network of horizontal and radial segments in D . A ‘minimal segment’ in a grid is either one of the horizontal segments σ_{ij} or a radial segment connecting an end point of σ_{ij} to that of σ_{ij+1} .

As always, if $c : [a, b] \rightarrow X$ and if $\phi : [a', b'] \rightarrow [a, b]$ is a strictly increasing continuous mapping then c and $c \circ \phi$ will be viewed as the same curve in X .

8.1. Definition. A collection $\mathcal{C}$ of curves in Σ is *admissible* if it is non-empty and if there is a grid Γ in the covering disk D such that every curve in $\mathcal{C}$ is made up of a finite number of segments of the form $q(C)$ for some minimal segment C in the grid Γ , and a finite number of constant segments (point curves). If $\mathcal{C}$ is a set of curves in a space X , we will write $|\mathcal{C}|$ to denote the subset of X covered by the curves $\mathcal{C}$ and the set of all connected components of $X \setminus |\mathcal{C}|$ will be denoted $\mathcal{R}(\mathcal{C})$. Figure 3 shows an example of an admissible collection of curves on a torus. $\square$

Remark. If $\{C_1, \dots, C_n\}$ is admissible then by Remark 7.8 there is a triangulation of Σ such that each C_i , after dropping constant segments, is the composite of a number of oriented 1-simplices of the triangulation. $\square$

Hypotheses for G : In this section G is a connected compact Lie group whose Lie algebra $\mathfrak{g}$ is equipped with an Ad-invariant metric. We also use the covering group G' from Notation 6.1. $\square$

We now need to define $g(C; \omega)$ as a random variable in $\omega \in \overline{\mathcal{C}_o^{\Sigma, h}}$, for C an admissible loop in Σ based at o . The pre-image (via q^{-1}) of C in D is a collection of curve segments (whose ends are identified by q). Since stochastic parallel translation along curves in D has been defined (Section 4) what is needed now is a systematic method of tying together end points identified by q . For this, the loops S_i need to be used.

Let S be a closed curve in Σ , based at o , which is of the form $q \circ S'$ for some closed curve S' in D based at O . From the gauge theory (for the gauge group G') on D described in Section 4, the holonomy $g(S')$ is a random variable on $(\Omega_1, \mathcal{F}_1) \times (\Omega_2, \mathcal{F}_2)$, with values in G' . We write $g(S) \in G$ for the projection of $g(S') \in G'$ onto the group G . We view $g(S)$ as a random variable on $\overline{\mathcal{C}_o^{\Sigma, h}} = \Omega_1 \times \Omega_2 \times G^p$ (by composing with the projection on $\Omega_1 \times \Omega_2$). It may be noted here that $g(S')$ is defined on $\Omega_1 \times \Omega_2$ as a product of random variables $g(S'_i)$ where S'_i are segments of S' lying in the left half or in the right half of D . Therefore, by Proposition 5.9, we see that $g(S)$ is indeed well-defined on $(\overline{\mathcal{C}_o^{\Sigma, h}}, \mathcal{F}, \mu_{YM}^{\Sigma, h})$. It is a consequence of Lemma A16 and Proposition 5.9 that the specific choice of S' for a

given S does not affect $(\mu_{\text{YM}}^{\Sigma, h}$ —almost surely) the value of $\underline{g(S)}$.

Now consider any S_i . Define $g(S_i) : \omega \mapsto g(S_i; \omega)$ on $\overline{\mathcal{C}_o^{\Sigma, h}}$ by :

$$g(S_i) : \left(\omega_1, \omega_2, (\alpha_1, \dots, \alpha_p) \right) \mapsto \alpha_i \quad (8.1)$$

We set $g(\overline{S}_i) = g(S_i)^{-1}$.

If a closed curve S in Σ based at o is either of the form $q \circ S'$ as described above or is some S_j or $\overline{S}_j$ then we say that it is *proper*. Thus we have defined $\underline{g(S)}$ on $(\overline{\mathcal{C}_o^{\Sigma, h}}, \mathcal{F}, \mu_{\text{YM}}^{\Sigma, h})$ for proper loops S .

8.2. *Definition.* Let C be an admissible loop in Σ based at o . Then by Lemma A15 there is an admissible curve $\tilde{C}$ in Σ which is the composite of proper loops, such that C can be obtained from $\tilde{C}$ by dropping successively certain bonds which are traversed in opposite directions consecutively by $\tilde{C}$; in particular, $x(C) = x(\tilde{C})$ for every H -field (as in Terminology 7.1) x on the bonds making up $\tilde{C}$. More precisely, $\tilde{C} = \tilde{C}_k \dots \tilde{C}_1$, where each $\tilde{C}_i$ is a proper loop, and $\tilde{C}$ can also be written as a composite

$$\tilde{C} = \overline{H}_{r+1} H_r C_r \overline{H}_r H_r C_{r-1} \dots \overline{H}_1 H_1 C_1$$

where $C_r \dots C_1$ can be further reduced in this way to C . We define :

$$g(C; \omega) = g(\tilde{C}_k; \omega) \dots g(\tilde{C}_1; \omega) \quad (8.2)$$

for almost every $\omega \in (\overline{\mathcal{C}_o^{\Sigma, h}}, \mathcal{F}, \mu_{\text{YM}}^{\Sigma, h})$. Then $g(C; \omega)$ does not depend on the choices made in constructing $\tilde{C}$ (Lemma A16). $\square$

8.3. *Terminology/Convention.* Let $\mathcal{C}$ be an admissible collection of curves in Σ . Then each $C_i \in \mathcal{C}$ can be written as a composite $C_i = C_{i1} \dots C_{i1}$ such that each C_{ab} is either constant or is one-to-one, and two such segments C_{ab} and C_{cd} (corresponding to the same or different curves in $\mathcal{C}$) are (i) either disjoint or (ii) intersect at one or both end points, or (iii) C_{ab} coincides with C_{cd} or its reverse $\overline{C_{cd}}$. To be definite, we work with the smallest set of such non-degenerate segments - i.e. those segments which are naturally defined by the intersections of the curves in $\mathcal{C}$. This fixes the terminology. We denote by $\Gamma_{\mathcal{C}}$ or $\Gamma(\mathcal{C})$ the graph etched out on Σ by the curves in $\mathcal{C}$ - vertices are the endpoints of the curve segments C_{ab} and bonds are the non-constant curves C_{ab} . By $\mathcal{R}(\mathcal{C})$ we mean the (finite) set of all connected components of $\Sigma \setminus |\mathcal{C}|$. $\square$

8.4. *Theorem.* Assume that at least one of the following holds :

- (1) G is semisimple ; or
- (2) $\partial\Sigma \neq \emptyset$; or
- (3) Σ is non-orientable.

Recall Notation 6.1 for G' and notice that in each of the above cases, G' is compact. Equip G' with Haar measure dx' of unit total mass. Let $h \in \ker(G' \rightarrow G)$. If Σ is orientable, choose that orientation which makes the map $q : D \rightarrow \Sigma$ orientation-preserving. Let $\mathcal{C} = \{C_1, \dots, C_n\}$ be an admissible collection of closed curves in Σ all based at a point m , and let om be a curve from o (the fixed base point in Σ) to m such that $C \cup \{om\}$ is admissible. (Recall that an 'admissible' collection of curves in Σ is one which, roughly speaking, cuts up Σ into a finite number of well-behaved surfaces.) Denote by $\Gamma(\mathcal{C})$ the

graph etched out on Σ by the curves in $\mathcal{C}$, and let $\mathcal{R}(\mathcal{C})$ be the set of all components of the complement of the subset $|\mathcal{C}|$ of Σ covered by the curves in $\mathcal{C}$. We denote by $\overline{om}$ the curve obtained by traversing om in reverse; so each $\overline{om} \cdot C_i \cdot om$ is a loop based at o . Then for every bounded (or non-negative) measurable function f on G^n ,

$$\begin{aligned} & \int_{\mathcal{C}_o^{\Sigma, h}} f(g(\overline{om} \cdot C_1 \cdot om; \omega), \dots, g(\overline{om} \cdot C_n \cdot om; \omega)) d\mu_{YM}^{\Sigma, h}(\omega) \\ &= \frac{1}{Z_{\Sigma}^h} \int f(x(C_1), \dots, x(C_n)) Z_{R_*, \Gamma(\mathcal{C}) \cap \partial \overline{R}_*}^h(x' | \Gamma(\mathcal{C}) \cap \partial \overline{R}_*) \cdot \\ & \quad \cdot \prod_{R \neq R_*} Z_{R, \Gamma(\mathcal{C}) \cap \partial \overline{R}}(x' | \Gamma(\mathcal{C}) \cap \partial \overline{R}) dx' \end{aligned} \quad (8.3)$$

(the product in the integrand on the right is over all $R \in \mathcal{R}(\mathcal{C})$ other than R_*). On the left :

- (a) $\mathcal{C}_o^{\Sigma, h}$ is the stochastic/quantum analog of the space of all gauge equivalence classes of connections on a principal G -bundle $P \rightarrow \Sigma$ of class $[P] = h \in \ker(G' \rightarrow G)$, and
 - (b) $g(C; \omega)$ is the (stochastic) holonomy of the 'random connection' ω around the loop C .
- On the right :
- (i) x' denotes G' -fields over $\Gamma(\mathcal{C})$ (i.e assignments $b \mapsto x'_b \in G'$, where b runs over directed bonds of $\Gamma(\mathcal{C})$, satisfying $x'_b = x'_{\bar{b}}^{-1}$) and $x'(C_i)$ is given by the product, in the appropriate order, of x'_b as b runs around the loop C_i (and x is the G -field obtained from x' by the covering projection $G' \rightarrow G$);
 - (ii) $x' | \Gamma(\mathcal{C}) \cap \partial \overline{R}$ denotes the restriction of the G' -field x' to the bonds of $\Gamma(\mathcal{C})$ on $\partial \overline{R}$;
 - (iii) $Z_{R_*, \Gamma(\mathcal{C}) \cap \partial \overline{R}}^h(x' | \Gamma(\mathcal{C}) \cap \partial \overline{R})$ is a number (introduced in Definition 7.6) which depends only on the topology of $(R, \mathcal{C} | \partial \overline{R})$, the area $|R|$, the element $h \in G'$, and the restriction $x' | \Gamma(\mathcal{C}) \cap \partial \overline{R}$;
 - (iv) the integration dx' is $dx'_{b_1} \dots dx'_{b_N}$ where $b_1, \bar{b}_1, \dots, b_N, \bar{b}_N$ are all the bonds of $\Gamma(\mathcal{C})$;
 - (v) R_* is any element of $\mathcal{R}(\mathcal{C})$.

Let ϕ be an area-preserving homeomorphism of Σ which takes the admissible collection $\mathcal{C} = \{C_1, \dots, C_n\}$ to an admissible collection $\phi(\mathcal{C}) = \{\phi C_1, \dots, \phi C_n\}$, and let f be a bounded (or non-negative) measurable function on G^n . If Σ is unorientable or if Σ is orientable and ϕ is orientation preserving then :

$$\begin{aligned} & \int_{\mathcal{C}_o^{\Sigma, h}} f(g(\overline{om} \cdot C_1 \cdot om; \omega), \dots, g(\overline{om} \cdot C_n \cdot om; \omega)) d\mu_{YM}^{\Sigma, h}(\omega) \\ &= \int_{\mathcal{C}_o^{\Sigma, h}} f(g(o\phi(m) \cdot \phi C_1 \cdot o\phi(m); \omega), \dots, g(o\phi(m) \cdot \phi C_n \cdot o\phi(m); \omega)) d\mu_{YM}^{\Sigma, h}(\omega) \end{aligned}$$

where om is as before and $o\phi(m)$ is a curve from o to $\phi(m)$ such that $\phi(\mathcal{C}) \cup \{o\phi(m)\}$ is admissible. If Σ is orientable and ϕ is orientation reversing then on the right hand side of the equation above, h is replaced by h^{-1} .

Remarks. The left side of equation (8.3) involves the Yang-Mills measure which we have constructed by *conditioning* the product of a Gaussian measure and Haar measure to

satisfy the constraints imposed by the topology of Σ and of the bundle P , as described by a special way of cutting up (and pasting) Σ . It also involves the stochastic holonomy variables which were defined by interpreting the equation of parallel transport by a connection as a stochastic differential equation. The right side of (8.3) is defined combinatorially and depends only on the topology of Σ with the system of curves $\mathcal{C}$ and the areas of the regions of Σ cut out by $\mathcal{C}$, and on the topology of the bundle P .

Proof: We pull the collection $\mathcal{C} \cup \{S_1, \dots, S_p\}$ up to D . We draw enough radial segments (including the two along the y -axis) so that we have a grid Γ_1 in D (as in 4.6 and 4.13) consisting of radial lines $M_0, \dots, M_n$ and curve segments running from M_i to M_{i+1} . Adding other segments if necessary we also satisfy 7.8 and thus we assume that Γ_1 gives a triangulation of D . We use the notation concerning grids introduced in Section 4 which we now recall quickly. We label the curves running between M_i and M_{i+1} (with $M_{n+1} = M_0$) by $\sigma_{i1}, \dots, \sigma_{im_i}$, so that $\sigma_{i1} \leq \dots \leq \sigma_{im_i} = 1$. Denote by L_{ij} the loop ('lasso') running from O to the initial point of σ_{ij} , followed by σ_{ij} , followed by the radial segment to the final point of σ_{ij-1} , followed by $\bar{\sigma}_{ij-1}$, followed by the radial segment back to O (convention: σ_{i0} is the constant curve at O). Recall that the lassos can be put in a sequence $L_1, \dots, L_m$ ($L_1 = L_{0m_0}$, $L_2 = L_{0m_0-1}, \dots, L_{m_0} = L_{01}$, $L_{m_0+1} = L_{1m_1}, \dots, L_m = L_{n1}$) such that the composite $L_m \dots L_1$ reduces to $\bar{M}_0 \partial D M_0$ after dropping all segments which are traversed in opposite directions consecutively.

Let $C'_i = \bar{om} \cdot C_i \cdot om$. Then C'_i is a reduction of a composite of proper loops (see the discussion in Definition 8.2), and since each loop in the grid Γ_1 is a reduction of a composite of lassos and reversed lassos and possibly also some loops of the form $\bar{M}KM$ with K a constant curve, it follows that $g(C'_i)$ is a product of $g(L_i)$'s and $g(S_i)$'s and their inverses.

Thus $f(g(C'_1), \dots, g(C'_n))$ is of the form

$$F(g(L_1), \dots, g(L_m), g(S_1), \dots, g(S_p))$$

for some bounded (or non-negative) measurable function F . Then from Definition 6.2 of $\mu_{\text{YM}}^{\Sigma, h}$ we have, upon using Theorem 4.8 and Proposition 5.10, (with, as usual, $y'_i \in G'$ covering $y_i \in G$)

$$\begin{aligned} & \int_{\mathcal{C}_o^{\Sigma, h}} f(g(C'_1; \omega), \dots, g(C'_n; \omega)) d\mu_{\text{YM}}^{\Sigma, h}(\omega) \\ &= \frac{1}{Z_{\Sigma}^h} \int_{G'^m \times G^p} F(y_1, \dots, y_m, \alpha_1, \dots, \alpha_p) \prod_{j=1}^m Q'_{|L_j|}(y'_j) \delta(y'_m \dots y'_1 W'_{\Sigma}(\alpha_1, \dots, \alpha_p)^{-1} h^{-1}) \cdot \\ & \hspace{25em} dy'_1 \dots dy'_m d'\alpha \end{aligned}$$

(see Notation 6.3 for $d'\alpha$). Here the interpretation of $\delta(\cdot)$ is as in Proposition 5.10 (i.e. the term $\delta(\cdot)$ along with dy'_j is dropped from the integral and in the rest of the integrand y'_j is given the value for which $y'_m \dots y'_1 = h W'_{\Sigma}(\alpha_1, \dots, \alpha_p)$.) We are using Haar measure of total mass 1 on both G' and G . This makes sense since our hypotheses concerning G and Σ guarantee that G' is compact. As before, $Q'_t(y')$ is the density at time t , with respect to dy' of standard Brownian motion on G' .

Then by Lemma 8.5,

$$\begin{aligned}
& \int_{\mathcal{C}_o^{\Sigma, h}} f(g(C'_1; \omega), \dots, g(C'_n; \omega)) d\mu_{\text{YM}}^{\Sigma, h}(\omega) \\
&= \frac{1}{Z_\Sigma^h} \int F(x(qL_1), \dots, hx(qL_m), x(S_1), \dots, x(S_p)) Q'_{|qL_m|}(hx'(qL_m)) \cdot \\
&\quad \cdot \prod_{j=1}^{m-1} Q'_{|qL_j|}(x'(qL_j)) dx' \\
&= \frac{1}{Z_\Sigma^h} \int f(x(C'_1), \dots, x(C'_n)) Q'_{|R_*|}(hx'(\partial R_*)) \prod_{R \in \mathcal{R}(\Gamma'), R \neq R_*} Q'_{|R|}(x'(\partial R)) dx'
\end{aligned}$$

where Γ' is the graph in Σ obtained by projecting Γ_1 onto Σ ; i.e., $\Gamma' = q(\Gamma_1)$, x' runs over G' -fields over Γ' , x is the G -field obtained from x' by projecting $G' \rightarrow G$, and R_* corresponds to L_m . In the last equation we have used the defining relationship of F to f to change the $F(\cdot)$ -term in the integrand back to $f(\cdot)$. Recall from 6.3 that

$$Z_\Sigma^h = \int_{G'^p} Q'_{|\Sigma|}(hW_\Sigma(\alpha'_1, \dots, \alpha'_p)) d'\alpha'.$$

Now since the function $Q'_i(x')$ is invariant under the replacement $x' \mapsto g'^{-1}x'g'$ (because the metric on G' is bi-invariant) and since h is in the center of G' , the right side of

$$\begin{aligned}
& \int_{\mathcal{C}_o^{\Sigma, h}} f(g(C'_1; \omega), \dots, g(C'_n; \omega)) d\mu_{\text{YM}}^{\Sigma, h}(\omega) \\
&= \frac{1}{Z_\Sigma^h} \int f(x(C'_1), \dots, x(C'_n)) Q'_{|R_*|}(hx'(\partial R_*)) \prod_{R \in \mathcal{R}(\Gamma'), R \neq R_*} Q'_{|R|}(x'(\partial R)) dx'
\end{aligned} \tag{8.4}$$

is invariant under the substitution $f \mapsto f^g$ for every $g \in G$, where $f^g(x_1, \dots, x_n) = f(g^{-1}x_1g, \dots, g^{-1}x_ng)$.

Averaging over all $g \in G$ we see that in the right side of equation (8.4) we can replace f by $\bar{f}$, where $\bar{f}(x) = \int_G f^g(x) dg$. Then, since $\bar{f}^g = \bar{f}$, for every $g \in G$, on the right side of (8.4) we can replace each C'_i by C_i (since $x(C'_i) = x(om)^{-1}x(C_i)x(om)$). Now, using the same reasoning as before, we can put back f in again in place of $\bar{f}$ to obtain (as usual, x is the G -field induced by the G' -field x' via the covering map $G' \rightarrow G$):

$$\begin{aligned}
& \int_{\mathcal{C}_o^{\Sigma, h}} f(g(C'_1; \omega), \dots, g(C'_n; \omega)) d\mu_{\text{YM}}^{\Sigma, h}(\omega) \\
&= \frac{1}{Z_\Sigma^h} \int f(x(C_1), \dots, x(C_n)) Q'_{|R_*|}(hx'(\partial R_*)) \prod_{R \in \mathcal{R}(\Gamma'), R \neq R_*} Q'_{|R|}(x'(\partial R)) dx'
\end{aligned} \tag{8.5}$$

Now consider any $R \in \mathcal{R}(\mathcal{C})$. The bonds of Γ' which lie in $\overline{R}$ decompose $\overline{R}$ into 2-simplices (projections of 2-simplices in the grid Γ_1 in D). Let $\mathcal{T}$ be the set of all such 2-simplices. Then (see Definition 7.6):

$$\int \prod_{\Delta \in \mathcal{T}} Q'_{|\Delta|}(x'(\partial\Delta)) dx' = Z_{\overline{R}, \Gamma' \cap \partial\overline{R}}(x'|\Gamma' \cap \partial\overline{R})$$

and

$$\int Q'_{|\Delta^*|}(hx'(\partial\Delta^*)) \prod_{\Delta \in \mathcal{T}, \Delta \neq \Delta^*} Q'_{|\Delta|}(x'(\partial\Delta)) dx' = Z_{\overline{R}, \Gamma' \cap \partial\overline{R}}^h(x'|\Gamma' \cap \partial\overline{R})$$

the integration dx' being $dx'_{b_1} \dots dx'_{b_n}$ where $[b_1], \dots, [b_n]$ are all the distinct bonds of Γ' lying in $\overline{R}$ but not on $\partial\overline{R}$. We are writing $\Gamma' \cap \partial\overline{R}$ for the set of bonds of Γ' which lie on $\partial\overline{R}$. Using this observation we can carry out the integration in equation (8.5) over all x'_b for which b lies in some $R \in \mathcal{R}(\mathcal{C})$ (to be precise, b lies in the closure $\overline{R}$ but not on $\partial\overline{R}$) to obtain :

$$\begin{aligned} & \int_{\mathcal{C}^{\Sigma, h}} f(g(C'_1; \omega), \dots, g(C'_n; \omega)) d\mu_{\text{YM}}^{\Sigma, h}(\omega) \\ &= \frac{1}{Z_{\Sigma}^h} \int f(x(C_1), \dots, x(C_n)) Z_{\overline{R}_*, \Gamma'' \cap \partial\overline{R}_*}^h(x'|\Gamma'' \cap \partial\overline{R}_*). \end{aligned} \quad (8.6)$$

$$\prod_{R \in \mathcal{R}(\Gamma''), R \neq R_*} Z_{\overline{R}, \Gamma'' \cap \partial\overline{R}}(x'|\Gamma'' \cap \partial\overline{R}) dx'$$

where Γ'' is the graph obtained by dropping from Γ' all bonds that do not lie on any bond of $\Gamma_{\mathcal{C}}$, and R_* is the element of $\mathcal{R}(\Gamma'')$ containing the R_* of equation (8.5). If $R \in \mathcal{R}(\mathcal{C})$ then the bonds of Γ'' which lie on $\partial\overline{R}$ are exactly the same as the bonds of Γ' which lie on $\partial\overline{R}$. Let $b \in E(\Gamma_{\mathcal{C}})$. Then $b = b_k \dots b_1$ for some $b_i \in E(\Gamma'')$. Define a G' -field x' on $\Gamma_{\mathcal{C}}$ by setting $x'_b = x'_{b_k} \dots x'_{b_1}$. Thus the $x(C_i)$ are products of certain of the x'_b , for b in $\Gamma_{\mathcal{C}}$. Examination of the definition of $Z_{\overline{R}, \Gamma_{\mathcal{C}} \cap \partial\overline{R}}^h$ now shows that the right side of equation (8.6) equals :

$$\begin{aligned} & \frac{1}{Z_{\Sigma}^h} \int f(x(C_1), \dots, x(C_n)) Z_{\overline{R}_*, \Gamma_{\mathcal{C}} \cap \partial\overline{R}_*}^h(x'|\Gamma_{\mathcal{C}} \cap \partial\overline{R}_*) \cdot \\ & \prod_{R \in \mathcal{R}(\Gamma_{\mathcal{C}}), R \neq R_*} Z_{\overline{R}, \Gamma_{\mathcal{C}} \cap \partial\overline{R}}(x'|\Gamma_{\mathcal{C}} \cap \partial\overline{R}) dx' \end{aligned} \quad (8.7)$$

and this proves the main result.

We note that the choice of R_* in expression (8.7) does not affect the value of the integral. This is seen by recalling the definition of $Z_{\overline{R}_*, \Gamma_{\mathcal{C}} \cap \partial\overline{R}_*}^h$ as the integral of a product of terms $Q'_{|\Delta|}(x'(\partial\Delta))$ along with a term $Q'_{|\Delta^*|}(hx'(\partial\Delta^*))$. Choosing any $R \in \mathcal{R}(\mathcal{C})$, $R \neq R_*$, sharing a bond b with R_* , we observe that $Z_{\overline{R}_*, \Gamma_{\mathcal{C}} \cap \partial\overline{R}_*}^h$ contains the term $Q'_{|\Delta|}(hx'(\partial\Delta))$ in its defining integrand while $Z_{\overline{R}, \Gamma_{\mathcal{C}} \cap \partial\overline{R}}$ contains a unique term $Q'_{|\Delta'|}(x'(\partial\Delta'))$ with Δ' also containing the bond b . Since $Q'_{|\Delta|}(g) = Q'_{|\Delta|}(g^{-1})$, for every $g \in G$, we may assume that

$x'(\partial\Delta^*)$ contains the term x'_b while $x'(\partial\Delta')$ contains $x'_b{}^{-1}$. Changing variables $x'_b \mapsto h^{-1}x'_b$ and using the fact that $hx'(C_j) \in G'$ covers $x(C_j) \in G$, since $h \in \ker(G' \rightarrow G)$, we see that in the integrand in (8.7), R_* can be replaced by the adjacent region R since the term $Q'_{|\Delta|}(hx'(\partial\Delta^*))$ becomes $Q'_{|\Delta|}(x'(\partial\Delta^*))$ after change of variables while $Q'_{|\Delta'|}(hx'(\partial\Delta'))$ becomes $Q'_{|\Delta'|}(x'(\partial\Delta'))$. Therefore, since Σ is connected, R_* can be replaced by any $R \in \mathcal{R}(\mathcal{C})$.

We also observe that there is an apparent ambiguity in a term of the type $Q'_t(hx'(\partial\Delta))$, in that it may depend on the choice of orientation of Δ used in defining $\partial\Delta$. To resolve this we note that the term ∂R in (8.4) came from some lasso L_i in the grid. Thus if Σ is orientable we should equip Σ with the orientation inherited from D via $q : D \rightarrow \Sigma$, and use this to define $\partial\Delta$ in defining $Z_{\overline{R}, \Gamma_C \cap \partial\overline{R}}^h$. If Σ is unorientable then the definition of G' implies that h^2 equals the identity in G' and therefore $h = h^{-1}$; this implies that the value of $Q'_t(hx'(\partial\Delta))$ is independent of the choice of orientation for $\partial\Delta$. Here we have used the property $Q'_t(g^{-1}) = Q'_t(g)$, and that $Q'_t(gx'g^{-1}) = Q'_t(x')$. This conjugation invariance along with the fact that h lies in the center of G' ensures us that whether $\partial\Delta$ is written as $b_k \dots b_1$ or some other cyclic permutation, such as $b_2 b_1 b_k \dots b_3$, does not affect the value of $Q'_t(hx'(\partial\Delta))$.

The conclusions concerning the action of ϕ follow from equation (8.3) and Fact 3 of Section 7. $\square$

8.5. Lemma. *Let Γ be a grid (Definition 4.6 and Convention 4.13) on the disk D and let $L_1, \dots, L_m$ be the usual sequence of lassos in Γ . Recall the covering $q : D \rightarrow \Sigma$ which maps arcs on ∂D onto parts of the loops $S_1, \dots, S_p$ which generate $\pi_1(\Sigma, o)$. Assume that G' (which covers G) is compact. Equip both G' and G with Haar measure of total mass 1. Recall that if $(\alpha_1, \dots, \alpha_p) \in G^p$ then $W'_\Sigma(\alpha_1, \dots, \alpha_p)$ means $W_\Sigma(\alpha'_1, \dots, \alpha'_p)$, where each $\alpha'_i \in G'$ covers $\alpha_i \in G$; then W'_Σ is a well-defined continuous function $G^p \rightarrow G'$. Consider elements $y'_1, \dots, y'_m \in G'$ and $\alpha_1, \dots, \alpha_p \in G$, satisfying $y'_m \dots y'_1 = hW'_\Sigma(\alpha_1, \dots, \alpha_p)$ for some h in the center of G' . If Σ is a sphere with $k(\geq 0)$ holes then we take $\alpha_p = e$. Let $q\Gamma$ be the graph obtained by projecting Γ onto Σ . Then there is a G' -field x' over $q\Gamma$ such that:*

- (1) $x(S_i) = \alpha_i$ (x is the G -field corresponding to the G' -field x') for every $i \in \{1, \dots, p\}$, and
- (2) $x'(q(L_i)) = y'_i$ for $i \in \{1, \dots, m-1\}$, and $x'(q(L_m)) = h^{-1}y'_m$.

For every bounded (or non-negative) measurable function f on $G'^m \times G^p$:

$$\begin{aligned} \int_{G'^m \times G^p} f(y'_1, \dots, y'_m, \alpha_1, \dots, \alpha_p) \delta(y'_m \dots y'_1 W'_\Sigma(\alpha_1, \dots, \alpha_p)^{-1} h^{-1}) dy'_1 \dots dy'_m d'\alpha \\ = \int f(x'(qL_1), \dots, x'(qL_{m-1}), hx'(qL_m), x(S_1), \dots, x(S_p)) dx' \end{aligned} \quad (8.8)$$

where, as explained in Notation 6.3, $d'\alpha = d\alpha_1 \dots d\alpha_p$ unless Σ is a sphere with $k(\geq 0)$ holes in which case $d'\alpha = d\alpha_1 \dots d\alpha_{p-1} \delta(e) d\alpha_p$, and the integration is over all G' -fields x' over $q\Gamma$.

Proof: Let the grid Γ be defined by radial lines $M_0, \dots, M_n$ and horizontal curves $\sigma_{i1}, \dots, \sigma_{ik_i}$ running from M_i to M_{i+1} (as usual $M_{n+1} = M_0$), and $0 = \sigma_{i0} \leq \dots \leq \sigma_{ik_i} = 1$ (recall that $\theta \mapsto (\sigma_{ij}, (\theta), \theta)$ describes σ_{ij} in polar coordinates). Recall that $\partial D = K_J \dots K_1$

(we used n instead of J in Section 2) and $K'_j = q(\overline{M}_0)q(K_j)q(M_0)$ is one of the loops $S_1, \dots, S_p$ which generate $\pi_1(\Sigma, o)$ in homotopy. In case Σ is a sphere with $k(\geq 0)$ holes, the loop S_p is of the form $\overline{a}a$.

Choose, for each $j \in \{1, \dots, p\}$, some $\alpha'_j \in G'$ covering $\alpha_j \in G$. We write $\alpha'(K_r) = \alpha'_j$ if $K'_r = S_j$, while if $K'_r = \overline{S}_j$ then we set $\alpha'(K_r) = \alpha'_j{}^{-1}$. Then :

$$\alpha'(K_J) \dots \alpha'(K_1) = W'_\Sigma(\alpha_1, \dots, \alpha_p) \stackrel{\text{def}}{=} W_\Sigma(\alpha'_1, \dots, \alpha'_p).$$

We shall first fix arbitrary values $x'_b \in G'$ for certain bonds b in Γ and then introduce variables $x'_b \in G'$ to replace the y'_i and α'_j . Of course, we always require that $x'_b = x'_{b'}{}^{-1}$.

Now for $[b]$ on the initial radius M_0 , fix $x'_b \in G'$ arbitrarily. Each K_r is made up of a sequence of bonds, say $b_{k_r} \dots b_1$. In case Σ is a sphere with $k(\geq 0)$ holes and $r = J$, then K_J is of the form $b'_1 \dots b'_j b_j \dots b_1$ where q identifies each b_i with $\overline{b'_i}$, and we specify all the x'_b , arbitrarily except for the requirement that $x'_b = x'_{b'}{}^{-1}$ if q identifies b with $\overline{b'}$. In all other cases we specify all *except one* of the x'_{b_j} on K_r arbitrarily except for the requirement that $x'_b = x'_{b'}{}^{-1}$ whenever $q(b) = \overline{q(b')}$. The Convention 4.13 about grids ensures that if $q(b)$ and $q(b')$, as point sets, have any interior points in common then q identifies b with b' or $\overline{b'}$. Notice that in case $\partial\Sigma \neq \emptyset$, and K_r is of the form $c'Bc$, with $q(c') = \overline{q(c)}$, the exceptional bond must lie on B and not on c or c' .

Next, for each $M_i \neq M_0$, fix x'_b arbitrarily for all *except one* of the bonds $[b]$ on M_i . Having specified these x'_b we can now proceed to defining the variables x'_b defined for all other bonds b .

We replace the α'_i by using x'_b where b' runs over all the 'exceptional' bonds on ∂D . We do this by requiring that the new x'_b , for these 'exceptional' b on ∂D , satisfy :

$$x'(M_0)^{-1} x'(K_r) x'(M_0) = \alpha'(K_r) \quad (*)$$

For instance, if $K_r = b_{k_r} \dots b_2 b_1$ and all x'_{b_j} except x'_{b_2} have been specified, then we set:

$$x'_{b_2} = x'_{b_3}{}^{-1} \dots x'_{b_{k_r}}{}^{-1} x'(M_0) \alpha'(K_r) x'(M_0)^{-1} x'_{b_1}{}^{-1}.$$

The definition of $\alpha'(K_r)$ ensures that with these assignments we still have $x'_b = x'_{b'}$ whenever q identifies b with b' .

Notice that if Σ is a sphere with or without holes then $\alpha(K_J) = e$ (see the definition of $\alpha(K_r)$ in an earlier paragraph and the definition $\alpha_p = e$ for such Σ) is consistent with $(*)$ since K_J is of the form $\overline{b_1} \dots \overline{b_j} b_j \dots b_1$ and the left side of $(*)$ is identically equal to e .

Suppose now, inductively, that $l \leq n$ (the case $l = n + 1$ will be dealt with later), that we have specified x'_b for all bonds $[b]$ on or between the radii M_i , for $1 \leq i \leq l$, and that $x'(L_i) = y'_i$ holds for all lassos L_i involving such bonds only. This statement holds trivially if $l = 0$. We proceed now to define x_b for all bonds running between M_l and M_{l+1} , and then we will specify x_{b^*} for the one bond b^* on M_{l+1} for which x_b was not specified.

Recall that $0 = \sigma_{l0} \leq \dots \leq \sigma_{lk_l} = 1$ are the horizontal curves running from M_l to M_{l+1} corresponding to the bonds between these radii. For now, we will write b_j for the bond corresponding to σ_{lj} . Let b^* be as above and let s be such that b^* runs from b'_j to

b_{s+1}^f (recall that b^f denotes the final point of the bond b). Here we are taking b_0^f to be the origin O . We define the variables x_{b_r} for $r \leq s$ and for $r > s$ separately. If v and w are vertices on some M_j we write vw for the shortest chain (i.e. involving the least number of bonds) in Γ from v to w . Recall that the lasso L_{lr} is :

$$L_{lr} = b_{r-1}^i O \cdot \bar{b}_{r-1} \cdot b_r^f b_{r-1}^f \cdot b_r \cdot Ob_r^i,$$

wherein b_r^i is the initial point of b_r . For $1 \leq r \leq s$ we define x'_{b_r} as follows. If $s = 0$ then there is no such r and so we work with $s > 0$. Start by defining x'_{b_1} by requiring that $x'(L_{11}) = y'_{L_{11}}$. Here we are using the notation

$$y'_{L_{ij}} = y'_k \text{ if } L_{ij} = L_k$$

(recall that we have indexed the lassos in two different but natural ways). That is, we set

$$x'_{b_1} = x'(Ob_1^f) y'_{L_{11}} x'(Ob_1^i)^{-1}$$

(here we take $x(vw) = e$ if $v = w$). This is well-defined because $Ob_1^f \neq b^*$ since $s > 0$, and so $x'(Ob_1^f)$ is known. Suppose $r \leq s$ and that we have defined the variables $x'_{b_1}, \dots, x'_{b_{r-1}}$, and that $x'(L_i) = y'_i$ holds whenever $x'(L_i)$ is defined. Then we specify x'_{b_r} by the requirement

$$x'(L_{lr}) = y'_{L_{lr}},$$

that is,

$$x'_{b_r} = x'(b_r^f b_{r-1}^f)^{-1} x'_{b_{r-1}} x'(Ob_{r-1}^f) y'_{L_{lr}} x'(Ob_r^i)^{-1}.$$

Thus we have defined $x'_{b_1}, \dots, x'_{b_s}$. To define $x'_{b_{s+1}}, \dots, x'_{b_{k_l}}$ we start by noticing that $x'_{b_{k_l}}$ is already specified since b_{k_l} lies on ∂D . If $s = k_l - 1$ we are done. Otherwise, let $r \geq s+1$ and suppose $x'_{b_{k_l}}, \dots, x'_{b_{r+1}}$ have been defined and that $x'(L_i) = y'_i$ holds whenever $x'(L_i)$ is defined. Now recall that

$$L_{lr+1} = b_r^i O \cdot \bar{b}_r \cdot b_{r+1}^f b_r^f \cdot b_{r+1} \cdot Ob_{r+1}^i.$$

We define the variable x'_{b_r} by the requirement that

$$x'(L_{lr+1}) = y'_{L_{lr+1}};$$

$$\text{i.e. } x'_{b_r} = x'_{b_r^f b_{r+1}^f}^{-1} \cdot x'_{b_{r+1}} \cdot x'(Ob_{r+1}^i) y'_{L_{lr+1}}^{-1} x'(Ob_r^i)^{-1}.$$

We have now defined all of the $x'_{b_1}, \dots, x'_{b_{k_l}}$. To complete the inductive process we need to define x'_{b_s} . For this, recall that $b^* = b_s^f b_{s+1}^f$ and consider the lasso :

$$L_{ls+1} = \overline{Ob_s^i} \cdot \bar{b}_s \cdot \bar{b}^* \cdot b_{s+1} \cdot Ob_{s+1}^i.$$

Define x_{b^*} by the requirement that

$$x'(L_{l\ s+1}) = y'_{L_{l\ s+1}}$$

which is equivalent to

$$x'_{b^*} = x'_{b_{s+1}} x'(Ob_{s+1}^i) y'_{L_{l\ s+1}}^{-1} x'(Ob_s^i)^{-1} x'_{b_s}^{-1}.$$

Thus we have now defined x'_b for all bonds b except for those between M_n and $M_{n+1} = M_0$. This is the case $l = n + 1$ that we had deferred earlier. The first specialty about this case is that there is no b^* since x'_b is already specified for all bonds b on M_0 . The other new element here is that h is now involved.

For $b = \sigma_{nj}$, with $j \geq 1$, we define x'_b as was done above in the case $r > s$. That is, observe that $x'_{b_{k_n}}$ is specified since b_{k_n} lies on ∂D ; assuming that x'_{b_j} is known for $k_n \geq j > r \geq 1$, define x'_{b_r} by requiring that

$$y'_{L_{n\ r+1}} = x'(L_{n\ r+1}).$$

So we have specified x'_b for all bonds b in the grid Γ and they satisfy $x'(L_{ij}) = y'_{L_{ij}}$ except possibly for the last lasso $L_m (= L_{n1})$.

Since $L_m \dots L_1$ reduces to $\overline{M_0} \partial D M_0$ after successively dropping every bond which is traversed in opposite directions consecutively, it follows that

$$x'(L_m) \dots x'(L_1) = x'(M_0)^{-1} x'(\partial D) x'(M_0).$$

Now $\partial D = K_J \dots K_1$ and so, using (*),

$$\begin{aligned} x'(L_m) \dots x'(L_1) &= \alpha'(K_J) \dots \alpha'(K_1) \\ &= W'_\Sigma(\alpha_1, \dots, \alpha_p) \\ &= h^{-1} y'_m \dots y'_1. \end{aligned}$$

So, since $x'(L_i) = y'_i$ for $1 \leq i < m$, it follows that

$$x'(L_m) = h^{-1} y'_m$$

as claimed.

Recall that we had set $x'_b = x'_{b'}$ if q identifies b with b' . So x' induces a G' -field, which we also denote x' , on the graph $q(\Gamma)$ on Σ , by setting $x'_{q(b)} = x'_b$. Then $x'(qL_i) = x'(L_i)$.

It follows then by (*), the definition $K'_r = \overline{q(M_0)} q(K_r) q(M_0)$, and by the definition of $\alpha(K_r)$, that :

$$x'(S_i) = \alpha'_i \quad \text{for } i = 1, \dots, p, \quad (**)$$

We turn to the second part (relating to integration). Notice that in the left side of equation (8.8), the integrand, in its dependence on the α_i , can be viewed as a function of $(\alpha'_1, \dots, \alpha'_n)$, and $d'\alpha$ replaced by $d'\alpha'$ (with $d\alpha'_i$ being Haar measure of mass 1 on G'). The transformation of variables $(\{y'_i\}, \{\alpha'_i\}) \mapsto \{x'_b\}$, we described above, is effected step

by step using conjugations (and possibly inverses) of previously defined variables only, and therefore, by the bi-invariance of Haar measure on G' , $dy'_1 \dots dy'_m d'\alpha'$ is taken into a product of dx'_b . The term $\delta(y'_m \dots y'_1 W'_\Sigma(\alpha_1, \dots, \alpha_p)^{-1} h^{-1})$ drops out because the way the x'_b have been defined automatically ensures that $y'_m \dots y'_1 = h W'_\Sigma(\alpha_1, \dots, \alpha_p)$ (when the y'_i and the α'_j are expressed in terms of the x'_b). Now integrating over all other x'_b , which appear as arbitrary ‘constants’ in the integrand, and using (**) and $|G'| = 1$ we obtain :

$$\begin{aligned} \int_{G'^m \times G^p} f(y'_1, \dots, y'_m, \alpha_1, \dots, \alpha_p) dy'_1 \dots dy'_m d'\alpha \\ = \int f(x'(L_1), \dots, x'(L_{m-1}), hx'(L_m), x(S_1), \dots, x(S_p)) dx'. \end{aligned}$$

Finally we observe that for each bond $[q(b)]$ of $q(\Gamma)$ there corresponds exactly one variable in the integration $\int \dots dx'$ above. That is, for each bond $q(b)$ of $q(\Gamma)$ we can change notation, replacing the dx'_b integral by a $dx'_{q(b)}$ integral. This gives :

$$\begin{aligned} \int_{G'^m \times G^p} f(y'_1, \dots, y'_m, \alpha_1, \dots, \alpha_p) dy'_1 \dots dy'_m d'\alpha \\ = \int f(x'(qL_1), \dots, x'(qL_{m-1}), hx'(qL_m), x(S_1), \dots, x(S_p)) dx' \end{aligned}$$

wherein x' runs over all G' -fields over $q(\Gamma)$. $\square$

9. Some Tools for the Abelian Case

In this section we prove a number of results, collected together in Proposition 9.4, to be used in Section 10.

9.1. *Hypotheses on G and Σ .* In this section G is a compact connected abelian Lie group, and Σ is oriented and without boundary. $\square$

As usual, we have Σ represented as a quotient of the closed unit disk D via $q : D \rightarrow \Sigma$. Recall the loops $S_1, \dots, S_p$, all based at $*$ = $q(O)$, where O is the center of the disk D . The homotopy classes of $S_1, \dots, S_p$ in $\pi_1(\Sigma, *)$ form a basis of this homotopy group.

9.2. *Remarks.* In this section we shall use the terms ‘chain’, ‘cochain’, ‘cycle’ and ‘boundary’ as in the theory of simplicial complexes (all cochains will be G -valued cochains). This conflicts with our previous graph-theoretic usage of the terms ‘chain’ and ‘cycle’. However, this will not cause any problems because in this section we will use only cochains with values in the abelian group G and, in this setting, ‘ $x(K)$ ’ has the same value whether one views K as a chain in the graph-theoretic sense and x as a G -field or if instead one views K as a 1-chain and x as a G -valued 1-cochain. $\square$

9.3. *Notation.* If S is a simplicial complex, then S_2 will denote the set of all 2-simplices of S , and S_1 will denote the set of all 1-simplices of S . $\square$

9.4. *Proposition.* Let S be a simplicial complex triangulating Σ . Equip S with the orientation from Σ . Assume that S is such that each of the curves $S_1, \dots, S_p$ is composed of 1-simplices of S . Let $C = \{C_1, \dots, C_n\}$ be a collection of closed curves in Σ , each composed of oriented 1-simplices of S . Let $\mathcal{R}(C)$ denote the set of all components of the complement of the set of points covered by the curves C_i .

- (i) Let $\{y_\Delta\}_{\Delta \in S_2}$ be a G -valued 2-cochain on S , with $\prod_{\Delta} y_\Delta = e$ (the product being over positively oriented Δ) and $(\alpha_1, \dots, \alpha_p) \in G^p$, with $\alpha_p = e$ if Σ is a sphere. Then there is a 1-cochain x on S such that $x(\partial\Delta) = y_\Delta$ for every $\Delta \in S_2$, and $x(S_i) = \alpha_i$ for every $i \in \{1, \dots, p\}$.
- (ii) Viewed as a 1-cycle, C_i is a linear combination of boundaries $\partial\Delta$ ($\Delta \in S_2$) and the cycles S_j . Consequently, if x is a 1-cochain then $x(C_i)$ is a product of certain $x(\partial\Delta)$ ’s and certain $x(S_j)^{\pm 1}$, in a manner independent of the choice of x .
- (iii) Let $\{y_\Delta\}_{\Delta \in S_2}$, $(\alpha_1, \dots, \alpha_p)$ and x be as in (i). Let f be a bounded measurable function on G^n . By (ii), $f(x(C_1), \dots, x(C_n))$ is a bounded measurable function of $\{y_\Delta\}_{\Delta \in S_2}$ and $(\alpha_1, \dots, \alpha_p)$. Then

$$\int f(x(C_1), \dots, x(C_n)) d'\alpha$$

depends only on the values $y_R \stackrel{\text{def}}{=} \prod_{|\Delta| \subset \overline{R}} y_\Delta$ (the product is over positively oriented 2-simplices Δ of S contained in $\overline{R}$).

- (iv) For any collection $\{y_R\}_{R \in \mathcal{R}(C)}$, with $\prod_{R \in \mathcal{R}(C)} y_R = e$, and $(\alpha_1, \dots, \alpha_p)$ (the latter as in (i)), there is a 1-cochain x on S for which $x(S_i) = \alpha_i$, for each i , and for which

- $\prod_{|\Delta| \in \bar{\mathcal{R}}} x(\partial\Delta) = y_R$ (the 2-simplices Δ here are positively oriented).
 (v) If $\{y_R\}_{R \in \mathcal{R}(\mathcal{C})}$, $(\alpha_1, \dots, \alpha_p)$ and x are as in (iv), then

$$\int f(x(C_1), \dots, x(C_n)) d'\alpha$$

is a bounded measurable function of the variable $\{y_R\}_{R \in \mathcal{R}(\mathcal{C})}$.

- (vi) Let S' be another triangulation of Σ , and $\phi : \Sigma \rightarrow \Sigma$ a homeomorphism for which $\phi(C_1), \dots, \phi(C_n), \phi(S_1), \dots, \phi(S_p), S_1, \dots, S_p$ are composites of oriented 1-simplices of S' . Let $\{y_R\}_{R \in \mathcal{R}(\mathcal{C})}$, $(\alpha_1, \dots, \alpha_p)$, and x be as in (iv). Denote by $\mathcal{R}(\mathcal{C})'$ the collection of components of the complement in Σ of the curves $\phi(C_i)$; thus each element of $\mathcal{R}(\mathcal{C})'$ is of the form $\phi(R)$ for some $R \in \mathcal{R}(\mathcal{C})$. Let x' be a 1-cochain on S' corresponding (as in (iv)) to $\{y_{R'}\}_{R' \in \mathcal{R}(\mathcal{C})'}$ and $(\alpha_1, \dots, \alpha_p)$, wherein $y'_{\phi(R)} = y_R^{\text{sgn } \phi}$ for each $R \in \mathcal{R}(\mathcal{C})$, where $\text{sgn } \phi = \pm 1$ according as ϕ is orientation preserving or not. Then :

$$\int f(x'(\phi C_1), \dots, x'(\phi C_n)) d'\alpha = \int f(x(C_1), \dots, x(C_n)) d'\alpha.$$

Proof: (i) By Lemma 8.5, this holds in the case S is obtained from a grid as in 7.8. For general S , the *Hauptvermutung* (see [Br]) implies that there is a subdivision which is isomorphic, with the C_i and S_j fixed, to a subdivision of a simplicial complex obtained from a grid. Thus the result holds for a subdivision of S . It follows directly that it holds for S as well.

(ii) Each curve C_i is homotopic to a composite of loops of the types S_j and $\bar{S}_j$. Therefore C_i , as a 1-cycle, is homologous to a linear combination of cycles S_j . Since 'boundaries' are generated by the cycles $\partial\Delta$, the result follows.

(iii) We have noted that there is a bounded measurable function F such that :

$$f(x(C_1), \dots, x(C_n)) = F(\{y_\Delta\}_{\Delta \in \mathcal{S}_2}, \alpha_1, \dots, \alpha_p).$$

Let $\Delta_1, \Delta_2 \in \mathcal{S}_2$ share a common edge $[b^*]$ not on any C_j , with $\partial\Delta_1$ containing b^* and $\partial\Delta_2$ containing $\bar{b}^*$. Let $\eta \in G$. Define a G -field x' by setting $x'_b = x_b$ for all bonds $[b] \neq [b^*]$, and set $x'_{b^*} = x_{b^*}\eta$. Then $x'(\partial\Delta) = x(\partial\Delta)$ unless Δ is Δ_1 or Δ_2 ; $x'(\partial\Delta_1) = x(\partial\Delta_1)\eta$, and $x'(\partial\Delta_2) = x(\partial\Delta_2)\eta^{-1}$. Moreover, $x'(S_i) = \alpha_i\eta^{\pm 1}$ or α_i (depending on whether b^* lies on the non-retraced part of S_i or not). Since $[b^*]$ does not lie on any C_j , $x'(C_j) = x(C_j)$.

So it follows that

$$F(\{y_\Delta\}_{\Delta \in \mathcal{S}_2 \setminus \{\Delta_1, \Delta_2\}}, y_{\Delta_1}\eta, y_{\Delta_2}\eta^{-1}, \alpha_1\eta^{\epsilon_1}, \dots, \alpha_p\eta^{\epsilon_p}) = f(x(C_1), \dots, x(C_n))$$

(wherein ϵ_i is ± 1 or 0 according to the criterion mentioned above).

Therefore, since $\alpha_i \mapsto \alpha_i\eta_i^{\epsilon_i}$ preserves Haar measure,

$$\int f(x(C_1), \dots, x(C_n)) d'\alpha = \int F(\{y_\Delta\}_{\Delta \notin \{\Delta_1, \Delta_2\}}, y_{\Delta_1}\eta, y_{\Delta_2}\eta^{-1}, \alpha_1, \dots, \alpha_i, \dots, \alpha_p) d'\alpha.$$

Consequently,

$$\begin{aligned} \int F(\{y_\Delta\}_{\Delta \in \mathcal{S}_2}, \alpha_1, \dots, \alpha_p) d'\alpha &= \int f(x(C_1), \dots, x(C_n)) d'\alpha \\ &= \int F(\{y_\Delta\}_{\Delta \notin \{\Delta_1, \Delta_2\}}, y_{\Delta_1} \eta, y_{\Delta_2} \eta^{-1}, \alpha_1, \dots, \alpha_p) d'\alpha. \end{aligned}$$

It follows now by Lemma A17 that $\int F(\{y_\Delta\}_{\Delta \in \mathcal{S}_2}, \alpha_1, \dots, \alpha_p) d'\alpha$ depends only on the products $\prod_{|\Delta| \subset \bar{R}} y_\Delta$, as R ranges over $\mathcal{R}(C)$, since (see Lemma A5 of [Se2] for a detailed argument) any two Δ_1, Δ_2 with $|\Delta_1|, |\Delta_2| \subset \bar{R}$ can be connected in the manner required by Lemma A17.

(iv) For $R \in \mathcal{R}(C)$, set $y_\Delta = e$ for all 2-simplices Δ with $|\Delta| \subset \bar{R}$ *except* for one such positively oriented Δ for which set $y_\Delta = y_R$. Then (iv) follows from (i).

(v) Given $\{y_R\}_{R \in \mathcal{R}(C)}$, satisfying $\prod_{R \in \mathcal{R}(C)} y_R = e$, and $(\alpha_1, \dots, \alpha_p)$, define $\{y_\Delta\}_{\Delta \in \mathcal{S}_2}$ as in (iv). Then, with x being a G -field corresponding to the $\{y_\Delta\}$ and $(\alpha_1, \dots, \alpha_p)$, each $x(C_i)$ is a product of certain $y_R^{\pm 1}$ and $\alpha_i^{\pm 1}$. This shows that

$$\{y_R\}_{R \in \mathcal{R}(C)} \mapsto \int f(x(C_1), \dots, x(C_n)) d'\alpha$$

is measurable. It is bounded since f is bounded and G is compact.

(vi) Since C_i is homologous to a linear combination of the cycles S_j , there is a 1-cycle B_i which is a linear combination of boundaries $\partial\Delta$, and there exist integers $k_1, \dots, k_p$ such that :

$$z(C_i) = z(B_i)z(S_1)^{k_1} \dots z(S_p)^{k_p}$$

for every 1-cochain z on $\mathcal{S}$ taking values in any abelian group. Using the *Hauptvermutung* we may and will assume that ϕ is a simplicial isomorphism (this may require replacing $\mathcal{S}$ and $\mathcal{S}'$ by subdivisions). Now ϕ induces an isomorphism of homology groups $H_1(\mathcal{S}) \rightarrow H_1(\mathcal{S}')$ and thereby induces an automorphism $(\alpha_1, \dots, \alpha_p) \mapsto (\alpha'_1, \dots, \alpha'_p)$ (this is $G^p \rightarrow G^p$ except in the case where Σ is a sphere in which case it is $G^{p-1} \times \{e\} \rightarrow G^{p-1} \times \{e\}$), via $H_1(\mathcal{S}) \otimes G \rightarrow H_1(\mathcal{S}') \otimes G$ and via the basis of $H_1(\mathcal{S})$ given by $S_1, \dots, S_p$.

Define a 2-cochain y' on $\mathcal{S}'$ by setting $y'_{\phi(\Delta)} = y_\Delta$. Let x' be a 1-cochain on $\mathcal{S}'$ associated to $\{y'_{\Delta'}\}_{\Delta' \in \mathcal{S}'_2}$ and $(\alpha_1, \dots, \alpha_p)$ by (i). If ϕ is orientation preserving then, with products being always over positively oriented 2-simplices :

$$\prod_{|\Delta| \subset \phi(\bar{R})} x'(\partial\Delta) = \prod_{|\Delta| \subset \bar{R}} y_\Delta = y_R;$$

while if ϕ is orientation reversing then, again with products being over *positively oriented* 2-simplices both in $\mathcal{S}'$ and in $\mathcal{S}$,

$$\prod_{|\Delta| \subset \phi(\bar{R})} x'(\partial\Delta) = \prod_{|\Delta| \subset \bar{R}} x'(\partial\phi\Delta)^{-1} = \prod_{|\Delta| \subset \bar{R}} y_\Delta^{-1} = y_R^{-1}.$$

So x' satisfies the requirements stated in (vi). We have :

$$x'(\phi C_i) = x'(\phi B_i) \alpha_1'^{k_1} \dots \alpha_p'^{k_p}.$$

Since $x'(\phi \partial \Delta) = x'(\partial \phi \Delta) = y'_{\phi \Delta} = y_{\Delta} = x(\partial \Delta)$, we have

$$x'(\phi B_i) = x(B_i).$$

So, with F being defined as in the proof of (iii), and F' the corresponding function associated to S' and $f(x'\phi(C_1), \dots, x'\phi(C_n))$, we have :

$$F'(\{y'_{\Delta}\}, \alpha_1, \dots, \alpha_p) = F(\{y_{\Delta}\}, \alpha'_1, \dots, \alpha'_p)$$

Notice that the right side is $f(\tilde{x}(C_1), \dots, \tilde{x}(C_n))$ where the $\tilde{g}$ -field $\tilde{x}$ corresponds to $\{y_{\Delta}\}$ and $\{\alpha'_1, \dots, \alpha'_p\}$. We integrate with respect to $d\alpha_1 \dots d\alpha_p$ (or, in the case where Σ is a sphere, with respect to $d\alpha_1 \dots d\alpha_{p-1}$), and use the fact that $(\alpha_1, \dots, \alpha_p) \mapsto (\alpha'_1, \dots, \alpha'_p)$ preserves Haar measure since this transformation is an automorphism (and a diffeomorphism). We obtain :

$$\int f(x'\phi(C_1), \dots, x'\phi(C_n)) d'\alpha = \int f(x(C_1), \dots, x(C_n)) d'\alpha.$$

Recall that $d'\alpha = d\alpha_1 \dots d\alpha_p$ except in the case when Σ is a sphere in which case $d'\alpha = d\alpha_1 \dots d\alpha_{p-1}$. $\square$

10. Loop Expectation Values II

In Section 8 we derived the effect of certain transformations $\phi : \Sigma \rightarrow \Sigma$ on loop expectation values under certain hypotheses on Σ and G . Here we cover other cases. In combination, Theorems 8.4 and 10.1 cover all cases in which Σ is a compact connected Riemannian two-manifold, possibly with boundary, and G is a compact connected Lie group.

10.1. Theorem. *Suppose that G is a compact connected Lie group (with Lie algebra equipped with an Ad -invariant metric), $h \in \tilde{G}$ (the universal cover of G), $\partial\Sigma = \emptyset$, and that Σ is orientable. Let $\mathcal{C} = \{C_1, \dots, C_n\}$ be an admissible collection of closed curves in Σ , all based at a point m , and let om be a curve from o (the fixed base point in Σ) to m such that $\mathcal{C} \cup \{om\}$ is admissible. Let $\phi : \Sigma \rightarrow \Sigma$ be an area-preserving homeomorphism. Assume that there is a curve $o\phi(m)$ from o to $\phi(m)$ such that*

$$\mathcal{C}' = \{o\phi(m), \phi \circ C_1, \dots, \phi \circ C_n, \phi \circ S_1, \dots, \phi \circ S_p\} \quad \text{is admissible.}$$

Let f be a bounded measurable function on G^n .

If ϕ is orientation preserving then :

$$\begin{aligned} \int_{\mathcal{C}^{\Sigma, h}} f\left(g(\overline{om} \cdot C_1 \cdot om; \omega), \dots, g(\overline{om} \cdot C_n \cdot om; \omega)\right) d\mu_{YM}^{\Sigma, h}(\omega) \\ = \int_{\mathcal{C}'^{\Sigma, h}} f\left(g(\overline{o\phi(m)} \cdot \phi C_1 \cdot o\phi(m); \omega), \dots, g(\overline{o\phi(m)} \cdot \phi C_n \cdot o\phi(m); \omega)\right) d\mu_{YM}^{\Sigma, h}(\omega). \end{aligned}$$

If ϕ is orientation reversing then :

$$\begin{aligned} \int_{\mathcal{C}^{\Sigma, h}} f\left(g(\overline{om} \cdot C_1 \cdot om; \omega), \dots, g(\overline{om} \cdot C_n \cdot om; \omega)\right) d\mu_{YM}^{\Sigma, h}(\omega) \\ = \int_{\mathcal{C}'^{\Sigma, h^{-1}}} f\left(g(\overline{o\phi(m)} \cdot \phi C_1 \cdot o\phi(m); \omega), \dots, g(\overline{o\phi(m)} \cdot \phi C_n \cdot o\phi(m); \omega)\right) d\mu_{YM}^{\Sigma, h^{-1}}(\omega). \end{aligned}$$

Proof: As in the proof of Theorem 8.4 we can construct a grid Γ_1 in D such that each curve in $\mathcal{C}$ is the composite of a finite number of segments $q(b)$, where $q : D \rightarrow \Sigma$ is the usual projection, and b runs over bonds of Γ_1 . For later convenience, we add a finite number of extra vertices and bonds so that Γ_1 yields a triangulation of Σ as explained in 7.8. Let $L_1, \dots, L_m$ be the sequence of lassos in Γ_1 . Then as in the proof of Theorem 8.4, for any bounded measurable function f on G^n :

$$\begin{aligned} \int_{\mathcal{C}^{\Sigma, h}} f\left(g(\overline{om} \cdot C_1 \cdot om; \omega), \dots, g(\overline{om} \cdot C_n \cdot om; \omega)\right) d\mu_{YM}^{\Sigma, h}(\omega) \\ = \frac{1}{Z_{\Sigma}^h} \int_{\tilde{G}^m \times G^p} F(y_1, \dots, y_M, \alpha_1, \dots, \alpha_p) \prod_{j=1}^M \tilde{Q}_{|L_j|}(\tilde{y}_j) \delta(\tilde{y}_M \dots \tilde{y}_1 W_{\Sigma}(\tilde{\alpha}_1, \dots, \tilde{\alpha}_p)^{-1} h^{-1}) d\tilde{y} d'\alpha \end{aligned}$$

wherein $d\tilde{y}_i$ is some bi-invariant Haar measure on $\tilde{G}$, $\tilde{Q}_t(\tilde{y})$ is the density, with respect to $d\tilde{y}$, at time t of standard Brownian motion on $\tilde{G}$ (governed by the metric $\langle \cdot, \cdot \rangle_{\tilde{g}}$), $d\alpha_i$ is Haar measure on G of unit total mass, and $y_i, \alpha_i \in G$ are covered by $\tilde{y}_i, \tilde{\alpha}_i \in \tilde{G}$, respectively, and F is the function defined as in the proof of Theorem 8.4.

Let $\tilde{K}$ be the connected component of the center of $\tilde{G}$. There is a semisimple subgroup $\tilde{S}$ of $\tilde{G}$ such that $\tilde{G}$ can be identified with $\tilde{S} \times \tilde{K}$, by $\tilde{S} \times \tilde{K} \rightarrow \tilde{G} : (y^{\tilde{S}}, y^{\tilde{K}}) \mapsto y^{\tilde{S}} y^{\tilde{K}}$. We use a Haar measure $d\tilde{y}$ on $\tilde{G}$ which is the product $dy^{\tilde{S}} dy^{\tilde{K}}$, wherein $dy^{\tilde{S}}$ is Haar measure of unit total mass on $\tilde{S}$. The Lie algebras of $\tilde{S}$ and $\tilde{K}$ have inner-products induced by that on $\tilde{g}$. We denote by S and K the projections of $\tilde{S}$ and $\tilde{K}$, respectively, onto G by the covering map $\tilde{G} \rightarrow G$.

The integral on the right in the equation above can be written as :

$$I(F) = \int_{\tilde{K}^M \times \tilde{S}^M \times K^p \times S^p} \square d'\alpha^K d'\alpha^S dy_1^{\tilde{S}} \dots dy_M^{\tilde{S}} dy_1^{\tilde{K}} \dots dy_M^{\tilde{K}}$$

wherein (with self-explanatory notation)

$$\square = F(y_1, \dots, y_M, \alpha_1, \dots, \alpha_p) \prod_{j=1}^M \tilde{Q}_{|L_j|}(\tilde{y}_j) \delta(y_M^{\tilde{S}} \dots y_1^{\tilde{S}} W_{\Sigma}(\alpha_1^{\tilde{S}}, \dots, \alpha_p^{\tilde{S}})^{-1} h_{\tilde{S}}^{-1}) \delta(\tilde{y}_M^{\tilde{K}} \dots \tilde{y}_1^{\tilde{K}} h_{\tilde{K}}^{-1})$$

with $\tilde{y}_i = y_i^K y_i^{\tilde{S}}$, and $h = h_{\tilde{S}} h_{\tilde{K}}$. The integrators $d\alpha^K$ and $d\alpha^S$ are Haar measures of unit mass on K and on S , respectively, and we have used $\int_G \phi(\alpha) d\alpha = \int_{K \times S} \phi(\alpha^K \alpha^S) d\alpha^K d\alpha^S$.

Since $\tilde{K}$ and $\tilde{S}$ are orthogonal with respect to the Ad -invariant metric on $\tilde{G}$,

$$\tilde{Q}_t(x^{\tilde{K}} x^{\tilde{S}}) = Q_t^{\tilde{K}}(x^{\tilde{K}}) \tilde{Q}_t^{\tilde{S}}(x^{\tilde{S}})$$

where $Q_t^{\tilde{K}}$ and $Q_t^{\tilde{S}}$ are the corresponding Brownian densities on $\tilde{K}$ and on $\tilde{S}$, respectively.

We view $I(F)$ as an iterated integral over the $\tilde{S}$ and S variables and over the $\tilde{K}$ and K variables. The invariance (under replacement of each curve C_i by ϕC_i) of the S and $\tilde{S}$ integration follows by Theorem 8.4, since S is semi-simple. This would essentially reduce the problem to the abelian case. We explain in more detail.

By Fubini's theorem we have :

$$I(F) = \int_{\tilde{K}^M \times K^p} I^S(F)(y_1^K, \dots, y_M^K, \alpha_1^K, \dots, \alpha_p^K) \prod_{j=1}^M \tilde{Q}_{|L_j|}^{\tilde{K}}(y_j^{\tilde{K}}) \delta(y_M^{\tilde{K}} \dots y_1^{\tilde{K}} h_{\tilde{K}}^{-1}) dy_1^{\tilde{K}} \dots dy_M^{\tilde{K}} d'\alpha$$

wherein (as usual, with tildes denoting covering elements)

$$\begin{aligned} I^S(F)(y_1^K, \dots, y_M^K, \alpha_1^K, \dots, \alpha_p^K) &= \int_{\tilde{S}^M \times S^p} F(y_1^K y_1^{\tilde{S}}, \dots, y_M^K y_M^{\tilde{S}}, \alpha_1^K \alpha_1^{\tilde{S}}, \dots, \alpha_p^K \alpha_p^{\tilde{S}}) \\ &\quad \cdot \prod_{j=1}^M \tilde{Q}_{|L_j|}^{\tilde{S}}(y_j^{\tilde{S}}) \delta(y_M^{\tilde{S}} \dots y_1^{\tilde{S}} W_{\Sigma}(\alpha_1^{\tilde{S}}, \dots, \alpha_p^{\tilde{S}})^{-1} h_{\tilde{S}}^{-1}) dy_1^{\tilde{S}} \dots dy_M^{\tilde{S}} d'\alpha^S. \end{aligned}$$

Let $\tilde{x} = x^{\tilde{K}} x^{\tilde{S}}$ be associated to $\tilde{y}_1, \dots, \tilde{y}_M$, and $\alpha_1, \dots, \alpha_p$, as the $\tilde{G}$ -field given by Lemma 8.5. Then, writing $C'_i = \overline{\partial m} \cdot C_i \cdot \partial m$, we have :

$$I^S(F)(y_1^K, \dots, y_M^K, \alpha_1^K, \dots, \alpha_p^K) = \int f\left(x^S(C'_1)x^K(C'_1), \dots, x^S(C'_n)x^K(C'_n)\right) \cdot Q_{|L_M|}^{\tilde{S}}\left(h_{\tilde{S}}x^{\tilde{S}}(L_M)\right) \prod_{j=1}^{M-1} Q_{|L_j|}^{\tilde{S}}\left(x^{\tilde{S}}(L_j)\right) dx^{\tilde{S}}$$

where $dx^{\tilde{S}}$ is a product of Haar measure on $\tilde{S}$ of total mass 1. Note that $x^K(C'_i) = x^K(C_i)$ since K is abelian.

Recall from Lemma 8.5 that $x^{\tilde{K}}(L_M) = h_{\tilde{K}}^{-1} y_M^{\tilde{K}}$, and

$$x^{\tilde{K}}(L_i) = y_i^{\tilde{K}} \quad \text{for } 1 \leq i < M$$

and

$$x^K(S_j) = \alpha_j^K \quad \text{for every } j \in \{1, \dots, p\}$$

(as always, $\alpha_p^K = e$ if Σ is a sphere with or without holes).

We can define a bounded measurable function H on K^n by :

$$H(a_1, \dots, a_n) = \int f\left(x^S(C'_1)a_1, \dots, x^S(C'_n)a_n\right) \cdot Z_{\overline{R}_*, \Gamma_C \cap \partial \overline{R}_*}^{h_{\tilde{S}}}(x^{\tilde{S}}|\Gamma_C \cap \partial \overline{R}_*) \prod_{R \in \mathcal{R}(C), R \neq R_*} Z_{\overline{R}, \Gamma_C \cap \partial \overline{R}}(x^{\tilde{S}}|\Gamma_C \cap \partial \overline{R}) dx^{\tilde{S}}$$

where $Z_{\overline{R}, \Gamma_C \cap \partial \overline{R}}$ and $Z_{\overline{R}_*, \Gamma_C \cap \partial \overline{R}_*}^{h_{\tilde{S}}}(x^{\tilde{S}}|\Gamma_C \cap \partial \overline{R}_*)$ are defined with respect to the group $\tilde{S}$.

If $C_1, \dots, C_n$ are replaced by $\phi C_1, \dots, \phi C_n$, where ϕ is an area-preserving homeomorphism, then by Theorem 8.4, the value of $H(a_1, \dots, a_n)$ does not change if ϕ is orientation preserving, while if ϕ is orientation reversing then in $H(\cdot)$ the $h_{\tilde{S}}$ becomes $h_{\tilde{S}}^{-1}$.

Now our original integral $I(F)$ is given by :

$$I(F) = \int H\left(x^K(C_1), \dots, x^K(C_n)\right) \prod_{j=1}^M Q_{|L_j|}^{\tilde{K}}(y_j^{\tilde{K}}) \delta(y_M^{\tilde{K}} \dots y_1^{\tilde{K}} h_{\tilde{K}}^{-1}) dy_1^{\tilde{K}} \dots dy_M^{\tilde{K}} d' \alpha^K$$

since

$$I^S(F)(y_1^K, \dots, y_M^K, \alpha_1^K, \dots, \alpha_p^K) = H\left(x^K(C_1), \dots, x^K(C_n)\right).$$

Recall again that $x^{\tilde{K}}(L_M) = h_{\tilde{K}}^{-1} y_M^{\tilde{K}}$, $x^{\tilde{K}}(L_i) = y_i^{\tilde{K}}$ for $1 \leq i < M$, and $x^K(S_j) = \alpha_j^K$ for $j = 1, \dots, p$. Recall also the way the function H is affected if the collection $\{C_1, \dots, C_n\}$

is replaced by $\{\phi C_1, \dots, \phi C_n\}$. Thus we are reduced to the case where we are dealing with the group K instead of with G . Therefore we may and will assume that G itself is abelian (i.e. $G = K$). This will simplify the notation.

The situation at this stage is this : G is compact, connected, abelian, and $\mathcal{C} = \{C'_1, \dots, C'_n\}$ is admissible ($C'_i = \overline{om} \cdot C_i \cdot om$). For each $R \in \mathcal{R}(\Gamma)$, where $\Gamma = q\Gamma_1$ is the system of curves on Σ obtained by projecting the grid Γ_1 onto Σ , we have $\tilde{y}_R \in \tilde{G}$. We also have $\alpha_1, \dots, \alpha_p \in G$ ($\alpha_p = e$ if Σ is a sphere with or without holes) and $\prod_{R \in \mathcal{R}(\Gamma)} \tilde{y}_R = h$. Since Σ is orientable with $\partial\Sigma = \emptyset$, and G is abelian, $W_\Sigma(\alpha_1, \dots, \alpha_p) = e$.

For a bounded measurable function f on G^n we set

$$I(f) = \int f(x(C_1), \dots, x(C_n)) \prod_{R \in \mathcal{R}(\Gamma)} \tilde{Q}_{|R|}(\tilde{y}_R) \delta\left(\prod_{R \in \mathcal{R}(\Gamma)} \tilde{y}_R h^{-1}\right) \prod_R d\tilde{y}_R d'\alpha$$

where x is a G -field on Γ , obtained from Lemma 8.5, satisfying $x(\partial R) = y_R$ and $x(S_j) = \alpha_j$ for all $R \in \mathcal{R}(\Gamma)$ and $j \in \{1, \dots, p\}$. Here $\tilde{G}$ is equipped with some Haar measure and $\tilde{Q}$ is defined relative to that measure; $d\alpha_j$ is Haar measure of unit total mass on G . Each $x(C_i)$ is a product of certain $y_R^{\pm 1}$ and $\alpha_j^{\pm 1}$, so $I(f)$ makes sense.

Let

$$J(f) = \int f(x(C_1), \dots, x(C_n)) d'\alpha.$$

By Proposition 9.4(iii), $J(f)$ depends only on the $y_R = \prod_{R' \in \mathcal{R}(\Gamma), R' \subset R} y_{R'}$ as R runs over $\mathcal{R}(\mathcal{C})$. Writing $\mathcal{R}(\mathcal{C}) = \{R_1, \dots, R_N\}$, we have by the convolution property of $\tilde{Q}_t$:

$$\begin{aligned} I(f) &= \int f(x(C_1), \dots, x(C_n)) \prod_{R \in \mathcal{R}(\Gamma)} \tilde{Q}_{|R|}(\tilde{y}_R) \delta\left(\prod_{R \in \mathcal{R}(\Gamma)} \tilde{y}_R h^{-1}\right) \prod d\tilde{y}_R d'\alpha \\ &= \int J(f) \left(\prod_{R \in \mathcal{R}(\Gamma), R \subset R_1} y_R, \dots, \prod_{R \in \mathcal{R}(\Gamma), R \subset R_N} y_R \right) \cdot \\ &\quad \cdot \prod_{R \in \mathcal{R}(\Gamma)} \tilde{Q}_{|R|}(\tilde{y}_R) \delta\left(\prod_{R \in \mathcal{R}(\Gamma)} \tilde{y}_R h^{-1}\right) \prod d\tilde{y}_R \\ &= \int J(f)(y_{R_1}, \dots, y_{R_N}) \prod_{R \in \mathcal{R}(\mathcal{C})} \tilde{Q}_{|R|}(\tilde{y}_R) \delta\left(\prod_{R \in \mathcal{R}(\mathcal{C})} \tilde{y}_R h^{-1}\right) d\tilde{y}_{R_1} \dots d\tilde{y}_{R_N} \end{aligned} \quad (*)$$

According to Proposition 9.4 (vi), if ϕ is orientation preserving then there is no change in $J(f)$ if the curves C_i are replaced by the curves ϕC_i . So since ϕ is also area-preserving, it is apparent that this transformation has no effect on the last integral (*).

Assume now that ϕ is orientation reversing. By 9.4 (vi), the effect on $J(f)(y_{R_1}, \dots, y_{R_N})$ is that each y_{R_i} is replaced by $y_{R_i}^{-1}$. Since ϕ is area-preserving, there is no effect on the

terms $\tilde{Q}_{|R|}(\tilde{y}_R)$. We change variables $\tilde{y}_{R_i} \mapsto \tilde{y}_{R_i}^{-1}$ in the last integral $(*)$ (since $\tilde{G}$ is abelian this preserves Haar measure), and use the fact that $\tilde{Q}_i(\tilde{y}^{-1}) = \tilde{Q}_i(\tilde{y})$. Combining all this we see that the effect of the transformation ϕ on $I(f)$ is to replace h by h^{-1} . $\square$

10.2. *Remark.* In view of Theorem 3.16, we can interpret Theorems 10.1 and 8.4 as saying roughly that an area-preserving homeomorphism $\phi : \Sigma \rightarrow \Sigma$ maps the Yang-Mills measure for a principal G -bundle P into the Yang-Mills measure for the pullback bundle ϕ^*P . Conceptually, this is what one may conjecture since the Yang-Mills action S and, formally, the ‘measure’ $[\mathcal{D}\omega]$ are invariant under area preserving diffeomorphisms of Σ . However, given that the actual construction of the Yang-Mills measure involves cutting up the surface Σ into the flat disk, thus making the construction lose topological invariance, it is amazing that in the end the loop expectation values are able ‘see where they came from’ and encode the topology of the bundle and of the surface. $\square$

Appendix

1. The quotient $q : D \rightarrow \Sigma$.

The goal of A1-A5 is to spell out the smoothness properties of the quotient map $q : D \rightarrow \Sigma$. Lemma A5 ensures that certain curves, such as the basic loops S_i , on Σ have reasonable piecewise smooth parametrizations and that thus parallel transport with smooth connections along such curves is meaningful.

A1. Definition of $q : D \rightarrow \Sigma$ (See Figure 1 and Notation 2.2): Let D be the unit disk in the plane R^2 with center at the origin O . Let $R \mapsto \partial D : t \mapsto x_t = (-\sin 2\pi t, \cos 2\pi t)$ be the curve tracing out the boundary ∂D of D anticlockwise, starting at the point x_0 at the 'top' of ∂D . For j any integer, we write $t_j = j/N$. We denote the arc $[t_{j-1}, t_j] \rightarrow \partial D : t \mapsto x_t$ by $\tilde{K}_j$. Thus $\partial D = \tilde{K}_N \dots \tilde{K}_1$ (as the composite of these curves). Let J be a certain set of pairs of integers (i, j) , with $1 \leq i, j \leq N$. Assume the following for J : (a) J contains all the pairs (i, i) for $1 \leq i \leq N$; (b) if $(i, j) \in J$ then $(j, i) \in J$; and (c) for each $i \in \{1, \dots, N\}$ there is at most one $j \neq i$ for which $(i, j) \in J$. For each $(i, j) \in J$, denote by $\phi_{j,i}$ the 'linear' map $[t_{i-1}, t_i] \rightarrow [t_{j-1}, t_j]$ with either: (a) $\phi_{j,i}(t_{i-1}) = t_{j-1}$ and $\phi_{j,i}(t_i) = t_j$, or (b) $\phi_{j,i}(t_{i-1}) = t_j$ and $\phi_{j,i}(t_i) = t_{j-1}$. Define Σ to be the quotient set of D obtained by identifying every pair of points $x_t, x_{t'} \in \partial D$ for which there exists $(i, j) \in J$ with $\phi_{j,i}(x_t) = x_{t'}$. That is, $\Sigma = (D \setminus \partial D) \cup \{(x_t, x_{t'}) : \phi_{j,i}(t) = t' \text{ for some } (i, j) \in J\}$. Equip Σ with the quotient topology, and denote by q the quotient map $q : D \rightarrow \Sigma$. $\square$

It is readily verified that *the space Σ is compact Hausdorff*.

Moreover, Σ is a two-dimensional topological manifold possibly with boundary. To see this, first note that $q : (D \setminus \partial D) \rightarrow \Sigma \setminus q(\partial D)$ is a homeomorphism. Therefore, each point in $\Sigma \setminus q(\partial D)$ has a neighborhood homeomorphic to an open disk in the plane R^2 . Now consider a point $y \in q(\partial D)$. By choosing small enough open disks (intersected with D) centered at the points in $q^{-1}(y)$, we see that y has an open neighborhood U in Σ such that $q^{-1}(U)$ is the union of a finite set of disjoint open disks intersected with D . Since each of these subsets of D is homeomorphic to a 'sector' S of a disk, where S includes both 'radii' but not the circular arc forming part of its boundary, and since q essentially identifies the radii of some pairs of these 'sectors', we see that these 'sectors' paste together, under q , to form a set homeomorphic to either an open disk or a 'sector'-type region again. Thus each $y \in q(\partial D)$ has a neighborhood homeomorphic either to an open disk or to a closed half plane (which is homeomorphic to a sector). Therefore Σ is a two dimensional manifold, possibly with boundary.

A2. Definition (Smooth structure on Σ): We define charts in different ways around points in $q(D \setminus \partial D)$, and around different types of points on $q(\partial D)$.

(i) Let $q|(D \setminus \partial D) : D \setminus \partial D \rightarrow \Sigma \setminus q(\partial D)$ define a chart over $\Sigma \setminus q(\partial D)$.

(ii) Now we define charts around points in the interiors of the arcs $\tilde{K}_i$. If $i \in \{1, \dots, N\}$ is such that there is no $j \neq i$ for which $(i, j) \in J$, then we define a chart around points in the interior of $q(\tilde{K}_i)$ (i.e. on $\{q(x_t) : t_{i-1} < t < t_i\}$) by using q^{-1} . More precisely, for such i we

require that the homeomorphism $(0, 1] \times (t_{i-1}, t_i) \rightarrow q(\{rx_t : (r, t) \in (0, 1] \times (t_{i-1}, t_i)\}) : (r, t) \mapsto q(rx_t)$ be a diffeomorphism. In this case, of course, $q(\tilde{K}_i)$ forms part of the boundary $\partial\Sigma$. Now consider the arc $\tilde{K}_i$ for which there is a $j \in \{1, \dots, N\}$, with $j \neq i$, for which $(i, j) \in J$. For this case we define a chart by pasting together (by ϕ_{ij}) the sector of D marked out by the arc $\tilde{K}_i$ with the one by $\tilde{K}_j$ and form an open disk in the plane. In more detail, we require that the map $\psi : (0, 1) \times (-1, 1) \rightarrow \Sigma$ defined by

$$\psi(t, y) = q((y+1)x_{t_{i-1}+\frac{1}{N}}) \quad \text{if} \quad (t, y) \in (0, 1) \times (-1, 0],$$

and for $(t, y) \in (0, 1) \times [0, 1)$ by

$$\psi(t, y) = \begin{cases} q((1-y)x_{t_{j-1}+\frac{1}{N}}) & \text{if } \phi_{j,i} \text{ is increasing;} \\ q((1-y)x_{t_{j-1}-\frac{1}{N}}) & \text{if } \phi_{j,i} \text{ is decreasing.} \end{cases}$$

It is straightforward to check that ψ is a homeomorphism onto an open subset of Σ . We define a chart by requiring that it be a diffeomorphism.

(iii) Charts around the points $q(x_{t_i})$, with $i \in \{0, \dots, N-1\}$: Let $i_1, \dots, i_m$ be the distinct integers in $\{0, \dots, N-1\}$ for which $q^{-1}(x_{t_i}) = \{x_{t_{i_1}}, \dots, x_{t_{i_m}}\}$. Define, for every $k \in \{1, \dots, m\}$, the map $\psi_k : (-\frac{1}{2}, \frac{1}{2}) \times [0, 1) \rightarrow \Sigma : (t, y) \mapsto q((1-y)x_{t_{i_k}+t/N})$. Viewing $(-\frac{1}{2}, \frac{1}{2}) \times [0, 1)$ as a subset of the complex plane $C = \mathbb{R}^2$, we use the mapping (in complex variable notation) $z \mapsto z^\alpha$, for some $\alpha > 0$ (α is of the form $1/j$ - or possibly $2/j$ in the case Σ has boundary - for some integer j , except in the case where Σ is a sphere with or without holes in which case α could be 2), to map $(-\frac{1}{2}, \frac{1}{2}) \times [0, 1)$ homeomorphically onto a subset of a closed sector (conical region with vertex at the origin) in the complex plane. Pasting together these sectors along their sides as identified via q (by the $\phi_{i,j}$), we obtain a homeomorphism χ of a neighborhood (the neighborhood given by $\cup_{k=1}^m \psi_k((-\frac{1}{2}, \frac{1}{2}) \times [0, 1))$) of $q(x_{t_i})$ onto either an open neighborhood of $(0, 0)$ in $\mathbb{R}^2$ or an open neighborhood of $(0, 0)$ in the closed half-plane $\{(x, y) : x \in \mathbb{R}, y \geq 0\}$. (If one wants Σ to have 'corners' then α may be chosen so that $\cup_{k=1}^m \psi_k((-\frac{1}{2}, \frac{1}{2}) \times [0, 1))$ is a 'conical' region with vertex at $(0, 0)$.) We take χ to be a chart around $q(x_{t_i})$.

It is readily verifiable that these different charts overlap smoothly (i.e. the coordinate transformation mappings are smooth diffeomorphisms).

Thus Σ is a smooth two-dimensional compact manifold, possibly with boundary. $\square$

The mapping q is smooth on the set $D' = D \setminus \{x_{t_0}, \dots, x_{t_{N-1}}\}$. In addition there is the following smoothness property:

A3. Lemma. For every $s \in [0, 1]$ there is a $\delta > 0$ and a continuous mapping $\phi : (s - \delta, s + \delta) \times [0, 1] \rightarrow D$ such that:

- (i) $q \circ \phi : (s - \delta, s + \delta) \times [0, 1] \rightarrow \Sigma$ is smooth;
- (ii) for each fixed $t \in (s - \delta, s + \delta)$, the path $\phi_t : [0, 1] \rightarrow D : r \mapsto \phi(t, r)$ traces out a radial path from O to a point on ∂D , and, for every $r \in [0, 1)$, the tangent vector $\phi'_t(r)$ is non-zero; and
- (iii) the image of $q \circ \phi$ is contained in a contractible open subset of Σ . Moreover, ϕ maps $(s - \delta/2, s + \delta/2) \times (\frac{1}{2}, 1]$ homeomorphically onto a neighborhood of x_s in D .

Proof: If $s \in [0, 1] \setminus \{t_0, \dots, t_{N-1}\}$ we can simply set $\phi(t, r) = rx_t$ and then (i)-(iv) are obvious. Now suppose $s \in \{t_0, \dots, t_{N-1}\}$. We will assume, without loss of generality that

$s = 0$. Let $h : R \rightarrow R$ be the function given by :

$$h(t) = \begin{cases} e^{-\frac{1}{t}}e & \text{if } t > 0; \\ 0 & \text{if } t = 0; \\ -e^{\frac{1}{t}}e & \text{if } t < 0. \end{cases}$$

We take $\delta = \frac{1}{2}$ and define ϕ by :

$$\phi : (-\frac{1}{2}, \frac{1}{2}) \times [0, 1] \rightarrow \Sigma : (t, r) \mapsto (1 - h(1 - r))x_{\frac{1}{N}h(t)}.$$

Since h is smooth, $q \circ \phi$ is smooth on $(-\frac{1}{2}, \frac{1}{2}) \times [0, 1)$. Recall, from our discussion in (A2), the mapping

$$\psi_0 : (-\frac{1}{2}, \frac{1}{2}) \times [0, 1) \rightarrow \Sigma : (t, y) \mapsto q((1 - y)x_{\frac{1}{N}})$$

which was used to define a chart around $q(x_0)$. Thus the map $q \circ \phi$ is smooth if and only if the map

$$(\psi_0^{-1}q\phi)^\alpha : (-\frac{1}{2}, \frac{1}{2}) \times (0, 1] \rightarrow C : (t, r) \mapsto (h(t) + ih(1 - r))^\alpha$$

(where C is the complex plane) is smooth. Since α is of the form j/N for some positive integer N and some integer $j \in \{1, 2, 2N\}$, it will suffice to prove that the function

$$H : \{(x, y) : y \geq 0\} \rightarrow R^2 : (x, y) \mapsto (h(x) + ih(y))^{\frac{1}{N}}$$

is smooth. Lemma A4 below says that H is indeed smooth. This proves (i). Since h is continuous, so is ϕ . To prove (ii), observe that as r increases from 0 to 1, the number $1 - h(1 - r)$ also increases from 0 to 1. Therefore, ϕ_t does trace out a radial path from O to ∂D . The second part of (ii) follows from the simple calculation $\frac{d}{dr}\{1 - h(1 - r)\} = (1 - r)^{-2}e^{-1/(1-r)}e$, valid for $r \in [0, 1)$. The statement (iii) is verified easily from the topology of Σ . Finally, we note that ϕ is one-to-one on $(s - \delta, s + \delta) \times (0, 1]$ and therefore maps compact subsets of this homeomorphically onto their images. Moreover, it is clear that ϕ maps $(s - \delta/2, s + \delta/2) \times (\frac{1}{2}, 1]$, which is contained in the compact set $[s - \delta/2, s + \delta/2] \times [1/2, 1]$, onto an open neighborhood of x_s in D . $\square$

A4. Lemma. *Let h be the function :*

$$h : R \rightarrow R : t \mapsto \begin{cases} e^{-\frac{1}{t}} & \text{if } t > 0; \\ 0 & \text{if } t = 0; \\ -e^{\frac{1}{t}} & \text{if } t < 0. \end{cases}$$

Define, for $y \geq 0$:

$$H(x, y) = \{h(x) + ih(y)\}^{\frac{1}{N}}$$

where N is a positive integer and the N -th root is defined by

$$(re^{i\theta})^{\frac{1}{N}} = r^{\frac{1}{N}}e^{\frac{i\theta}{N}}$$

for $r \geq 0$ and $0 \leq \theta \leq \pi$.

Then H is smooth on the closed upper half-plane $\{(x, y) : x \in R, y \geq 0\}$.

Proof: We have, inductively, that for every integer $j \geq 0$:

$$h^{(j)}(t) = h(t)P_j\left(\frac{1}{t}\right)$$

where P_j is a polynomial, separately on $t > 0$ and on $t < 0$, of degree $2j$.

Let

$$g(x, y) = H(x, y)^N = h(x) + ih(y).$$

For $x \neq 0$ and $y > 0$, we may verify inductively that for every pair of integers $a, b \geq 0$:

$$\begin{aligned} & \left(\frac{\partial}{\partial x}\right)^a \left(\frac{\partial}{\partial y}\right)^b H(x, y) \\ = & H(x, y) \sum_{\{m_j\}_{j=1}^{max(a,b)}, \{n_k\}_{k=1}^{max(a,b)}} C(\cdot, \cdot) g(x, y)^{-\sum_{j=1}^{max(a,b)} (m_j + n_j)} \prod_{j=1}^{max(a,b)} h^{(j)}(x)^{m_j} h^{(j)}(y)^{n_j} \end{aligned}$$

where $\{m_j\}$, $\{n_j\}$ are sequences of non-negative integers, and the $C(\cdot, \cdot)$ are constants depending on the $\{m_j\}$ and $\{n_j\}$. We write $m = \sum_{j=1}^{max(a,b)} m_j$ and $n = \sum_{j=1}^{max(a,b)} n_j$. If $a, b > 0$ then $m, n > 0$. The arguments below apply under the assumption that $a, b > 0$. The cases where a or b is zero are very similar and require minor simplifications of the estimates below.

We have then, for $x \neq 0$ and $y > 0$,

$$\begin{aligned} & |H(x, y)g(x, y)^{-\sum_{j=1}^{max(a,b)} (m_j + n_j)}| \\ = & |H(x, y)|^{\frac{1}{2}} \prod_{j=1}^{max(a,b)} \left(\frac{1}{|g(x, y)|^{1 - \frac{1}{4N^j m}}}\right)^{m_j} \prod_{k=1}^{max(a,b)} \left(\frac{1}{|g(x, y)|^{1 - \frac{1}{4N^k n}}}\right)^{n_k} \\ \leq & |H(x, y)|^{\frac{1}{2}} \prod_{j=1}^{max(a,b)} \left(\frac{1}{|h(x)|^{1 - \frac{1}{4N^j m}}}\right)^{m_j} \prod_{k=1}^{max(a,b)} \left(\frac{1}{|h(y)|^{1 - \frac{1}{4N^k n}}}\right)^{n_k} \\ = & |H(x, y)|^{\frac{1}{2}} |h(x)|^{\frac{1}{4N}} |h(y)|^{\frac{1}{4N}} \prod_{j=1}^{max(a,b)} \frac{1}{|h(x)|^{m_j}} \frac{1}{|h(y)|^{n_j}}. \end{aligned}$$

Combining all of our observations above we have for $x \neq 0$ and $y > 0$ (and $a, b > 0$):

$$\frac{\left|\left(\frac{\partial}{\partial x}\right)^a \left(\frac{\partial}{\partial y}\right)^b H(x, y)\right|}{\sqrt{x^2 + y^2}}$$

$$\begin{aligned}
&\leq C \frac{|H(x, y)|^{\frac{1}{2}}}{\sqrt{x^2 + y^2}} |h(x)|^{\frac{1}{4N}} |h(y)|^{\frac{1}{4N}} \sum_{\{m_j\}, \{n_j\}} \prod_{j=1}^{\max(a, b)} |P_j(\frac{1}{x})|^{m_j} |P_j(\frac{1}{y})|^{n_j} \\
&\leq C' \frac{h(|x| + |y|)^{\frac{1}{4N}}}{|x| + |y|} |h(x)|^{\frac{1}{4N}} P(\frac{1}{|x|}) |h(y)|^{\frac{1}{4N}} Q(\frac{1}{y}) \rightarrow 0 \quad \text{as } (x, y) \rightarrow (0, 0)
\end{aligned}$$

wherein C and C' are constants, P and Q polynomials, and we have used the facts that $h(t)$ is increasing on $t > 0$, and that $(x^2 + y^2)^{1/2} \geq (|x| + |y|)/\sqrt{2}$. Moreover, for the limit, we have used the fact that for any $\delta > 0$ and M ,

$$\lim_{t \rightarrow 0+} \frac{h(t)^\delta}{t^M} = 0$$

which follows by noting that for $t > 0$, $h(t) < kt^k$ for every integer k .

Since H is smooth in $\{(x, y) : x \in R, y \geq 0\} \setminus \{(0, 0)\}$, it follows that (the limit $(x, y) \rightarrow (0, 0)$ being over $\{(x, y) : x \in R, y \geq 0\}$:

$$\lim_{(x, y) \rightarrow (0, 0)} \frac{|\left(\frac{\partial}{\partial x}\right)^a \left(\frac{\partial}{\partial y}\right)^b H(x, y)|}{\sqrt{x^2 + y^2}} = 0.$$

Therefore H has derivatives of all orders at $(0, 0)$ and they are all equal to 0. Moreover, the above limit also implies that

$$\lim_{(x, y) \rightarrow (0, 0)} \left(\frac{\partial}{\partial x}\right)^a \left(\frac{\partial}{\partial y}\right)^b H(x, y) = 0.$$

Therefore, H is smooth at $(0, 0)$. Smoothness of H at other points in the closed upper half-plane is obvious. $\square$

A5. Lemma. *The path $t \mapsto q(x_t)$ can be reparametrized to be smooth (having zero derivatives at the t_i). To be more precise, there is a monotone increasing homeomorphism $\psi : [0, 1] \rightarrow [0, 1]$, which is a diffeomorphism away from the t_i , such that $[0, 1] \rightarrow \Sigma : t \mapsto q(x_{\psi(t)})$ is smooth. It makes sense to speak of parallel-translating by a smooth connection along segments of the curve $t \mapsto q(x_t)$.*

Proof : By Lemma A3. The statement concerning parallel translation follows from Lemma A8. $\square$

2. Parallel Translation.

A6. Lemma. *Suppose ω and ω' are smooth connections on $\pi : P \rightarrow \Sigma$ for which $g_u(C; \omega) = g_u(C; \omega')$ holds for the following classes of loops C in Σ based at $q(O)$ (uis a fixed point on the fiber $\pi^{-1}(q(O))$) : (i) $C \in \{S_1, \dots, S_p\}$, or (ii) C is piecewise smooth and is of the form $q(c(1)O \cdot c \cdot Oc(0))$ for some continuous path $c : [0, 1] \rightarrow D$ for which $c^{-1}(D \setminus D')$ is a finite set (i.e. c hits $D \setminus D'$ finitely often). Recall that D' is D minus the endpoints of the arcs $\bar{K}_j$. As always, $Oc(t)$ and $c(t)O$ denote radial segments in D parametrized so that their projections onto Σ by q are smooth.*

Then there is a smooth bundle automorphism $\psi : P \rightarrow P$, taking each fiber $\pi^{-1}(x)$ onto itself, which fixes each point on the fiber $\pi^{-1}(q(O))$, and for which $\psi^*\omega = \omega'$.

Conversely, the existence of a ψ , with the properties described in the preceding paragraph, implies that $g_u(C; \omega) = g_u(C; \omega')$ for every piecewise smooth loop C in Σ based at $q(O)$.

Proof: The converse statement follows directly from $\tau_{\psi \cdot \omega}(C)u = \psi^{-1}(\tau_\omega(C)u)$, valid for every piecewise smooth path C starting at $q(O)$ (see the argument following equation (A3) below). Recall that $\tau_\omega(C)u$ is the parallel translate of u along C by ω .

Assume now that $g_u(C; \omega) = g_u(C; \omega')$ for all loops C of types (i) and (ii).

Define $\psi : P \rightarrow P$ by setting

$$\psi(y) = \tau_\omega(L)\tau_{\omega'}(\bar{L})y \quad (A1)$$

for any piecewise smooth path L from $*$ to $\pi(y)$ which is the composite of a finite number of curves, each of which is either one of the loops S_i (or $\bar{S}_i$) or is the projection $q \circ K$ of a path K in D which hits ∂D at most finitely often (the existence of such a path is readily verified : for instance, $L = q(O\pi(y))$ where $O\pi(y)$ is the radial segment, suitably parametrized). Assuming for the moment that ψ is well-defined (i.e. independent of the choice of L), it is clear that it maps each fiber of P onto itself G -equivariantly. Thus we need to prove the following three statements : (a) ψ is well-defined in the sense that it is independent of the choice of the paths L ; (b) ψ is smooth; and (c) $\psi^*\omega = \omega'$.

Proof of (a) : Let L_1 and L_2 be two paths, of the type used in equation (A1), from $*$ to y . Then we wish to show that

$$\tau_\omega(L_1)\tau_{\omega'}(\bar{L}_1)y = \tau_\omega(L_2)\tau_{\omega'}(\bar{L}_2)y$$

which is equivalent to

$$\tau_{\omega'}(\bar{L}_1) = \tau_\omega(\bar{L}_1 L_2)\tau_{\omega'}(\bar{L}_2).$$

In view of this, it will suffice to prove that

$$\tau_{\omega'}(\bar{L}_1 L_2) = \tau_\omega(\bar{L}_1 L_2),$$

and this is equivalent to

$$g_u(\bar{L}_1 L_2; \omega') = g_u(\bar{L}_1 L_2; \omega) \quad (A2)$$

Equation (A2) is a consequence of the hypothesis concerning equality of ω -holonomies and ω' -holonomies. To see this, first observe that by the properties of L_1 and L_2 described (for L) immediately after equation (1), the loop $\bar{L}_1 L_2$ is the composite of a finite number of curves each of which is either one of the S_i (or $\bar{S}_i$) or of the form $q \circ K$ where K is a path in D which hits $D \setminus D'$ at most finitely often. It follows that $g_u(\bar{L}_1 L_2; \omega)$ is a finite product of terms of the form $g_u(S_i; \omega)$, $g_u(S_i; \omega)^{-1}$, and $g_u(C; \omega)$ where C is a loop of type (ii) in the statement of this Lemma (for instance we can draw radii from O to the points where K intersects ∂D). Putting all this together with the given hypotheses of this Lemma, we see that equation (A2), and hence statement (a), is proved.

Proof of (b) : The smoothness of ψ follows by using : (i) local trivialization; (ii) the fact that around every $x \in \Sigma$ there is a chart $\xi : U \rightarrow U'$, where U is an open neighborhood of x in Σ and U' is the planar open unit disk (or that part of it within a closed half-plane) centered at $\xi(x)$, such that all the ‘radial lines’ in U' , emanating from its center $\xi(x)$, correspond to curves of the type used in equation (A1); and (iii) the (smooth version of the) argument from [KN] cited in the proof of Lemma A8 below in establishing continuity of Θ_ω .

Proof of (c) : Finally, to show that $\psi^*\omega = \omega'$, it will suffice to show that for any piecewise smooth path L starting at $*$, of the type used in equation (A1), the parallel-translates $\tau_{\psi^*\omega}(L)u$ and $\tau_{\omega'}(L)u$ are equal. To see why this is sufficient, note first that if $y \in P$ and $Y \in T_yP$ then, by using a suitable chart in Σ around $\pi(y)$, it is possible to choose a smooth curve $\tilde{C} : [0, 1] \rightarrow P$ whose final tangent vector $\tilde{C}'(1)$ is Y and whose projection $\pi \circ \tilde{C}$ onto Σ is a path of the type $q \circ K$, starting at $*$, used in equation (A1). Then, from the defining properties of connections and parallel translation, we have $\omega(Y) = -\frac{dg_t}{dt}|_{t=1}g_1^{-1}$, where ω is any smooth connection on P and $g_t \in G$ is defined by requiring that $\tau_\omega(\pi \circ \tilde{C}|[0, t])u = \tilde{C}(t)g_t$ for every $t \in [0, 1]$. This clearly implies our claim that it will suffice to prove that $\tau_{\psi^*\omega}(L)u$ and $\tau_{\omega'}(L)u$ are equal for every piecewise smooth path L starting at $*$, of the type $q \circ K$ described after equation (A1). Now, for such L , equation (A1) implies :

$$\psi^{-1}(\tau_\omega(L)u) = \tau_{\omega'}(L)u \quad (\text{A3})$$

But the left side of equation (A3) equals $\tau_{\psi^*\omega}(L)u$ because the curve $[0, 1] \rightarrow P : t \mapsto \psi^{-1}(\tau_\omega(L|[0, t])u)$ begins at u (since we have assumed that $\psi(u) = u$), projects under π to the curve L in Σ , and is horizontal with respect to the connection $\psi^*\omega$ because

$$(\psi^*\omega)\left(\frac{d}{dt}\psi^{-1}(\tau_\omega(L|[0, t])u)\right) = \frac{d}{dt}\omega\left(\tau_\omega(L|[0, t])u\right) = 0.$$

This observation along with equation (A3) implies that $\tau_{\psi^*\omega}(L)u$ and $\tau_{\omega'}(L)u$ are indeed equal, thereby completeing the proof of (c). $\square$

A7. Lemma. Let $c : [0, 1] \rightarrow D : t \mapsto c_t$ be a continuous map such that :

- (1) $q \circ c : [0, 1] \rightarrow \Sigma$ is piecewise smooth;
- (2) $c^{-1}(D \setminus D')$ is finite set (i.e. c hits the endpoints of the arcs $\tilde{K}_i$ at most finitely often).

For ω a smooth connection on $\pi : P \rightarrow \Sigma$, and fixed $u \in \pi^{-1}(q(O))$, define $g_t \in G$ by:

$$g_t = g_u\left(q(c_t O \cdot c|[0, t] \cdot Oc_0); \omega\right)$$

wherein, as usual, $c_t O$ is the radial segment from c_t to O (suitably parametrized), and Oc_0 is the radial segment from O to c_0 .

Then :

- (a) $[0, 1] \rightarrow G : t \mapsto g_t$ is continuous;
- (b) if ω' is another smooth connection on $\pi : P \rightarrow \Sigma$, such that

$$f_\omega = f_{\omega'}$$

(see Definition 3.7 for f_ω) then

$$g_t = g'_t$$

for every $t \in [0, 1]$, where g'_t is defined for ω' as g_t is for ω .

Proof: Recall the map $\Theta_\omega : D \rightarrow P : x \mapsto \tau_\omega(q(Ox))u$. By Lemma A8 below, Θ_ω is continuous. The definition of g_t easily implies that :

$$\Theta_\omega(c_t)g_t = \tau_\omega(q \circ c|[0, t])\Theta_\omega(c_0) \quad (*)$$

This equation shows that $[0, 1] \rightarrow P : t \mapsto \Theta_\omega(c_t)g_t$ is continuous. Combining this with the continuity of $[0, 1] \rightarrow P : t \mapsto \Theta_\omega(c_t)$, and using local trivality of P we see that $[0, 1] \rightarrow G : t \mapsto g_t$ is continuous. This proves (a).

To prove (b) we use the equation (to be justified below) :

$$\frac{dg_t}{dt}g_t^{-1} + A^\omega(c'(t)) = 0 \quad (A4)$$

valid for every $t \in c^{-1}(D')$, wherein A^ω is the g -valued 1-form on D' given by $\Theta_\omega^*\omega$. Now it is readily verified that the 1-form A^ω is the unique smooth g -valued 1-form on D' which vanishes on radial vectors and for which $dA^\omega = f_\omega d\sigma'$ (see Notation 3.6 for $d\sigma'$). This and equation (A4) then imply that if ω' is a smooth connection on P with $f_\omega = f_{\omega'}$ on D' then

$$\frac{dg_t}{dt}g_t^{-1} = \frac{dg'_t}{dt}g'^{-1}_t$$

for every $t \in c^{-1}(D')$.

Hence

$$\frac{d}{dt}(g_t^{-1}g'_t) = -g_t^{-1}\frac{dg_t}{dt}g_t^{-1}g'_t + g_t^{-1}\frac{dg'_t}{dt} = -g_t^{-1}\frac{dg_t}{dt} + g_t^{-1}\frac{dg'_t}{dt} = 0.$$

Therefore $g_t g'^{-1}_t$ is constant on each component of $c^{-1}(D')$. Since $[0, 1] \rightarrow G : t \mapsto g_t g'^{-1}_t$ is continuous and $[0, 1] \setminus c^{-1}(D')$ is, by hypothesis, a finite set, it follows that $g_t g'^{-1}_t$ is constant on all of $[0, 1]$. Since $g_0 = g'_0 = e$, we can now conclude that, for every $t \in [0, 1]$,

$$g_t = g'_t.$$

It remains only to verify equation (A4).

We have from (*), in view of the definition of parallel translation,

$$\omega\left(\frac{d}{dt}\{\Theta_\omega(c_t)g_t\}\right) = 0$$

where, since $q \circ c$ is piecewise smooth, $\frac{d}{dt}$ is taken as a right-hand/left-hand derivative at the finite number of points where $q \circ c$ is not differentiable.

So, from the defining properties of a connection, the above equation is equivalent to :

$$g_t^{-1}\omega\left((\Theta_\omega \circ c)'(t)\right)g_t + g_t^{-1}\frac{dg_t}{dt} = 0$$

(where we have used the notation that for every $g \in G$ and $X \in \underline{g} = T_e G$, we write $gX = \frac{d}{dt}|_{t=0} g \exp(tX) \in T_g G$ and analogously for Xg .)

Therefore, for $t \in c^{-1}(D')$ (so that then c' exists and is continuous in a neighborhood of t in $[0, 1]$):

$$A^\omega(c'(t))g_t + \frac{dg_t}{dt} = 0$$

where $A^\omega = \Theta_\omega^* \omega$. This proves equation (A4). $\square$

A8. Lemma. *If ω is a smooth connection on P then the map $\Theta_\omega : D \rightarrow P : x \mapsto \tau_\omega(q(Ox))u$ is continuous, wherein Ox is the radial segment parametrized so that $q(Ox)$ is smooth and u is a fixed point on the fiber $\pi^{-1}(q(O))$.*

Proof: Continuity on D' follows from the fact the $\tau_\omega(q(Ox))$ is obtained by solving a differential equation (defining parallel transport) having smooth coefficients (a detailed argument may be constructed from the Lemma in Section 3 of Chapter II in [KN]).

The continuity of Θ_ω at points in $D \setminus D'$ is seen by choosing coordinates around such points and transforming the problem to a form in which the argument of the preceding paragraph is applicable. The details are as follows.

Consider a point $x_s \in D'$, where we are using the notation $x_t = (-\sin 2\pi t, \cos 2\pi t)$, for any real number t , in usual Cartesian coordinates in R^2 . Then, according to Lemma A3, for some $\delta > 0$, there is a continuous mapping $\phi : [0, 1] \times (s - \delta, s + \delta) \rightarrow D$ such that : (i) $q \circ \phi : [0, 1] \times (s - \delta, s + \delta) \rightarrow \Sigma$ is smooth; (ii) for each fixed $t \in (s - \delta, s + \delta)$, the path $\phi_t : [0, 1] \rightarrow D : r \mapsto \phi(r, t)$ traces out the radius from O to x_t , and, for every $r \in [0, 1]$, the tangent vector $\phi'_t(r)$ is non-zero and points in the direction Ox_t ; (iii) the image of $q \circ \phi$ is contained in a contractible open subset of Σ ; and (iv) ϕ maps $(\frac{1}{2}, 1] \times (s - \frac{\delta}{2}, s + \frac{\delta}{2})$ homeomorphically onto its an open (in D) neighborhood of x_s . Thus, by (ii), for any $(r, t) \in [0, 1] \times (s - \delta, s + \delta)$, the point $\Theta_\omega(\phi(r, t))$ is obtained by parallel translating $u \in \pi^{-1}(*)$ along the path $q \circ \phi_t|_{[0, r]}$. Let ξ be a smooth section of P over a neighborhood of the image of $q \circ \phi$ (existence guaranteed by (iii)), with $\xi q(0, 0) = u$, and denote by A the $\underline{g}$ -valued 1-form on $[0, 1] \times (s - \delta, s + \delta)$ given by $A = (\xi \circ q \circ \phi)^* \omega$. Since ω , ξ and (by (i)) $q \circ \phi$ are smooth, so is A . Define $\gamma : [0, 1] \times (s - \delta, s + \delta) \rightarrow G$ by requiring that $\Theta_\omega(\phi(r, t)) = \xi q \phi(r, t) \gamma(r, t)$. Since $\Theta_\omega \phi(r, t) = \tau_\omega(q \phi_t|_{[0, r]})u$, the latter has a continuous derivative with respect to r , and ω vanishes on this derivative. From this and the fact that $\gamma(0, t) = e$ for every $t \in (s - \delta, s + \delta)$, it follows that for $(r, t) \in [0, 1] \times (s - \delta, s + \delta)$ (the left side exists and) :

$$\frac{\partial \gamma(r, t)}{\partial r} \gamma(r, t)^{-1} = -A|_{(r, t)} \frac{\partial}{\partial r}.$$

Since A is smooth it follows (see the [KN] reference cited above) that γ is smooth. From the definition of $\gamma(r, t)$ it now follows immediately that $\Theta_\omega \circ \phi$ is smooth and, in particular, continuous. Since ϕ^{-1} is continuous in a neighborhood of x_s we conclude that Θ_ω is continuous in a neighborhood of x_s . $\square$

The following result guarantees that while dealing with parallel translation along paths we need not worry too much about parametrizations of the paths. Our notation (involving $\tau_\omega(C)$) in many places has relied on this observation.

A9. Lemma. *Let $c_1 : [a_1, b_1] \rightarrow \Sigma$ and $c_2 : [a_2, b_2] \rightarrow \Sigma$, where the $[a_i, b_i] \subset R$, be piecewise smooth (C^1) paths and assume that : (i) $\{t \in [a_1, b_1] : c'_1(t) = 0\}$ is the union*

of a finite number of closed intervals, and (ii) there is a strictly increasing continuous map $\phi : [a_2, b_2] \rightarrow [a_1, b_1]$, with $\phi(a_2) = a_1$ and $\phi(b_2) = b_1$, for which $c_2 = c_1 \circ \phi$. Let ω be a smooth connection on P . Then $\tau_\omega(c_1) = \tau_\omega(c_2)$ (these are the parallel translation maps) as maps $\pi^{-1}(c_1(a_1)) \rightarrow \pi^{-1}(c_1(b_1))$.

Proof: First we notice that we may assume without loss of generality that $[a_1, b_1] = [a_2, b_2] = [a, b]$, and that both c_1 and c_2 are smooth. It will suffice to consider two cases : (1) c_1 is constant or (2) c'_1 is non-zero on (a, b) .

In the first case, the result is obvious since the hypothesis (ii) then implies that c_2 is also constant. So we turn to the case (2); i.e. we assume that c'_1 is non-zero on (a, b) . By partitioning (a, b) into two subintervals (and also considering reversed paths) if necessary, we may and will assume that c'_1 is non-zero on $[a, b]$. Then, for any $t \in [a, b]$, by choosing, as we may, a smooth real-valued function f in a neighborhood of $c_1(t)$, with $(f \circ c_1)'(t) \neq 0$, we see that the hypotheses (i) and (ii) imply that the derivative $\phi'(t)$ exists and equals $(f \circ c_2)'(t)/(f \circ c_1)'(t)$. This shows that ϕ' exists and is continuous on $[a, b]$. In this case a routine change of variable argument shows that $\tau_\omega(c_1|[a, t]) = \tau_\omega(c_2|[a, t])$ for every $t \in [a, b]$. By letting $t \rightarrow b$, and using the fact that, for every y on the fiber over $c_1(a)$, the parallel transport map $\tau_\omega(c_i|[a, t])y$ is continuous on $[a, b]$ for both $i = 1$ and $i = 2$, we conclude that $\tau_\omega(c_1) = \tau_\omega(c_2)$. $\square$

A10. Lemma. Suppose ω is a smooth connection on a principal G -bundle $\pi : P \rightarrow \Sigma$. With notation as above (for instance, the projection $q : D \rightarrow \Sigma$, and $o = q(O)$), fix a point $u \in \pi^{-1}(o)$. Define the points $s_0 = 0 < s_1 < \dots < s_n = 1$ by requiring that, for every $j \in \{1, \dots, n\}$, the arc K_j (Notation 2.2) be described by $[s_{j-1}, s_j] \rightarrow \partial D : s \mapsto x_s = (-\sin 2\pi s, \cos 2\pi s)$. Let $t \in [s_{i-1}, s_i]$, and $t' \in [s_{i'-1}, s_{i'}]$, be such that q identifies the arc $x_{s_{i-1}}x_t$ with the arc $x_{s_{i'-1}}x_{t'}$ where $j \in \{i'-1, i'\}$ (i.e the arc $[s_{i-1}, t] \rightarrow \partial D : u \mapsto x_u$ with the arc $x_{s_j}x_{t'}$ (note that in case $j = i'$ this is the reverse of the arc $[t', s_{i'}] \rightarrow \partial D : v \mapsto x_v$). Then :

$$g_u(q(x_{t'}O)q(Ox_t); \omega) = \gamma(t')\{g_u(K'_j; \omega) \dots g_u(K'_1; \omega)\}^{-1}\{g_u(K'_{i-1}; \omega) \dots g_u(K'_1; \omega)\}\gamma(t)^{-1} \quad (A5)$$

wherein $\gamma : [0, 1] \rightarrow G : s \mapsto \gamma_s$ is the continuous path, with initial condition $\gamma_0 = e$, satisfying the differential equation :

$$\frac{d\gamma_s}{ds} \gamma_s^{-1} = -\frac{d}{ds} \int_{D_s} f_\omega d\sigma' \quad (A6)$$

when $s \in [0, 1] \setminus \{t_0, \dots, t_{N-1}\}$, and we have used the notation $D_s = \{rx_u : (r, u) \in [0, 1] \times [0, s]\}$, and $d\sigma'$ is as in Notation 3.6. Recall that if Σ is a sphere then $n = 1$ and $g_u(K'_1; \omega) = e$, and therefore equation (A5) has $\gamma(t')\gamma(t)^{-1}$ on the right side.

If $\tau_\omega(c) : \pi^{-1}(c_0) \rightarrow \pi^{-1}(c_1)$ denotes the parallel transport map, then :

$$\Theta_\omega(x') = \tau_\omega(q(Ox'))u = \tau_\omega(q(Ox))ug_u(q(xO).q(Ox'); \omega) = \Theta_\omega(x)g_u(q(xO)q(Ox'); \omega).$$

Proof: This is essentially Lemma 3.8. The only new observation is equation (A6). This equation follows by using Stokes' theorem on (A4) and the fact, noted after (A4), that A^ω vanishes on radial vectors and satisfies $dA^\omega = f_\omega d\sigma'$. $\square$

A11. *Notation.* $h(x_{i'}, x_t)$. Define $s_0 = 0 < s_1 < \dots < s_n = 1$ by requiring that, for every $j \in \{1, \dots, n\}$, the arc K_j (see Notation 2.2) be described by $[s_{j-1}, s_j] \rightarrow \partial D : s \mapsto x_s$. Let $t \in [s_{i-1}, s_i]$, and $t' \in [s_{i'-1}, s_{i'}]$ be such that q identifies the arc $x_{s_{i-1}}x_t$ with the arc $x_{s_{i'-1}}x_{t'}$ where $j \in \{i' - 1, i'\}$. If $f : D \rightarrow \underline{g}$ is a continuous function and $\alpha_1, \dots, \alpha_p \in G$ then we define $h(x_{i'}, x_t)$ for points $x_{i'}, x_t \in \partial D$ with $q(x_t) = q(x_{i'})$ by :

$$h(x_{i'}, x_t) = \gamma(t') \{g(K'_j) \dots g(K'_1)\}^{-1} g(K'_{i-1}) \dots g(K'_1) \gamma(t)^{-1}$$

where $g(K'_i) = \alpha_{i'}$ or $g(K'_i) = \alpha_{i'}^{-1}$ according as whether $K'_i = S_{i'}$ or $K'_i = \bar{S}_{i'}$ (recall that $K'_i = q(x_0 O) q(K_i) q(O x_0)$), and $\gamma : [0, 1] \rightarrow G$ is the continuous map with $\gamma(0) = e$ and satisfying

$$\frac{d\gamma_s}{ds} \gamma_s^{-1} = -\frac{d}{ds} \int_{D_s} f d\sigma' \quad (A7)$$

with $D_s = \{rx_t : r \in [0, 1], t \in [0, s]\}$ (assuming that a unique such γ exists). The value $h(x_{i'}, x_t)$ is independent of the choices of i and i' (such a choice might exist in case t itself is some s_j), as may be verified without difficulty. $\square$

A12. *Notation.* For $i \in \{1, \dots, N\}$, let $V_i = \{q(rx_t) : t \in (t_{i-1}, t_i), r \in (0, 1)\}$. If $\tilde{K}_i$ is part of the boundary $\partial \Sigma$ then V_i is an open subset of Σ . If, instead, q identifies $\tilde{K}_i$ with $\tilde{K}_{i'}$ or with $\bar{\tilde{K}}_{i'}$, then $V_{i,i'} = V_i \cup V_{i'}$ is open in Σ . Recall that $D' = D \setminus \{x_{t_0}, \dots, x_{t_{N-1}}\}$. $\square$

A13. *Remarks for Proposition A 14.* A principal bundle G -bundle P over the surface Σ and a connection ω on P can be specified by : (i) a covering of Σ with open sets U_i ; (ii) a $\underline{g}$ -valued 1-form A_i on each U_i ; (iii) if U_i intersects U_j , then a 'transition function' $\phi_{ij} : \bar{U}_i \cap U_j \rightarrow G$, such that $\phi_{ij}\phi_{jk} = \phi_{ik}$ whenever defined; (iv) A_i and A_j are related via ϕ_{ij} . The forms A_i are obtained as $s_i^* \omega$, for some sections s_i over the U_i . If these sections are chosen to be in radial gauge for ω , i.e. if they are 'parallel' along 'radial lines', then A_i can be recovered from the function $f_i : U_i \rightarrow \underline{g}$ given by $dA_i = f_i d\sigma$, where $d\sigma$ is some non-zero 2-form. A third condition can be added to the statement of Proposition A14, which would make a converse possible; this has been alluded to in 3.11; the three conditions, together with the covering of Σ described in A2, would yield transition functions (relative to radial gauge) and the f_i , thereby leading to a bundle along with the appropriate connection. $\square$

A14. *Proposition.* Let ω be a smooth connection on a principal G -bundle $\pi : P \rightarrow \Sigma$, where G is a connected Lie group. Let $\alpha_i = g_u(S_i; \omega)$, for $i = 1, \dots, p$, and let $f = f_\omega : D \rightarrow \underline{g}$, and let $\gamma : [0, 1] \rightarrow G$ be defined as in Lemma A10. If $x, x' \in \partial D$, with $q(x) = q(x')$ then we use the notation $h(x, x') \in G$ explained in Notation A 11.

Then for the function $f : D \rightarrow \underline{g}$ and the elements $\alpha_1, \dots, \alpha_p \in G$ the following Smoothness-Compatibility (SC) conditions hold :

SC1. If $x, x' \in \partial D$ satisfy $q(x) = q(x')$, then

$$f(x') = \pm h(x', x) f(x) h(x', x)^{-1};$$

the ' $\pm$ ' sign is '+' if a neighborhood of $q(x)$ in Σ can be equipped with an orientation such that q is orientation-preserving both in a neighborhood of x and in a neighborhood

of x' ; the sign is $'-'$ if this does not hold. (Orientation and orientation-preserving should be interpreted in the topological sense at points where q is not differentiable.)

SC2. As before, let $t_0 = 0 < t_1 < \dots < t_N = 1$ define the endpoints of the arcs $\tilde{K}_i$ (i.e. $[t_{i-1}, t_i] \rightarrow \partial D : t \mapsto x_t$ describes the arc $\tilde{K}_i$); suppose q identifies the arc $\tilde{K}_i$ with $\tilde{K}_{i'}$ (or with $\overline{\tilde{K}_{i'}}$). Define the g -valued function $\bar{f}$ on the open set $V_{i,i'} = \{q(rx_t) : r \in (0, 1], t \in (t_{i-1}, t_i) \cup (t_{i'-1}, t_{i'})\}$, by :

$$\bar{f}(q(rx_t)) \stackrel{\text{def}}{=} f(rx_t) \quad (\text{A9})$$

$$\bar{f}(q(rx_{t'})) \stackrel{\text{def}}{=} \pm h(x_{t'}, x_t) f(rx_t) h(x_{t'}, x_t)^{-1} \quad (\text{A10})$$

where $r \in (0, 1]$ and $t \in (t_{i-1}, t_i)$, $t' \in (t_{i'-1}, t_{i'})$ are such that $q(x_t) = q(x_{t'})$, and the sign $'\pm'$ is chosen according to the rule described in SC1 applied to x_t and $x_{t'}$. Then $\bar{f}$ is smooth. (Note that SC1 guarantees that $\bar{f}$ is well-defined.)

Proof: SC1. Let $x \in \partial D$, and let $(e_1, e_2) \in T_{\Theta_\omega(x)}P$ be such that (π_*e_1, π_*e_2) is an orthonormal basis of $T_{q(x)}\Sigma$. Suppose that (π_*e_1, π_*e_2) is positively oriented relative to the orientation in a neighborhood of $q(x)$ obtained via q^{-1} from the usual orientation of D in a neighborhood x . Then $f(x) = \Omega^\omega(e_1, e_2)$. Now $e_1h(x, x')$ and $e_2h(x, x')$ are vectors in $T_{\Theta_\omega(x')}P$, since by Lemma A10 we have $\Theta_\omega(x') = \Theta_\omega(x)h(x, x')$. The orientation of the projected basis $(\pi_*e_1h(x, x'), \pi_*e_2h(x, x'))$ in $T_{q(x)}\Sigma$ is the same as that of (π_*e_1, π_*e_2) , because G is connected. Therefore, $f(x') = \pm \Omega^\omega(e_1h(x, x'), e_2h(x, x')) = \pm h(x', x)f(x)h(x', x)^{-1}$, with the sign being chosen according to the rule stated in SC1.

SC2. The basic point is that $\bar{f}d\bar{\sigma}$ (for a suitable locally-defined area-form $d\bar{\sigma}$) is $s^*\Omega^\omega$ for a certain smooth section s of the bundle P over the open set $V_{i,i'}$. Consider $x = rx_t$ with $r \in (0, 1]$ and $t \in (t_{i-1}, t_i)$. We set $s(x) = \tau_\omega(q(Ox))u$; i.e. $s(x) = \Theta_\omega(x)$ for such x . Now consider $x' = r'x_{t'}$, with $r' \in (0, 1]$ and $t' \in (t_{i'-1}, t_{i'})$. We set $s(x') = \tau_\omega(q(x_{t'}x')q(Ox_t))u$, where $t \in (t_{i-1}, t_i)$ is such that $q(x_t) = q(x_{t'})$, and Ox_t and $x_{t'}x'$ are the radial segments in D . From the way the smooth structure on Σ is defined, it is clear that s is smooth and that therefore $s^*\Omega^\omega$ is smooth. Let $d\bar{\sigma}$ be the usual area-form $d\sigma'$ on V_i , and $\pm d\sigma'$ on $V_{i'}$, with the sign chosen according as q identifies $\tilde{K}_i$ with $\overline{\tilde{K}_{i'}}$ or with $\tilde{K}_{i'}$. Define the smooth g -valued function $\bar{f}$ on $V_{i,i'}$ by requiring $s^*\Omega^\omega = \bar{f}d\bar{\sigma}$. Then, since $s^*\Omega^\omega = f d\sigma'$ on V_i , we have $\bar{f}(q(rx_t)) = f(rx_t)$. Next, we have by Lemma A10,:

$$\begin{aligned} s(x') &= \tau_\omega(q(Ox')q(x_{t'}O)q(Ox_t))u = \tau_\omega(q(Ox'))u g_u(q(x_{t'}O)q(Ox_t); \omega) \\ &= \Theta_\omega(x')h(x_{t'}, x_t). \end{aligned}$$

For $e_1, e_2 \in T_{\Theta_\omega(x')}P$, we have :

$$\Omega^\omega(e_1h(x_{t'}, x_t), e_2h(x_{t'}, x_t)) = h(x_{t'}, x_t)^{-1} \Omega^\omega(e_1, e_2) h(x_{t'}, x_t).$$

By the reasoning used in SC1, this is equivalent to

$$\bar{f}(q(rx_{t'})) = \pm h(x_{t'}, x_t)^{-1} f(rx_t) h(x_{t'}, x_t).$$

Thus $\bar{f}$ is indeed as specified in (SC2). $\square$

We have used the following results from [Se2] :

Lemma A15 (Lemma A3 of [Se2]). *Let C be an admissible closed curve in Σ based at $o (=q(O))$. Then C is a reduction of a composite of proper loops (i.e. C can be obtained from a composite of proper loops by dropping successively segments which are traversed in opposite directions consecutively). $\square$*

Lemma A16 (Lemma A2 of [Se2]). *Let z be a G -field (G being any group) over a grid Γ in D and $\alpha_1, \dots, \alpha_p \in G$, such that $z(L_m) \dots z(L_1) = W_\Sigma(\alpha_1, \dots, \alpha_p)$, where $L_1, \dots, L_m$ is the usual sequence of lassos in Γ . We take $\alpha_p = e$ if Σ is a sphere with $k(\geq 0)$ holes. Let C be a loop in $\Gamma' = q(\Gamma)$, based at o , such that C is a reduction of $\tilde{C}_k \dots \tilde{C}_1$ and also of $\tilde{C}'_1 \dots \tilde{C}'_l$, where every $\tilde{C}_j$ and every $\tilde{C}'_j$ is a proper loop. If $\tilde{C}_j$ is the projection by q of some loop in D then we denote that (not necessarily unique) loop by C_j ; i.e. $q(C_j) = \tilde{C}_j$. On the other hand, if $\tilde{C}_j$ is some S_k (or $\bar{S}_k$) we set $C_j = \tilde{C}_j$ and write $z(C_j)$ for α_k (or α_k^{-1} , respectively). We introduce a similar notation C'_j and $z(C'_j)$ with respect to $\tilde{C}'_j$. Then :*

$$z(C_k) \dots z(C_1) = z(C'_l) \dots z(C'_1). \quad \square$$

Lemma A17 (Lemma A4 of [Se2]). *Denote by A either $(0, \infty)$ or an abelian group (with operation written additively). Let $\mathcal{G}$ be a connected finite graph, $V_{\mathcal{G}}$ the set of all vertices of $\mathcal{G}$, and X the set of all maps $t : V_{\mathcal{G}} \rightarrow A : v \mapsto t_v$. Suppose f is a function on X such that $f(t) = f(t')$ whenever there exist two vertices $v_1, v_2 \in V_{\mathcal{G}}$ connected by an edge such that $t_v = t'_v$ for all v other than v_1 and v_2 , and :*

$$t_{v_1} + t_{v_2} = t'_{v_1} + t'_{v_2}.$$

Then there is a function F on A such that $f(t) = F(\sum_{v \in V_{\mathcal{G}}} t_v)$. $\square$

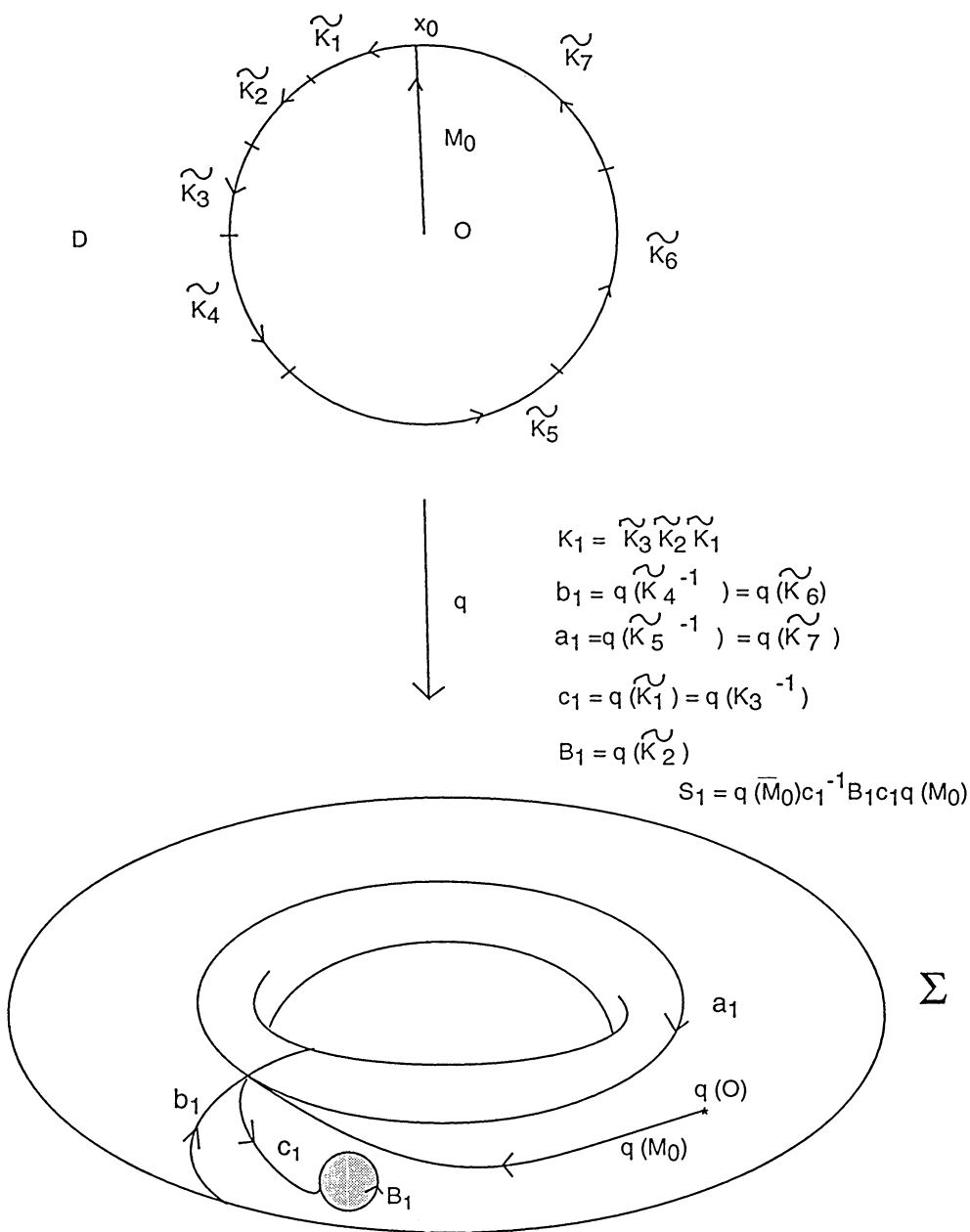


Figure 1. The torus with one hole

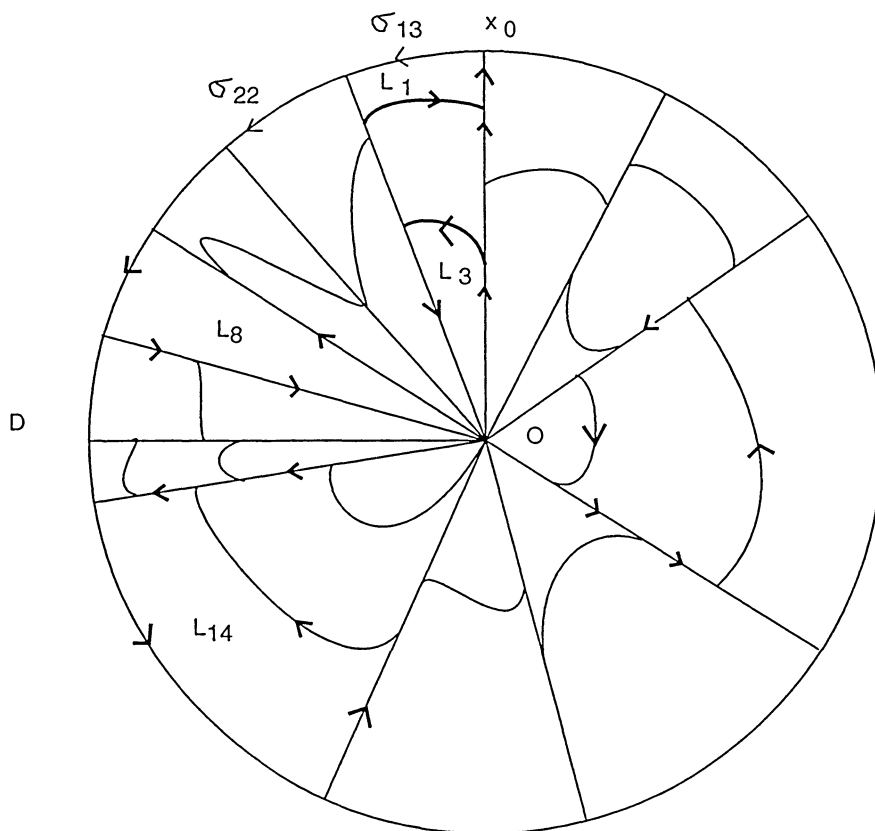


Figure 2. A 'grid' and some of its 'lassos'

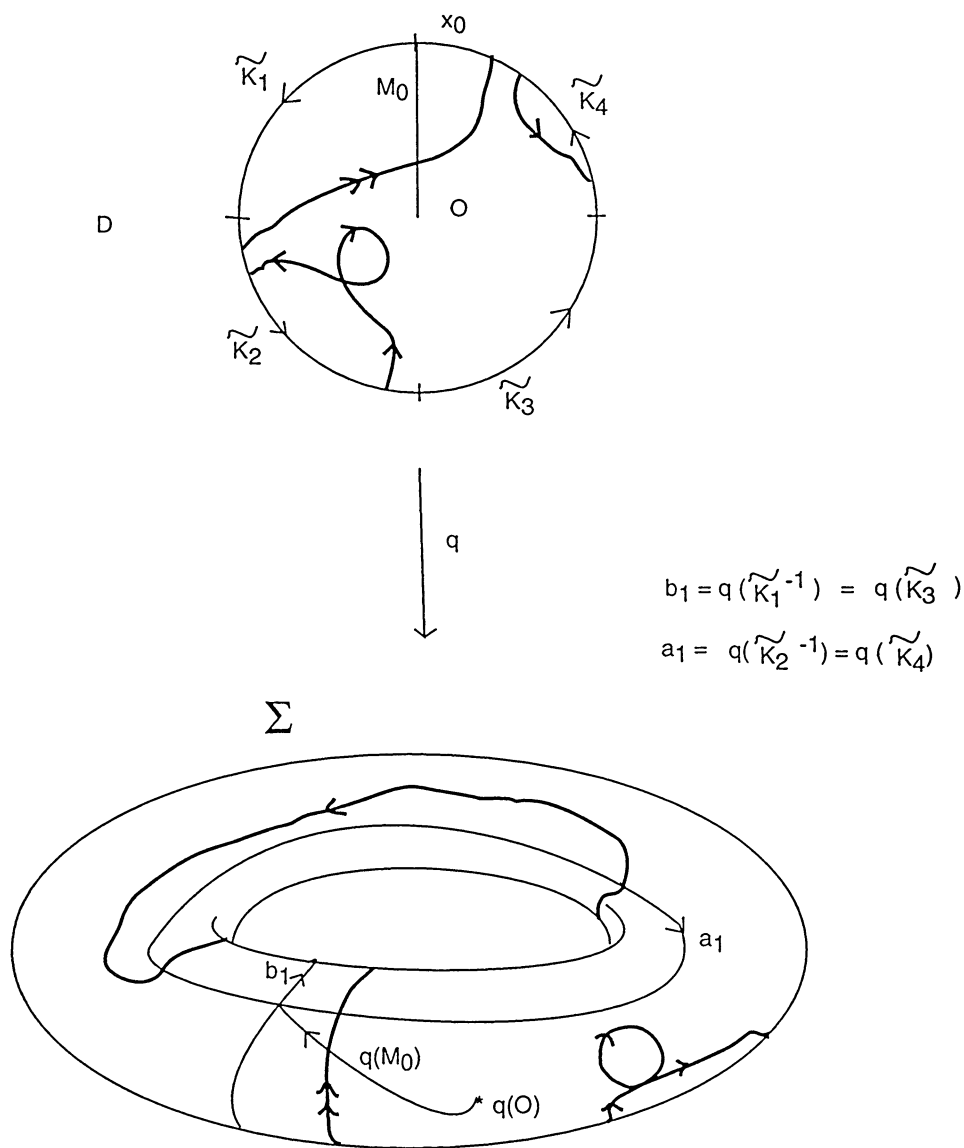


Figure 3. An 'admissible collection' of curves on the torus

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